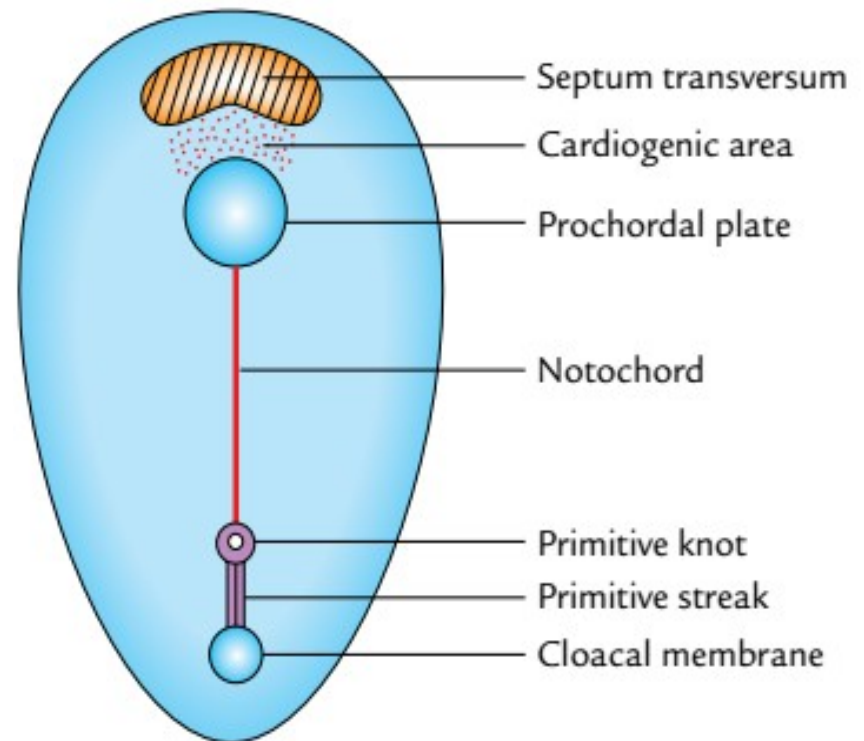


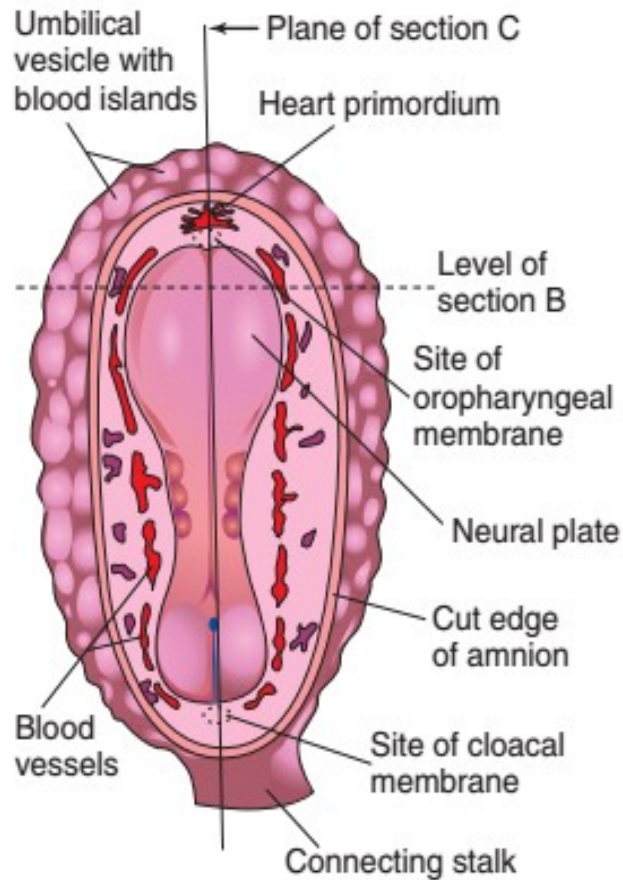
Development of Heart

Development of Heart

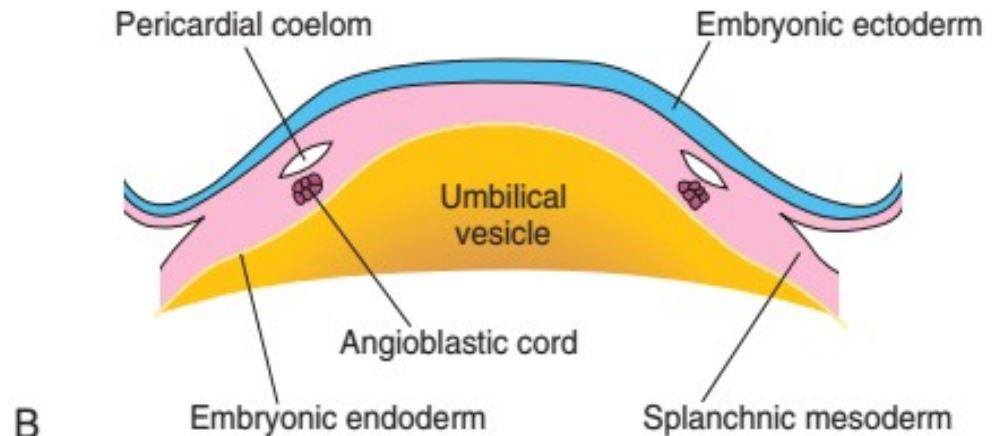
- Cardiovascular system is the first major system to function in the embryo
 - heart starts functioning at the end of the third week of intrauterine life
 - The blood flow begins - 4th week of intrauterine life
- The heart is **mesodermal** in origin.
- It develops from primitive heart tube, which forms from mesenchyme in the **cardiogenic area** of the embryo.



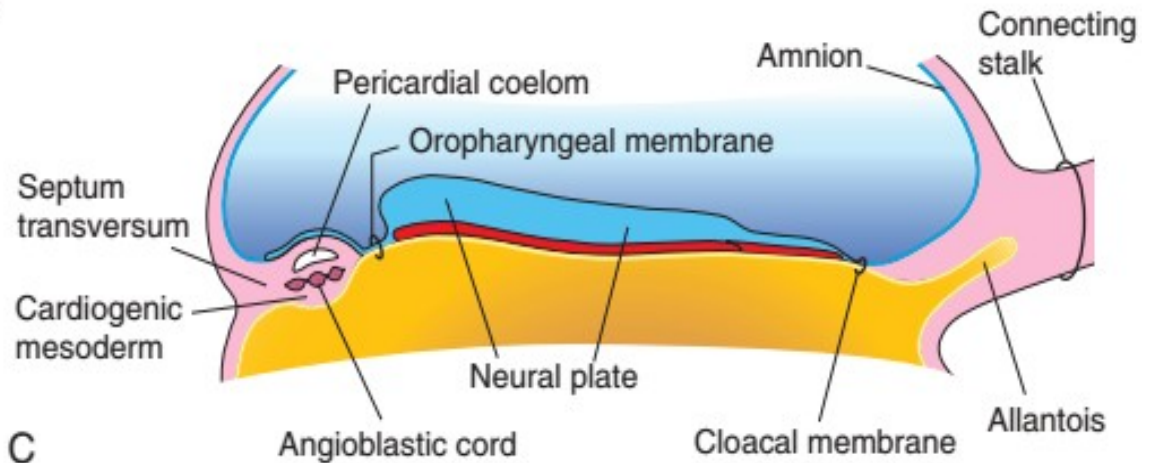
Development of Heart



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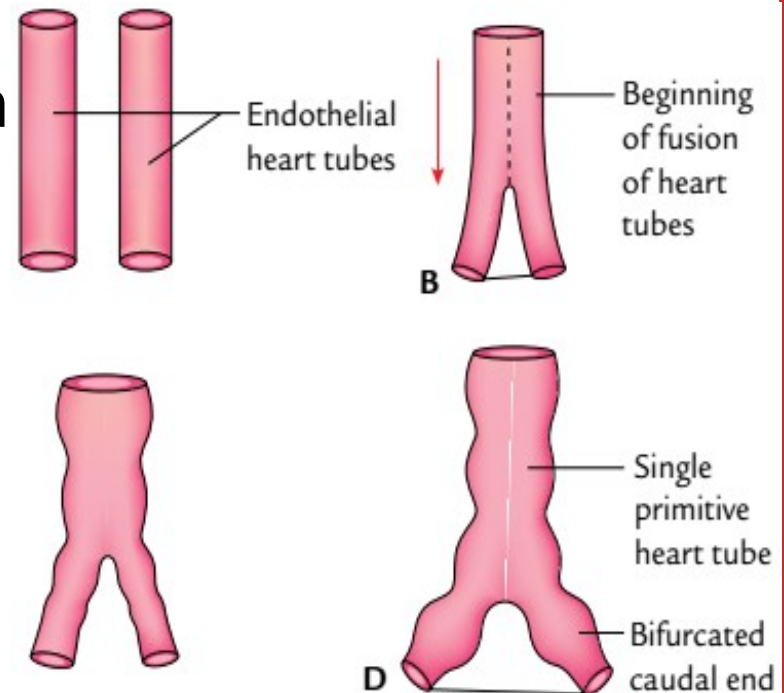
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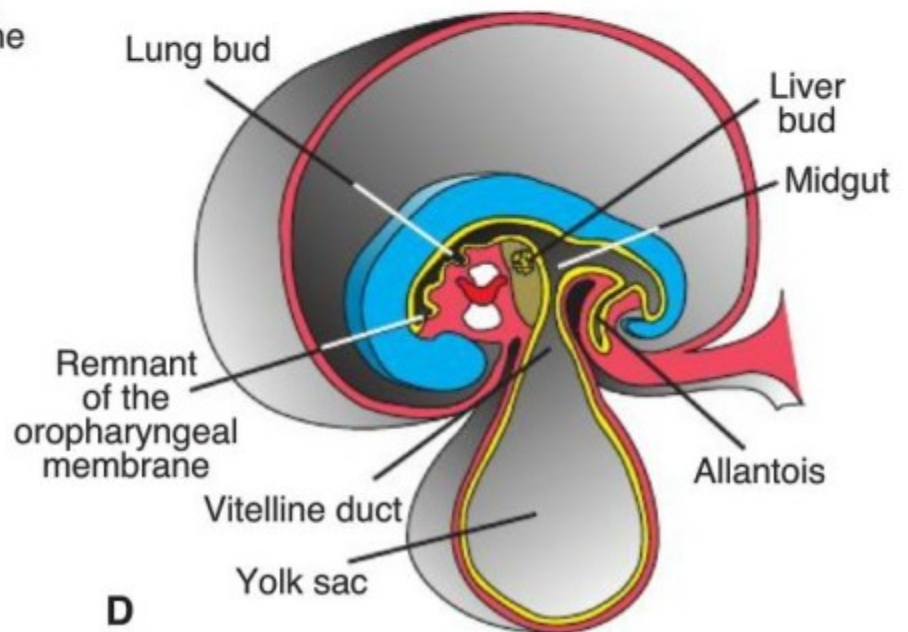
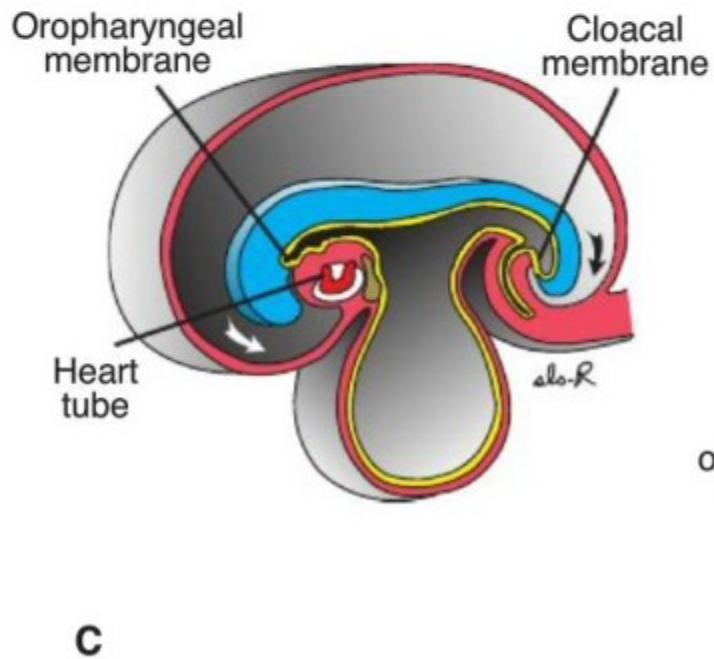
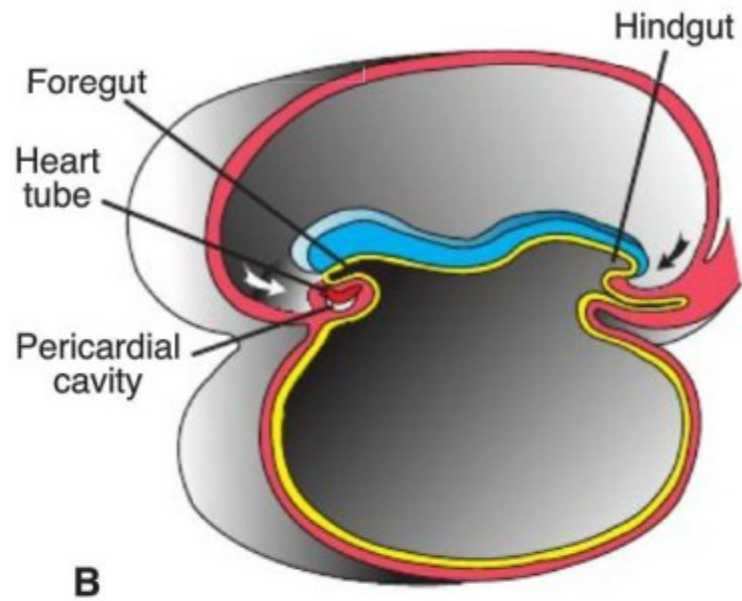
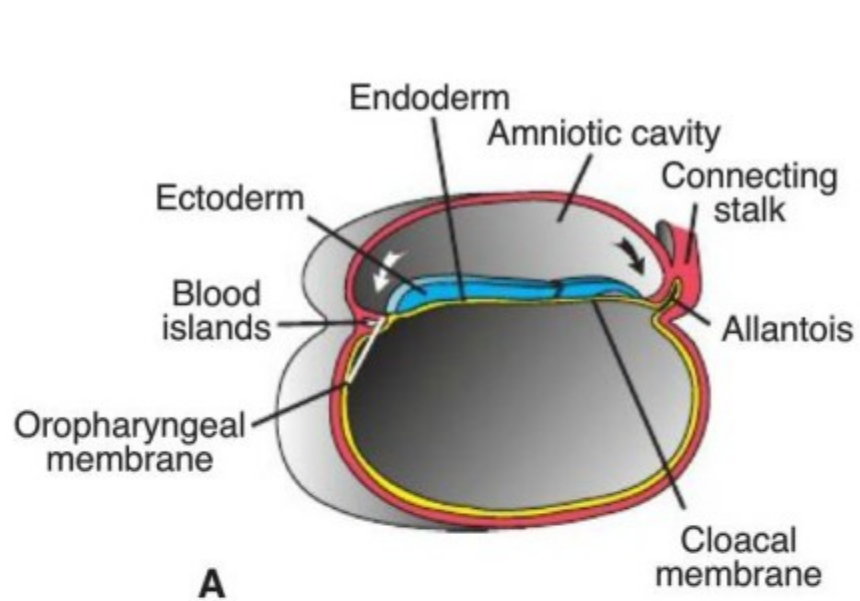


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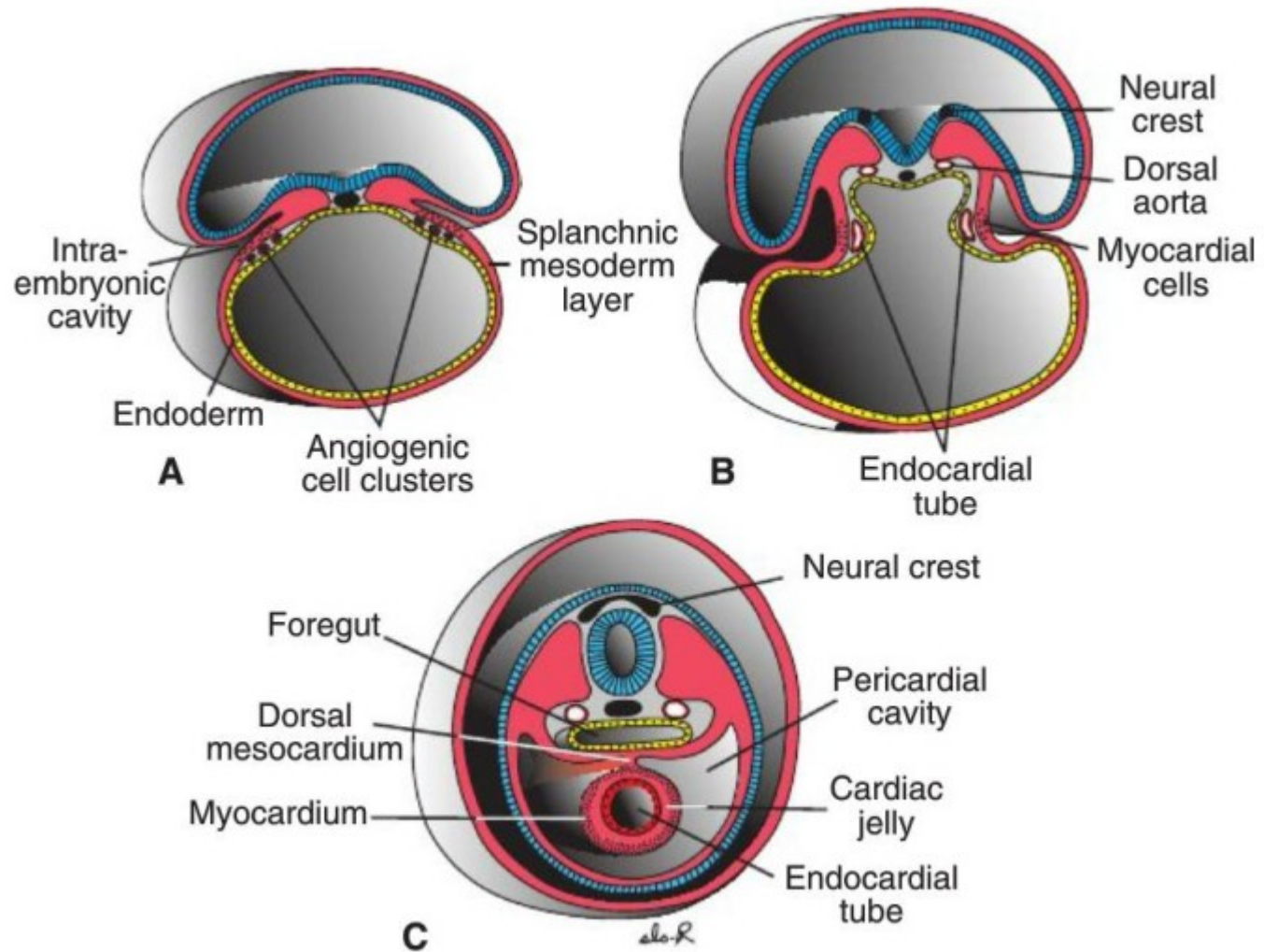
Formation of Heart Tube

- The mesenchymal cells in the cardiogenic area condense to form **two angioblastic cords** (cardiogenic cords)
 - These cords get canalized to form **two endothelial (endocardial) heart tubes**
 - These tubes fuse with each other in a craniocaudal direction to form a **single primitive heart tube**
- caudal ends of two heart tubes fail to fuse with each other (remains bifurcated)



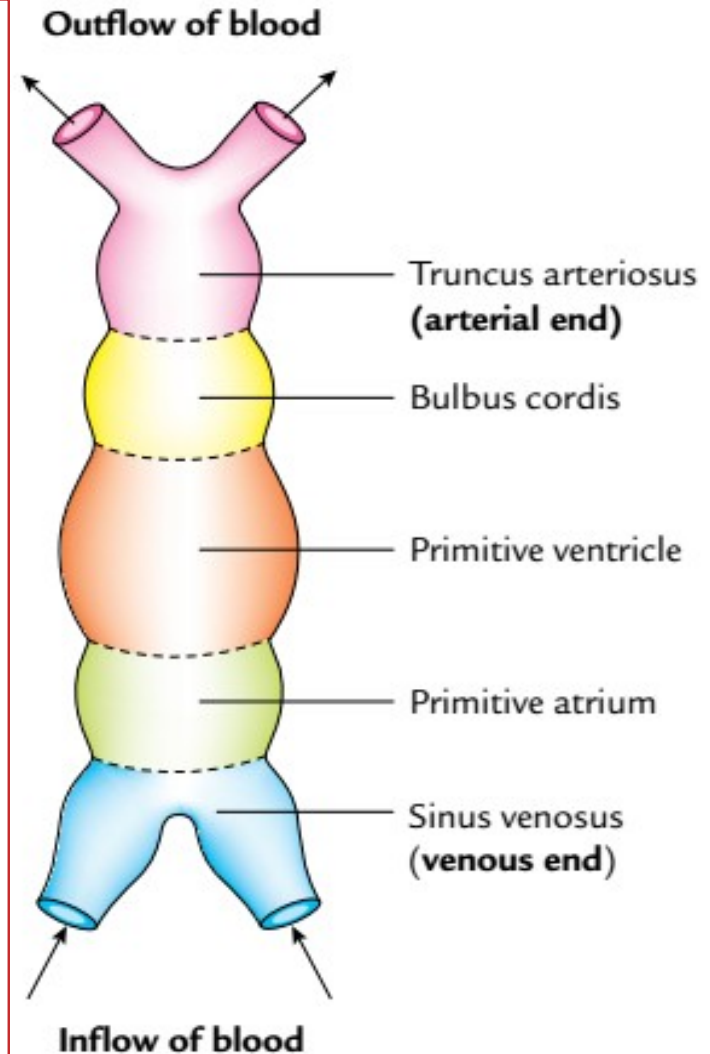


Cross sections showing formation of a single heart tube



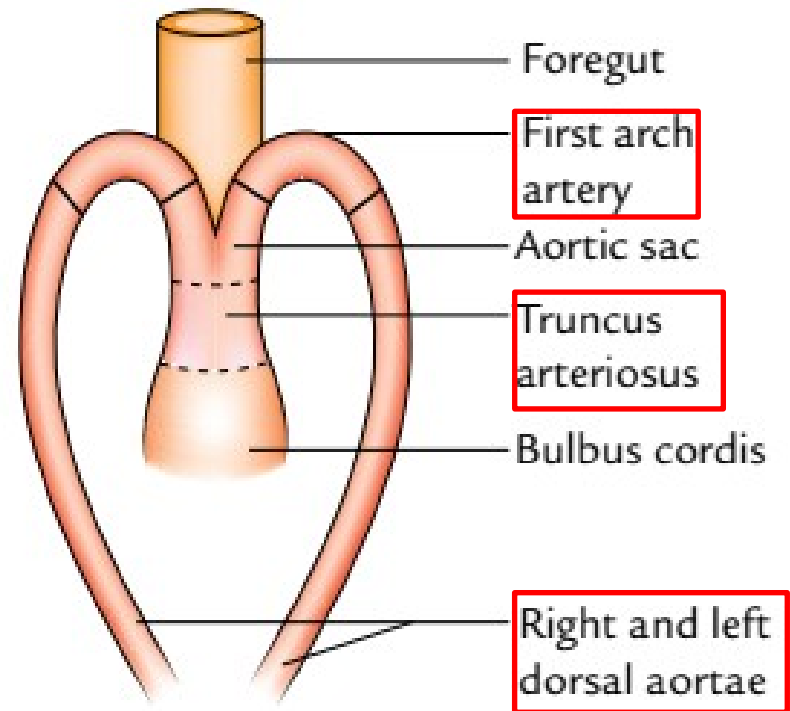
Primitive Heart Tube

- The heart tube forms five dilatations
- From cranial-to-caudal end, these are as follows:
 1. Truncus arteriosus (TA)
 2. Bulbus cordis (BC)
 3. Primitive ventricle (PV)
 4. Primitive atrium (PA)
 5. Sinus venosus (SV)
- TA is the arterial end of the primitive heart tube
- Sinus venosus is the venous end of this tube



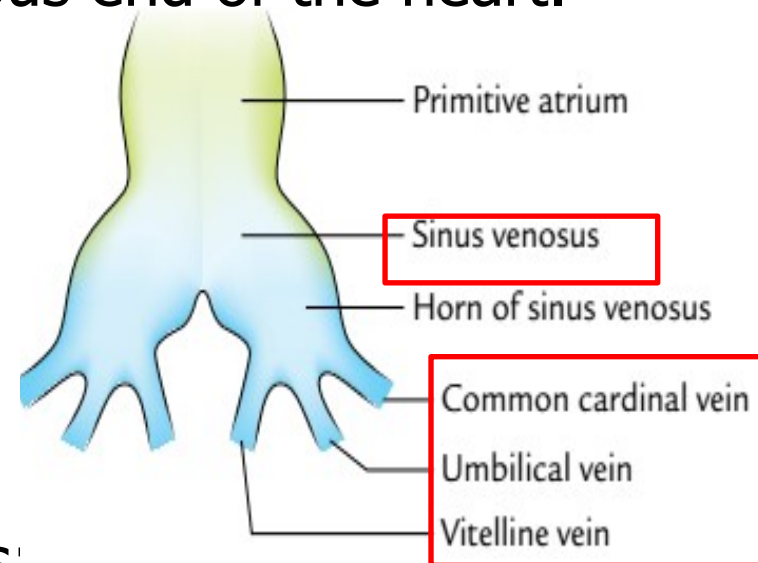
Primitive Heart Tube: **Arterial end**

- **Truncus arteriosus** represents the arterial end of the heart
 - Cranially it is **continuous with aortic sac** having right and left horns.
 - From each horn of aortic sac, the **first pharyngeal arch artery** arises
 - 1st pharyngeal arch artery passes backwards to become continuous with the respective **dorsal aorta**

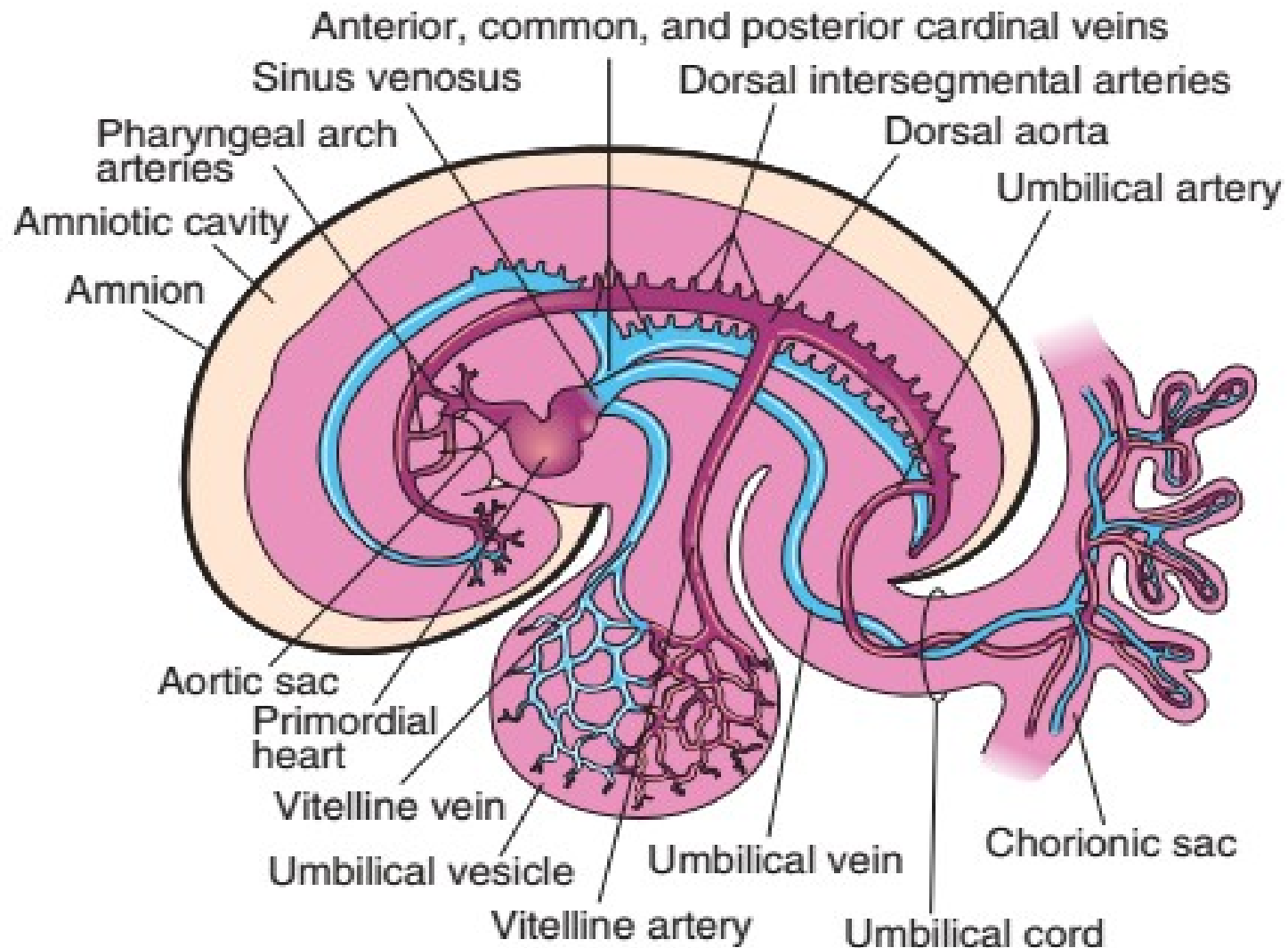


Primitive Heart Tube: Venous end

- **Sinus venosus** represents the venous end of the heart.
- Sinus venosus has
 - central part that communicates with the primitive atrium
 - right and left horns
- Each horn of sinus venosus receives **three** paired primitive veins:
 - **Vitelline vein** from the **yolk sac (umbilical vesicle)** - poorly oxygenated blood
 - **Umbilical vein** from the **placenta** - well-oxygenated blood
 - **Common cardinal vein** from the **body wall of the embryo to the heart** - poorly oxygenated blood

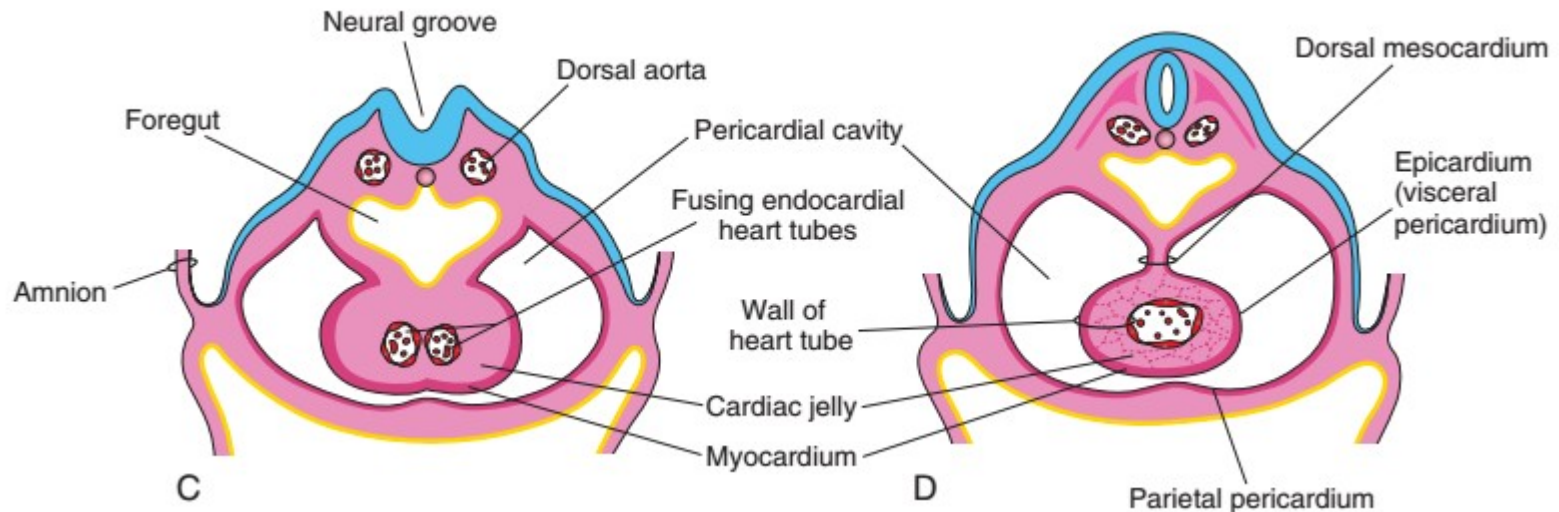


Embryonic cardiovascular system – day 26



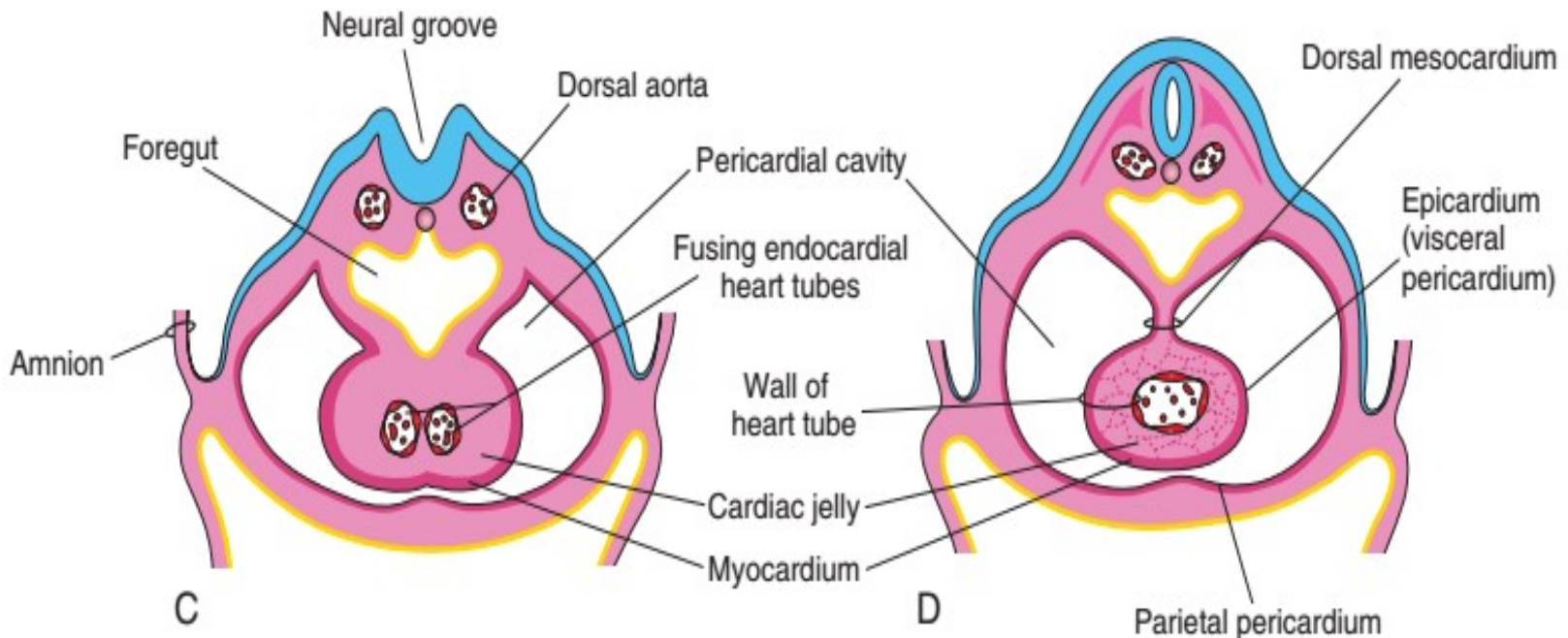
Formation of Heart Wall

- The splanchnic mesoderm surrounding the pericardial cavity forms the **primordial myocardium**, the external layer of the embryonic heart tube.
- The developing heart is composed of a thin endothelial tube, separated from a thick myocardium by a gelatinous (rich in hyaluronic acid) matrix of connective tissue, **cardiac jelly**.



Formation of Heart Wall

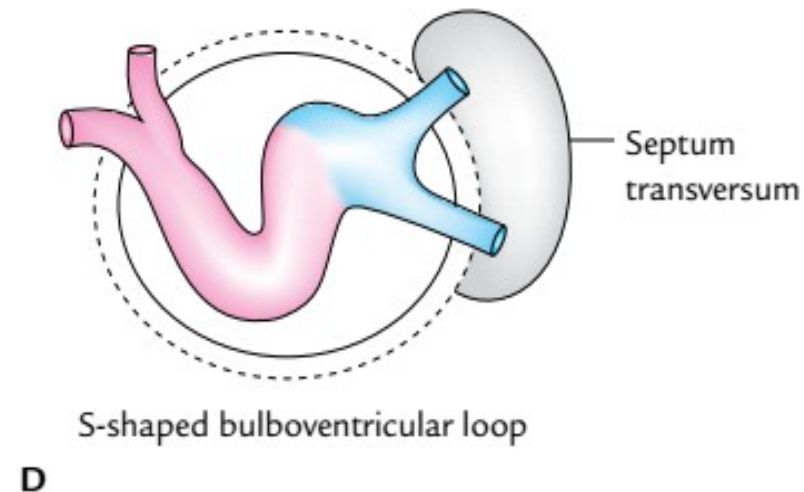
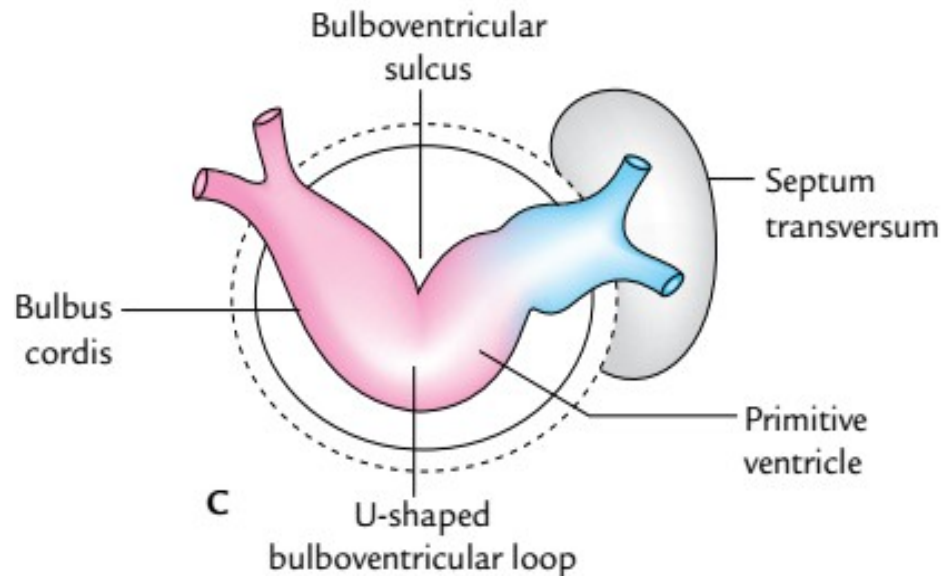
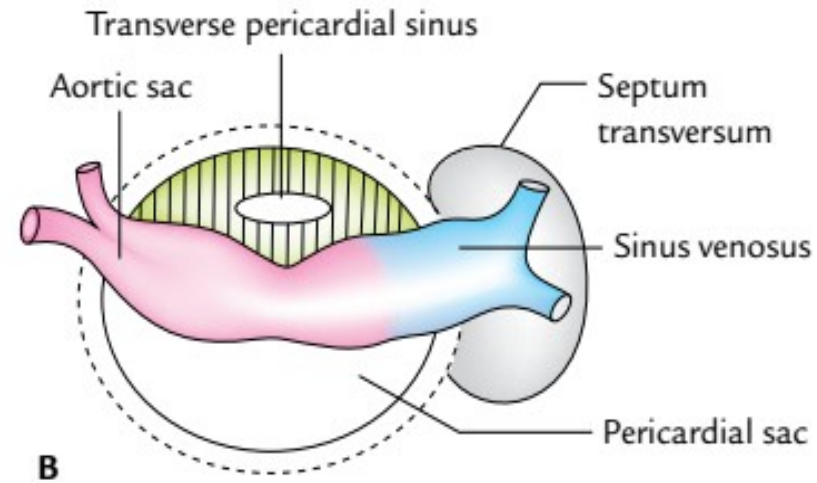
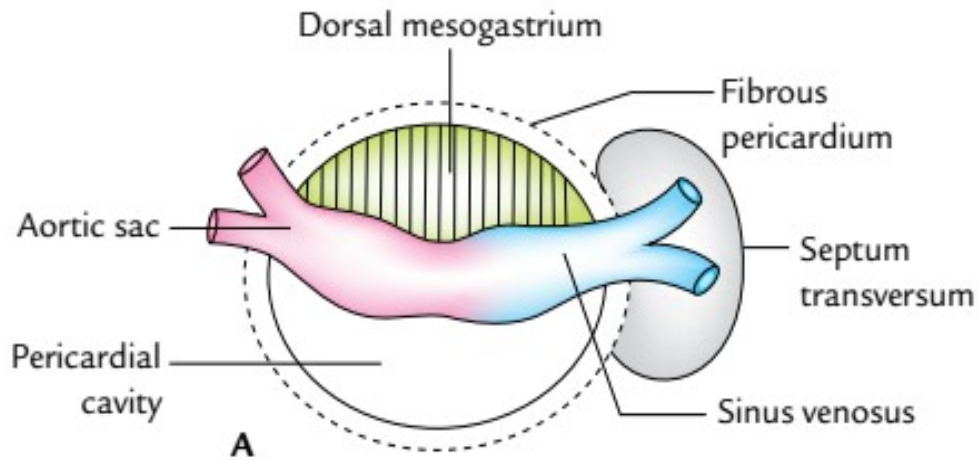
- The endothelial tube becomes **endocardium**
- The primordial myocardium becomes **myocardium**.
- The **epicardium**, is derived from mesothelial cells that arise from the external surface of the sinus venosus and spread over the myocardium



Formation of cardiac loop

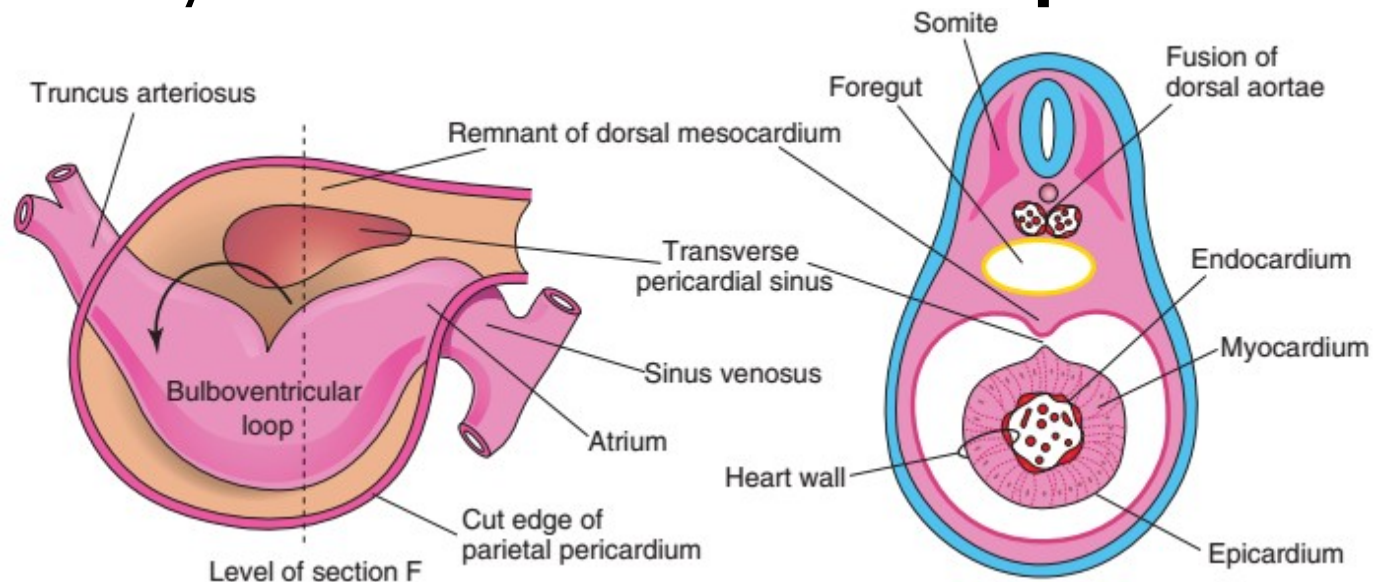
- The arterial and venous ends of the heart are fixed by the pharyngeal arches and septum transversum, respectively.
- Bulbus cordis and primitive ventricle grows rapidly in a limited space dorsal to the pericardial cavity.
- The heart tube gradually invaginates into the pericardial cavity in a 'U' shaped manner; forming a **cardiac loop** or **bulboventricular loop**
- The cardiac tube begins to bend (looping) on day 23 and completes by day 28.
- As the primordial heart bends, the atrium and sinus venosus come to lie dorsal to the TA, bulbus cordis, and ventricle.

Formation of cardiac loop



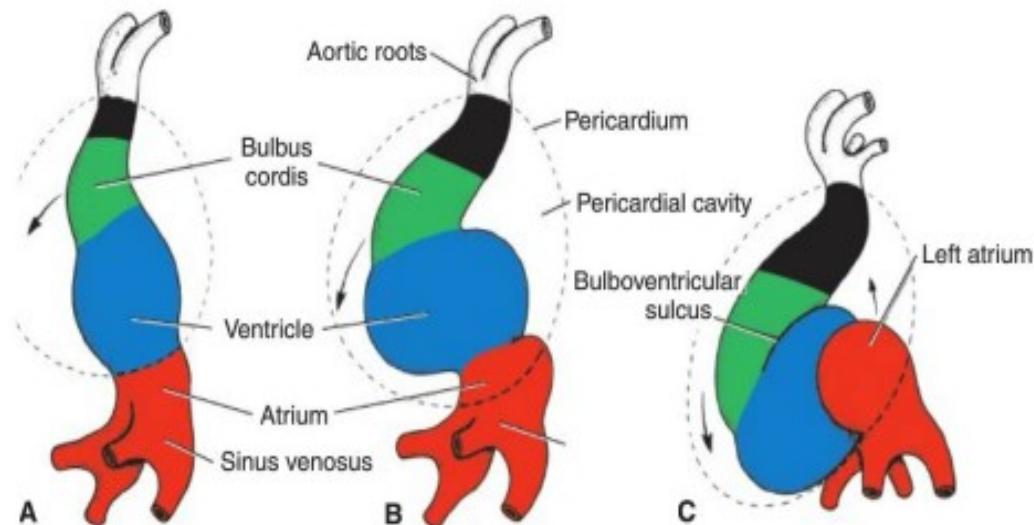
Formation of cardiac loop

- As the heart elongates and bends, it gradually invaginates into the pericardial cavity
- The heart is initially suspended from the dorsal wall of the pericardial sac by the **dorsal mesocardium**
- the central part of this mesentery soon degenerates, forming a communication between the right and left sides of the pericardial cavity & becomes the **transverse pericardial sinus**

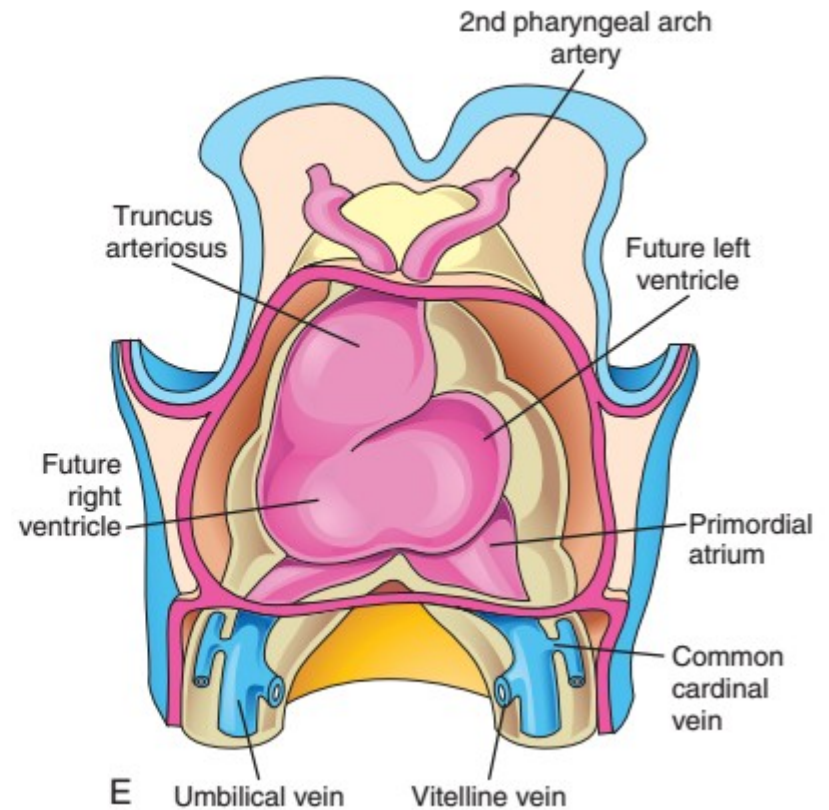
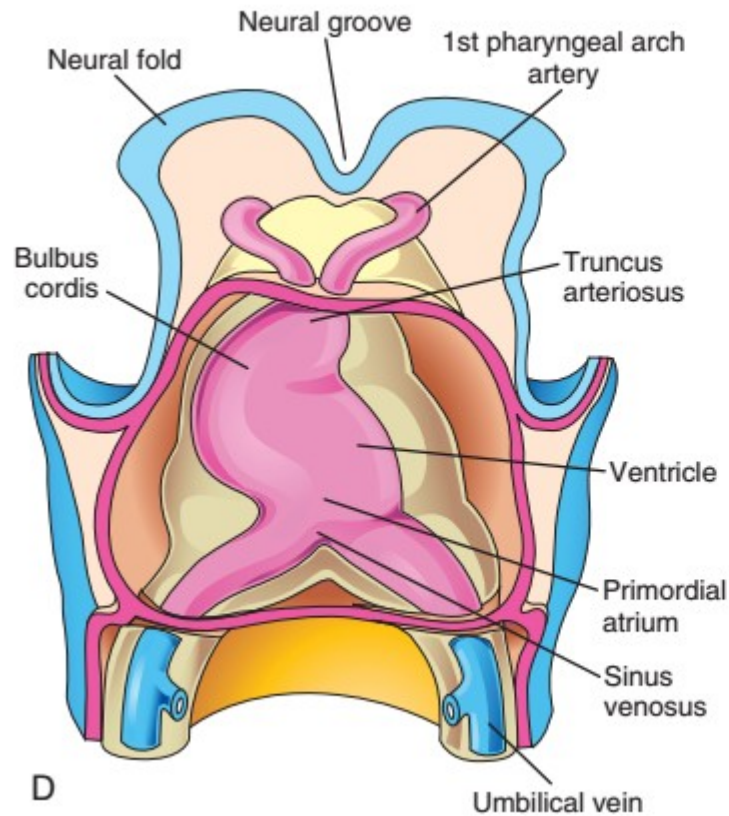


Formation of cardiac loop

- Primitive atrium and sinus venosus occupy position posterosuperior to the ventricle in the pericardial cavity.
- The heart tube now becomes **S shaped**.
- The bulboventricular sulcus disappears, and bulbus cordis and primitive ventricle together form a common chamber.
- The atrial chamber behind the truncus arteriosus expands & projects on either side of truncus arteriosus as **auricles**



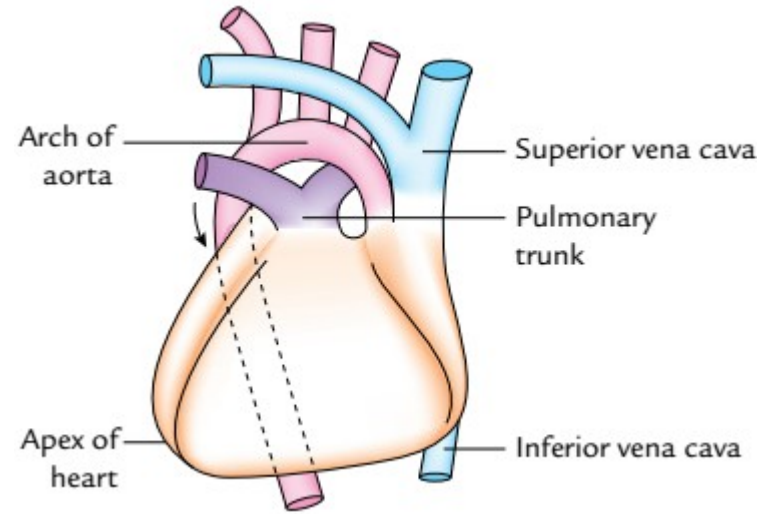
Formation of cardiac loop



Clinical

- **Dextrocardia**
 - heart tube bends to the left instead of to the right, the heart is displaced to the right
- **Ectopia cordis**

Heart exposed to the exterior



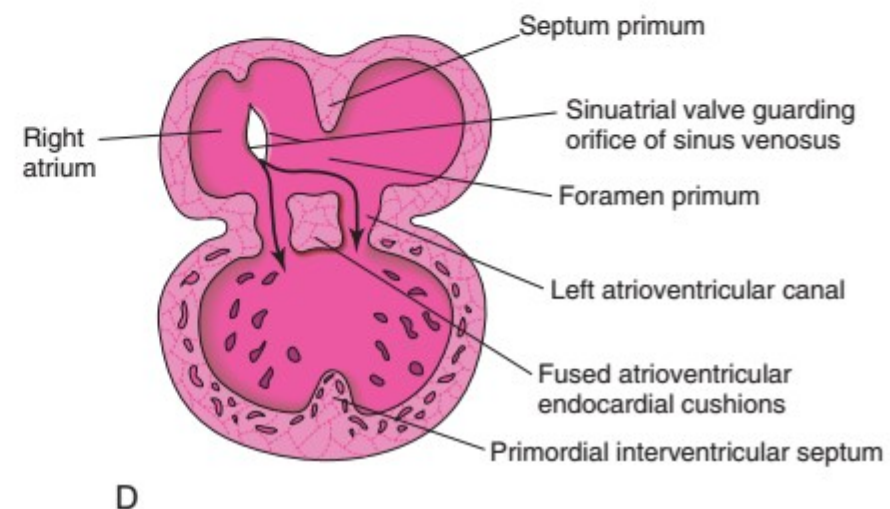
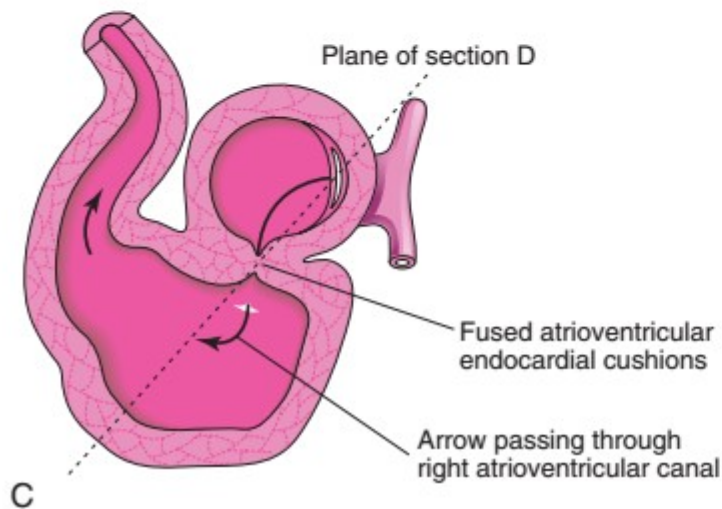
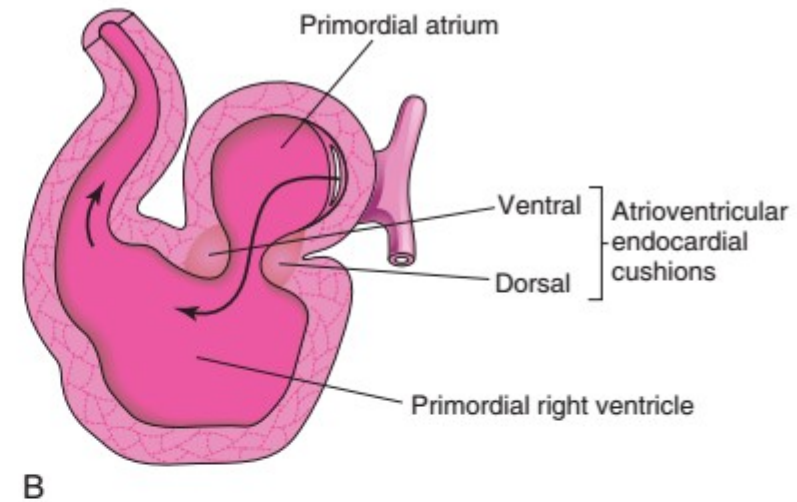
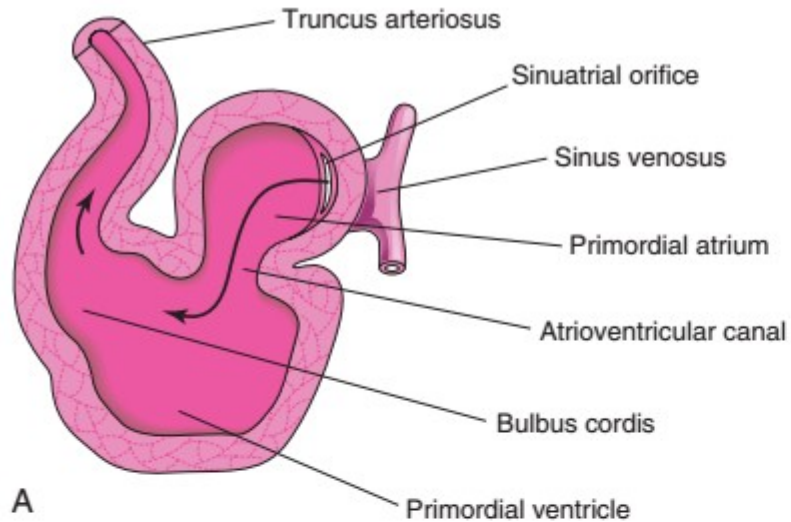
Partitioning of the primordial Heart

- The primitive heart tube has a **single lumen**.
 - AV canal, primordial atrium, ventricle, and outflow tract
- Partitioning begins during the middle of 4th week
 - Completed by the end of 8th week
- Partitioned into **four definitive** chambers by the septa
- These septa are:
 - Atrioventricular septum
 - Interatrial septum
 - Interventricular septum
 - Aorticopulmonary septum

Partitioning of Atrioventricular Canal

- **AV endocardial cushions**
 - form on the dorsal and ventral walls of the AV canal by the end of the 4th week
 - develop from a cardiac jelly and neural crest cells
 - The AV endocardial cushions approach each other and fuse
 - AV cushions divide the AV canal into **right** and **left canals**
 - These canals partially separate the primordial atrium from the primordial ventricle
 - The endocardial cushions function as AV valves.
 - The septal valves are derived from the fused superior and inferior endocardial cushions.
 - The mural leaflets are mesenchymal in origin

Partitioning of the primordial Heart



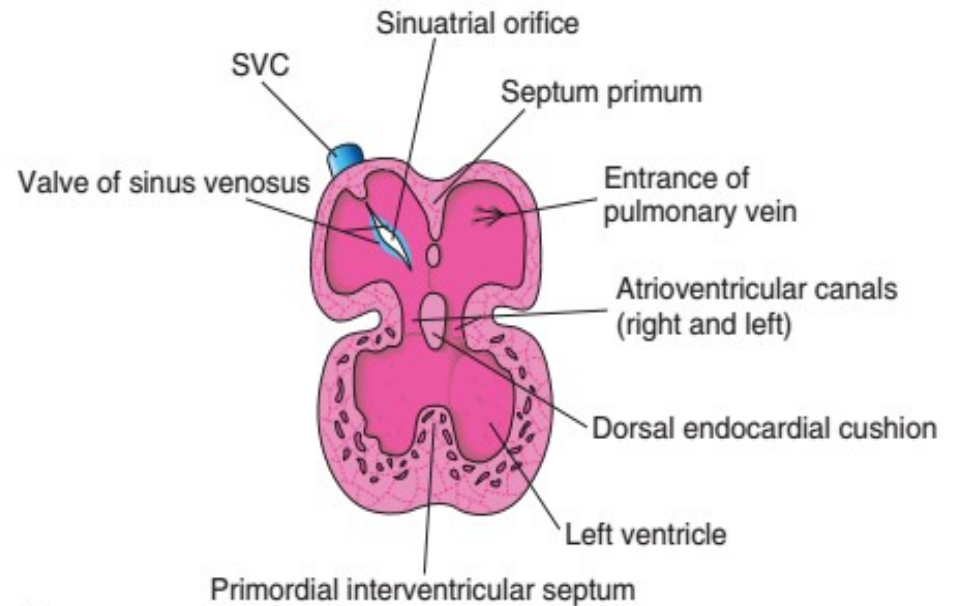
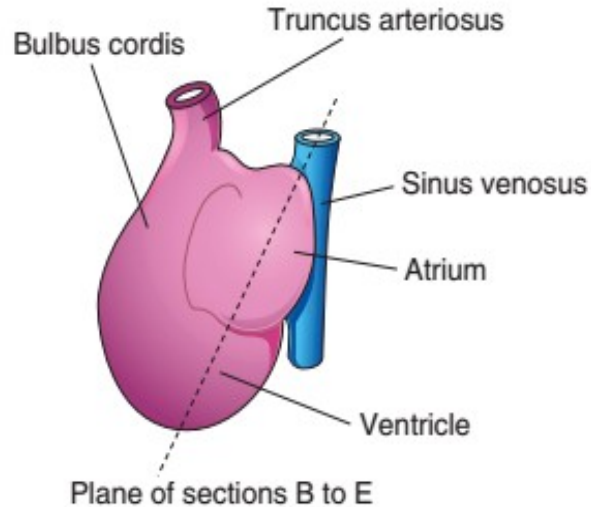
Partitioning of Primordial Atrium

- The primordial atrium is divided into right and left atria by the formation of **two septa** (at the end of the 4th week)
 - septum primum and septum secundum
- The **septum primum** grows toward the fusing endocardial cushions from the roof of the primordial atrium
 - partially divides the common atrium into right & left halves
- **foramen primum** is a large opening located between the free edge of septum primum and the endocardial cushions
 - enables oxygenated blood to pass from the right to the left atrium
 - The foramen becomes smaller and disappears as the mesenchymal cap of the septum primum fuses with the fused AV endocardial cushions to form a primordial AV septum

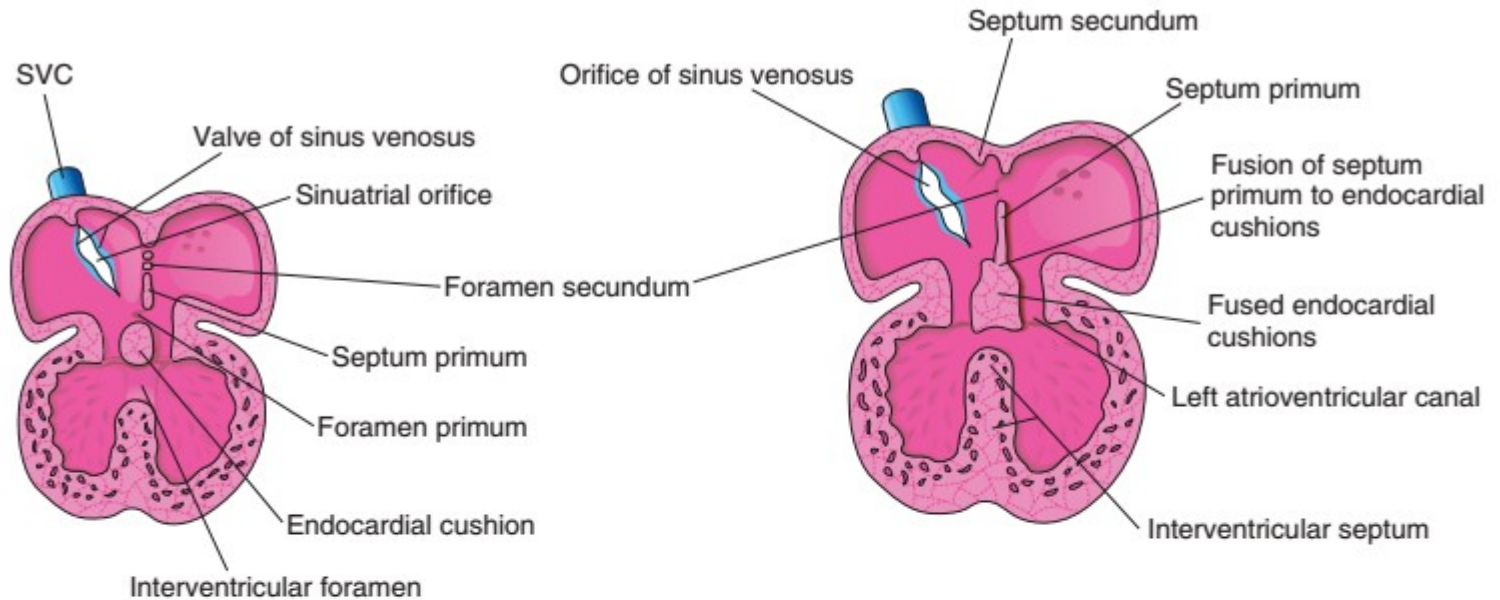
Partitioning of Primordial Atrium

- Before septum primum completes fusion with AV endocardial cushion, perforations (due to cell death) appear in the central part of the septum primum
 - Perforations coalesce together to form **foramen secundum**
 - the free edge of the septum primum fuses with the left side of the fused endocardial cushions, obliterating the foramen primum.
- The foramen secundum ensures continued shunting of oxygenated blood from the right to the left atrium.
- A second crescent-shaped septum now arises from the roof of right atrium immediately adjacent to the septum primum & is called **septum secundum**.

Partitioning of Primordial Atrium

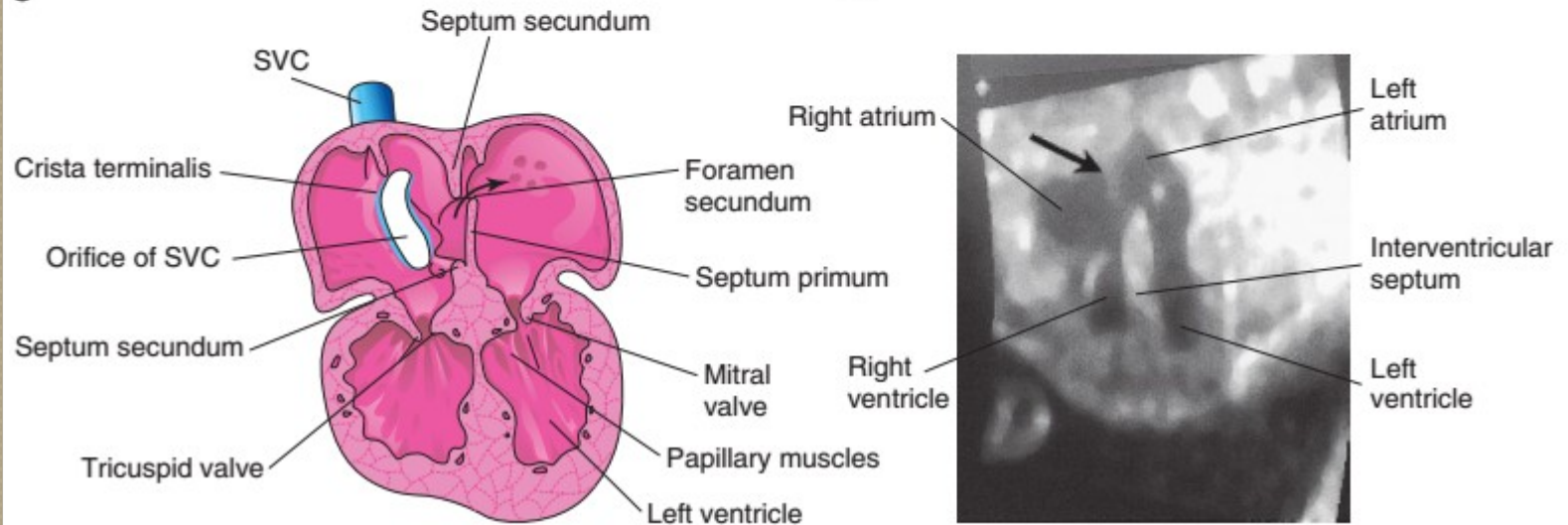


Partitioning of Primordial Atrium



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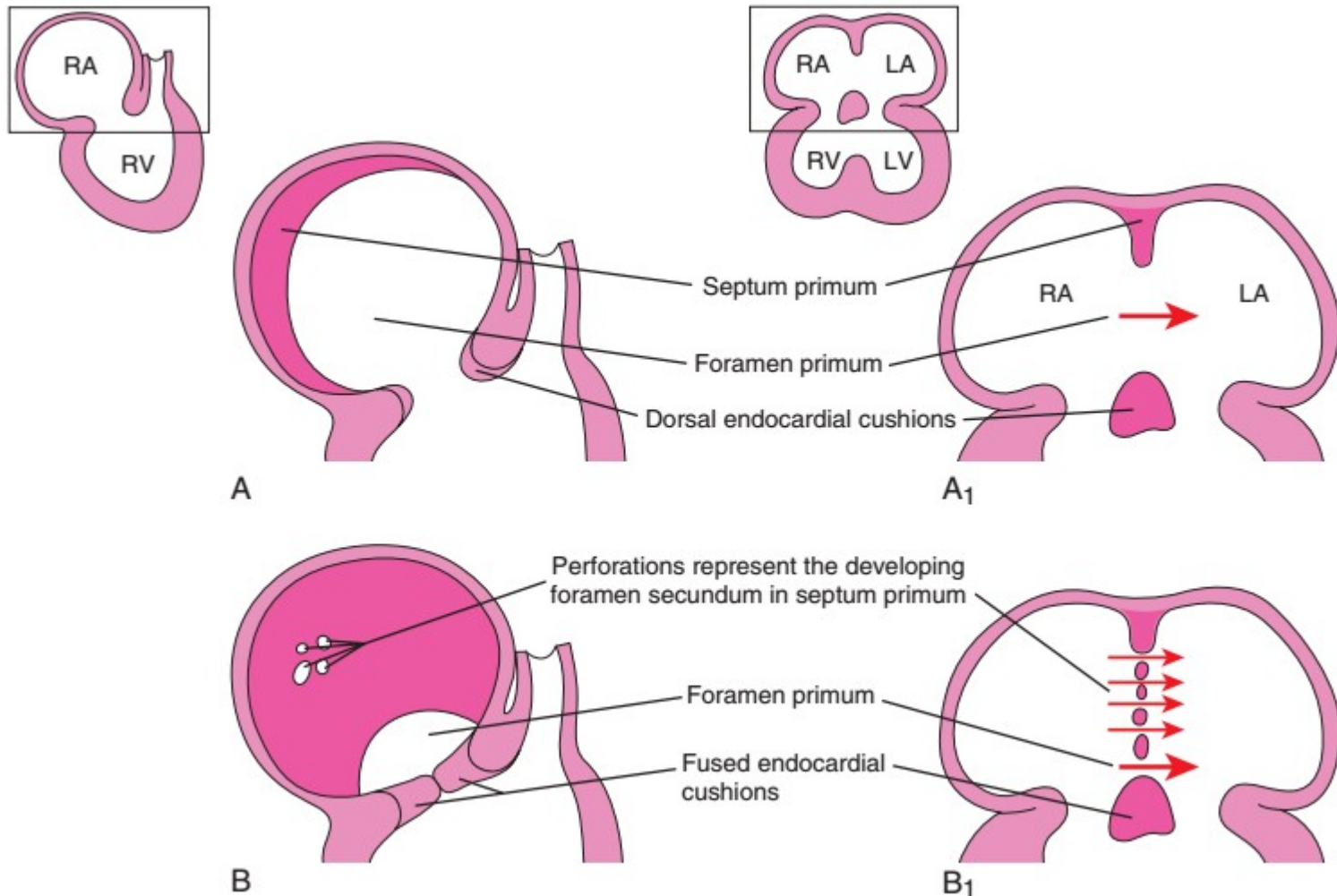
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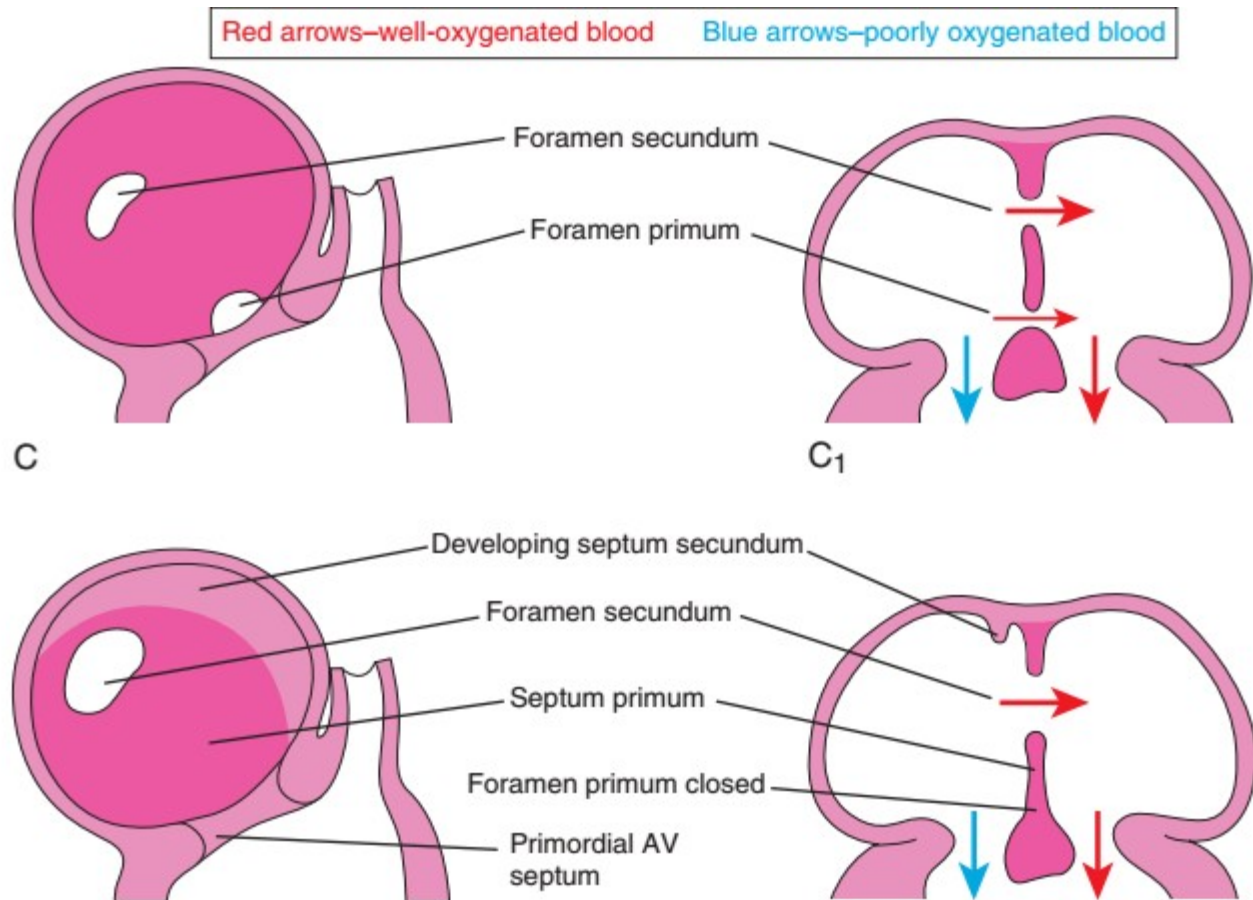
Partitioning of Primordial Atrium

- As **septum secundum** grows, it overlaps the foramen secundum forming an incomplete partition between the atria
 - The right and left atria communicate with each other through an oblique valvular passage between the upper margin of septum primum and lower margin of septum secundum.
 - This passage is called **foramen ovale**
- The cranial part of the septum primum gradually disappears.
- The lower part of the septum primum forms the flap-like valve of the foramen ovale
 - prevent blood from moving from the left atrium into the right atrium

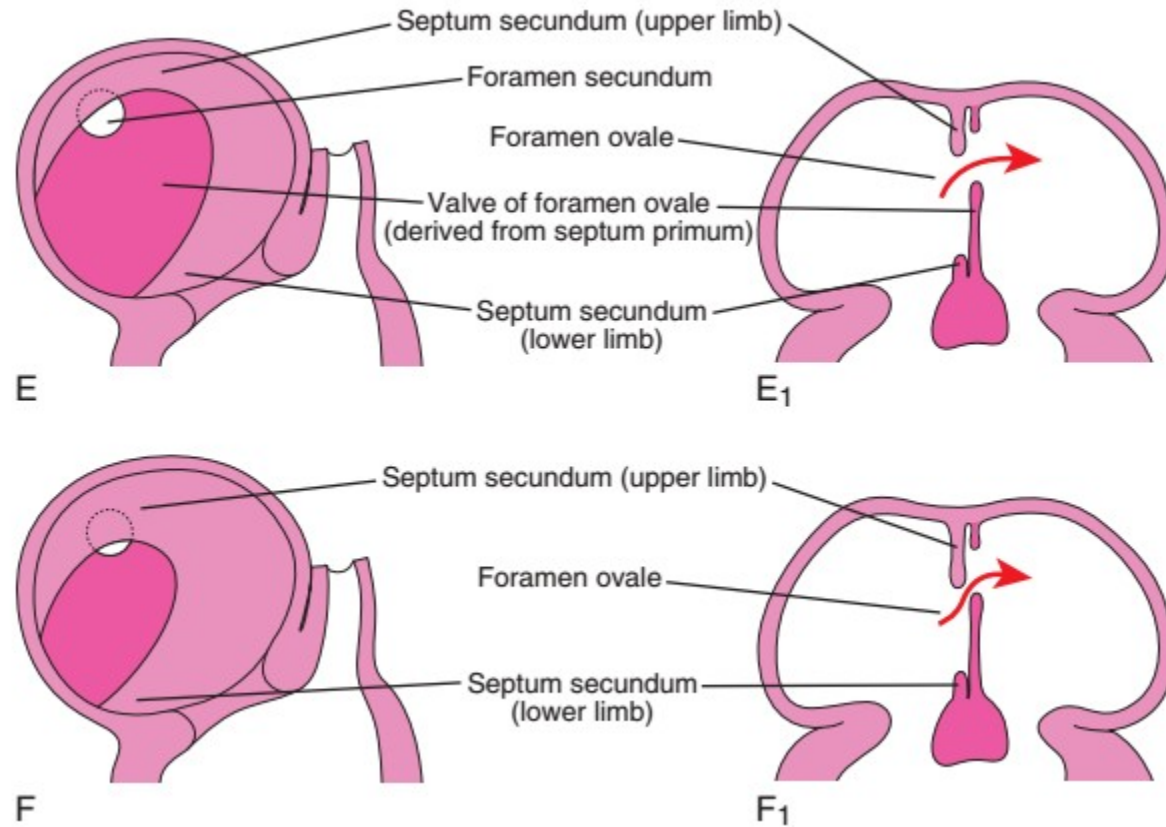
Partitioning of Primordial Atrium



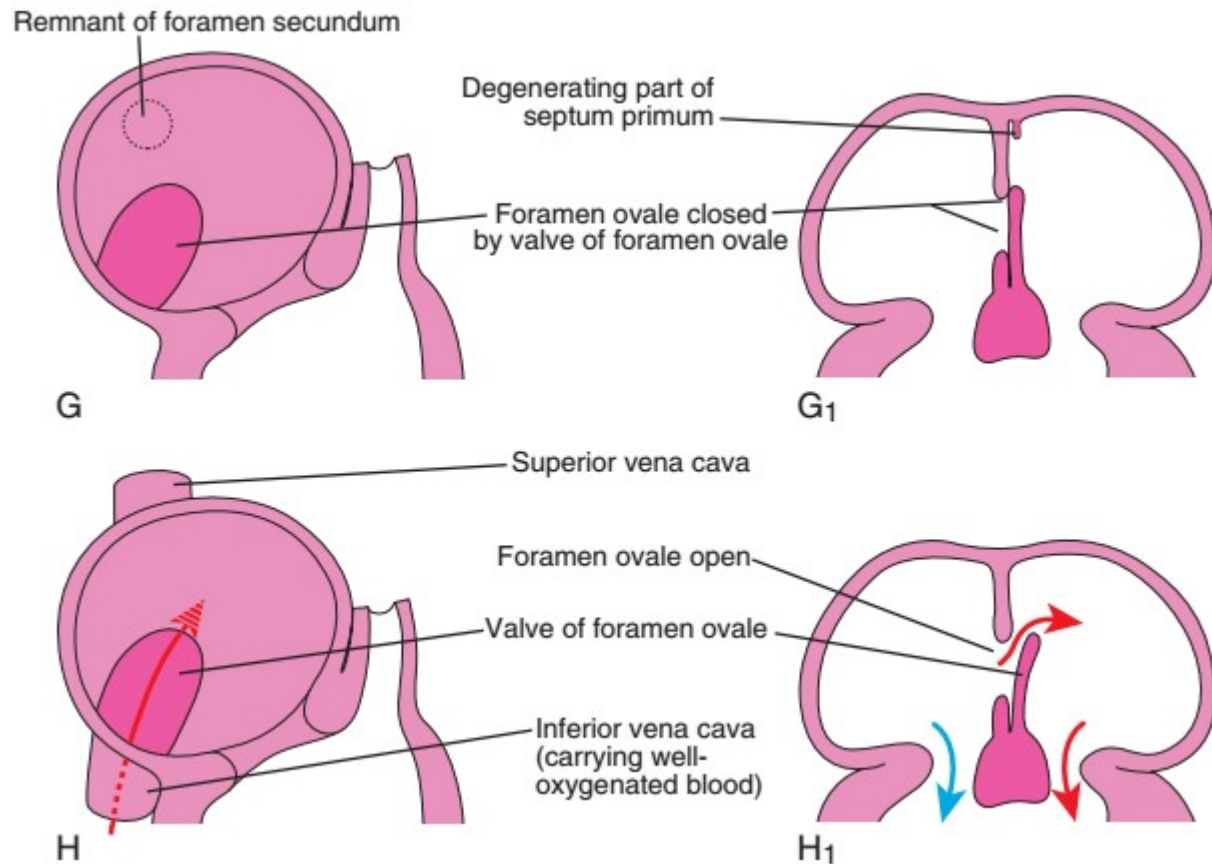
Partitioning of Primordial Atrium



Partitioning of Primordial Atrium

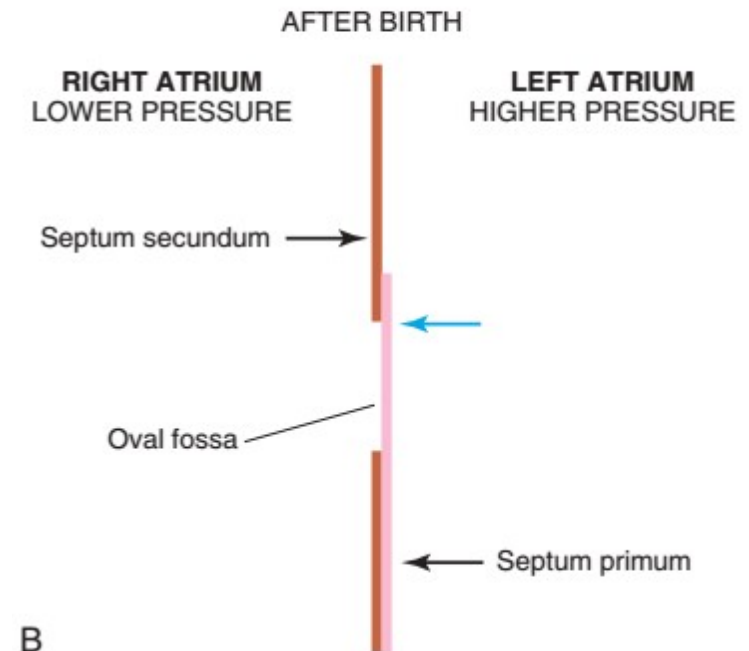
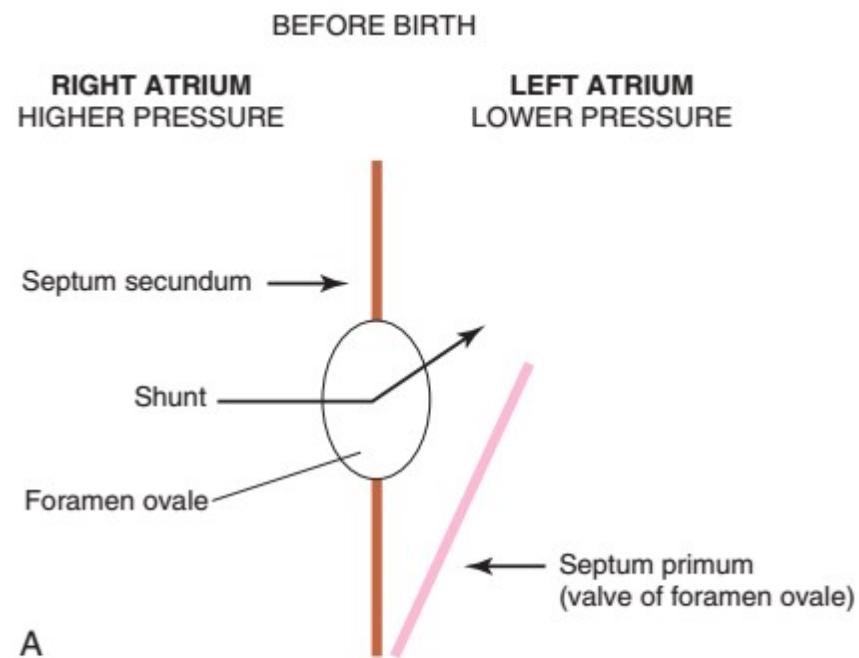


Partitioning of Primordial Atrium

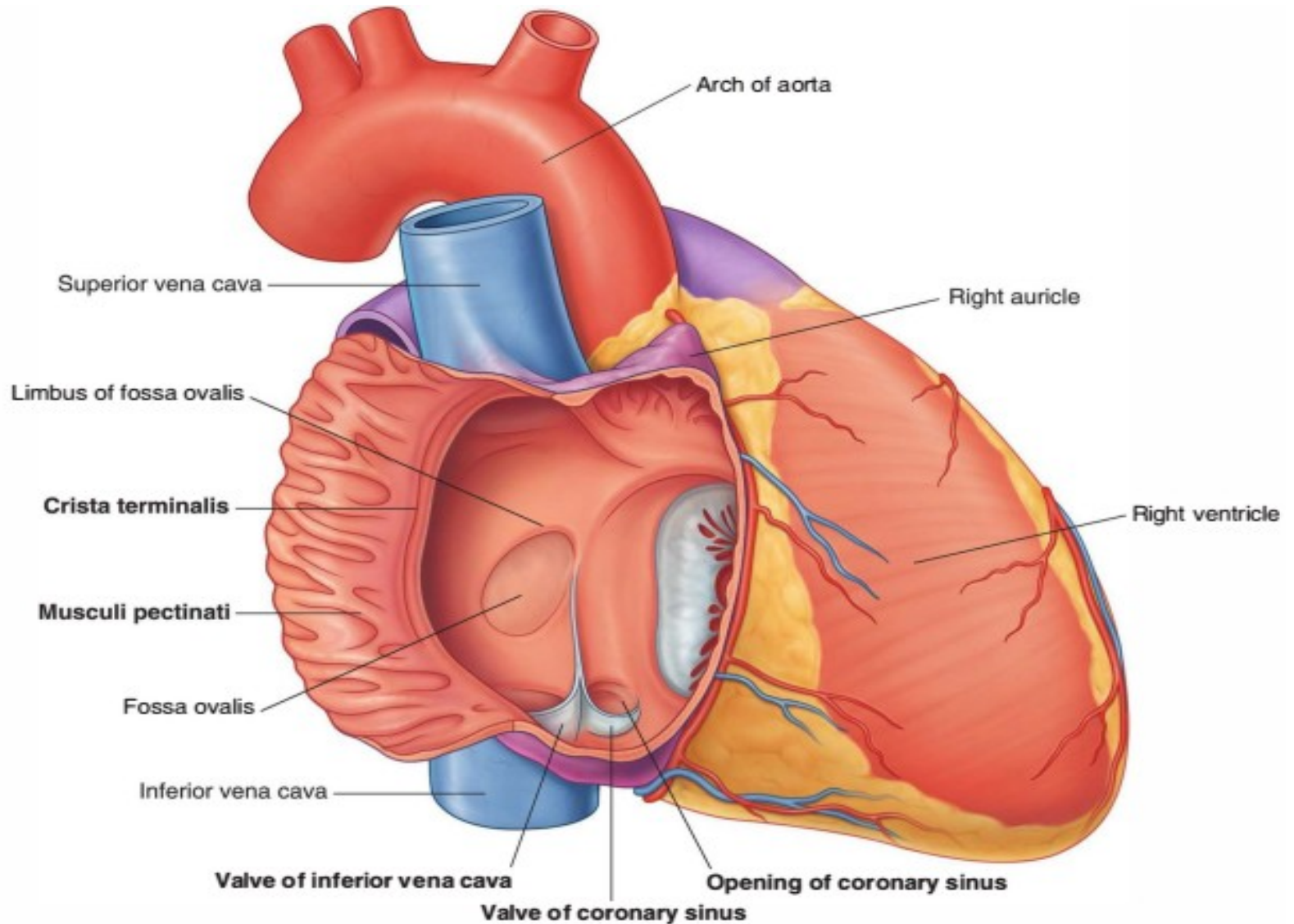


Foramen ovale...

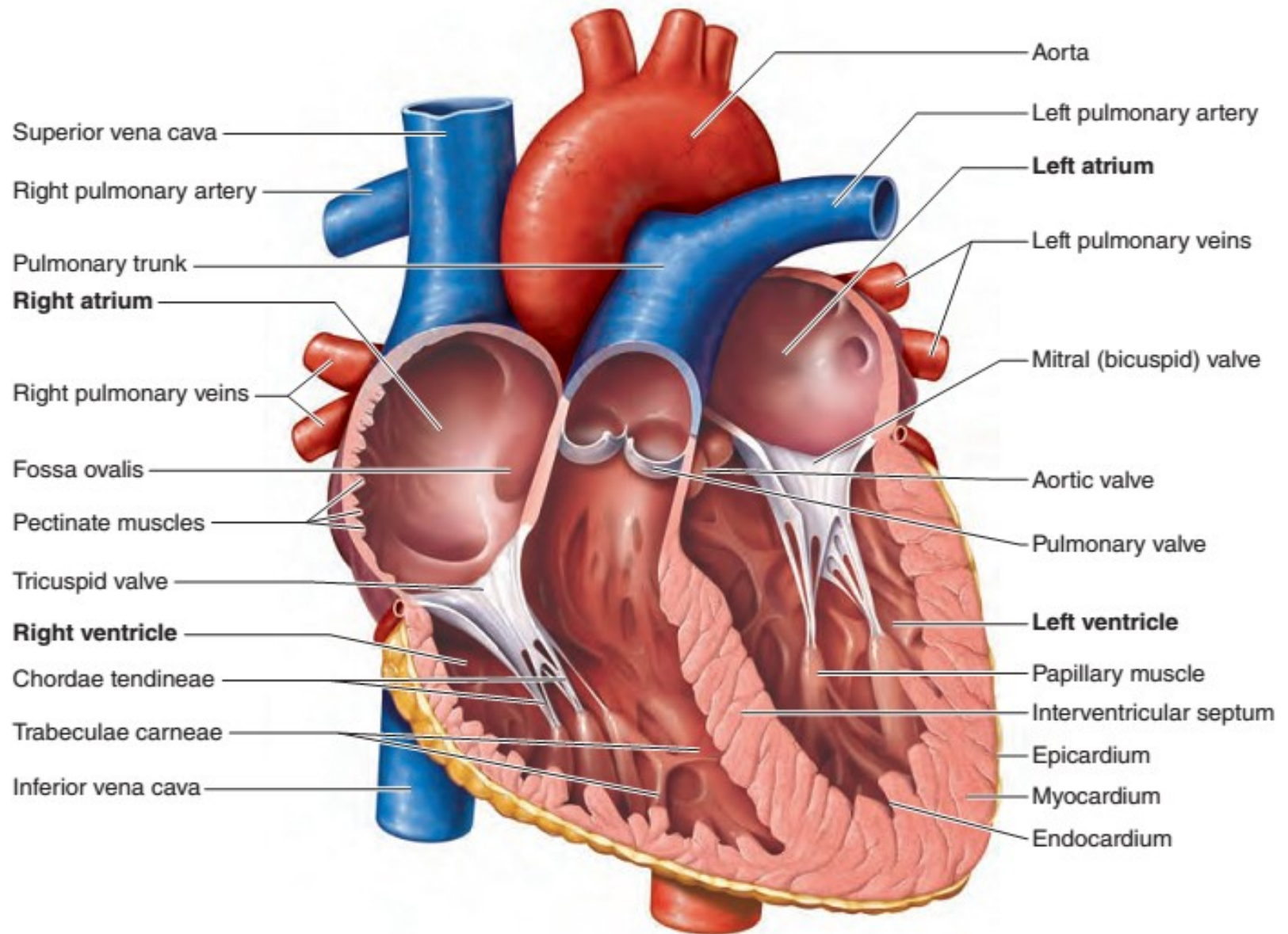
- Before birth, the foramen ovale allows oxygenated blood entering the right atrium from the IVC to pass into the left atrium
- At birth, owing to raised blood pressure in the left atrium, the septum primum is pressed against the septum secundum and fuses with it, and the foramen ovale is closed.
- At approximately 3 months, the valve of the foramen ovale fuses with the septum secundum, forming the **oval fossa (fossa ovalis)**



Right atrium: internal features

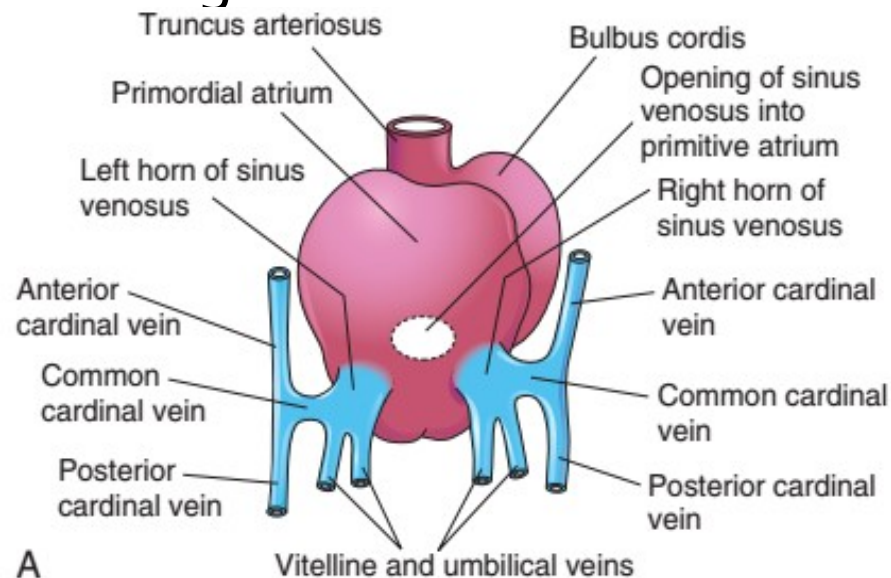


Heart: internal features



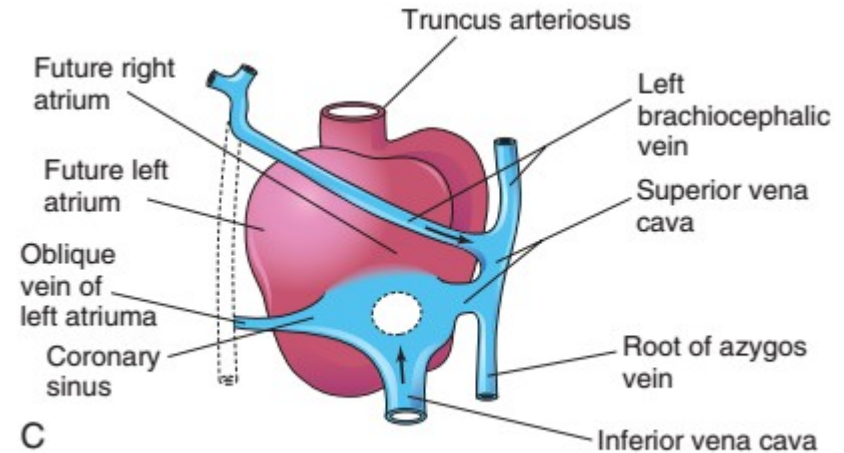
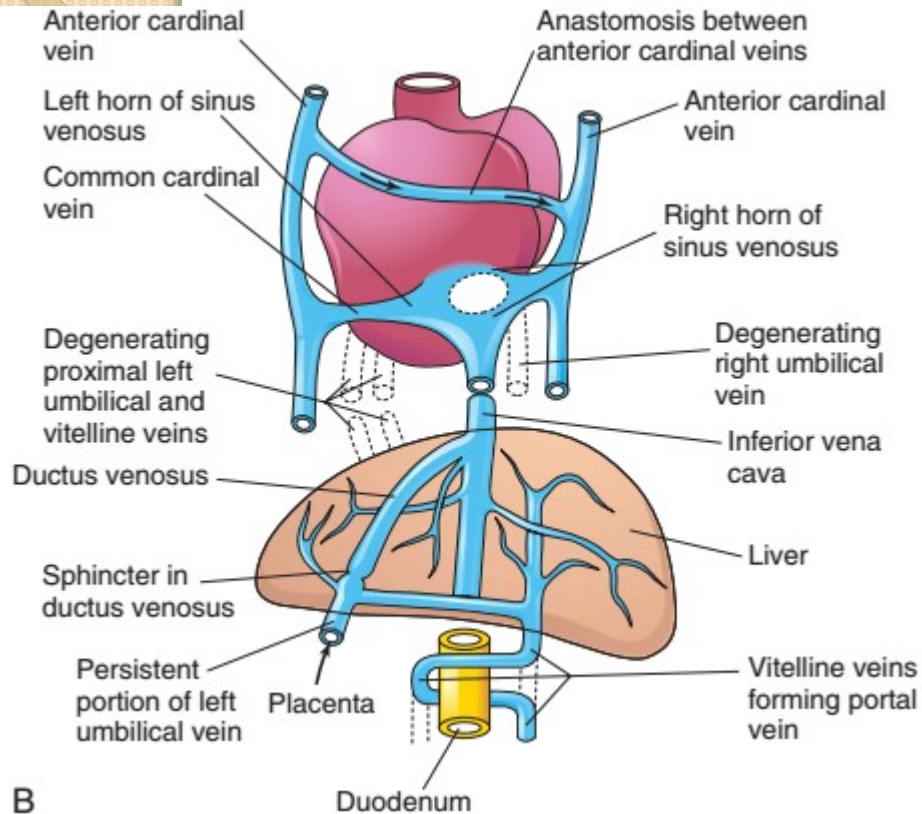
Fate of Sinus Venosus

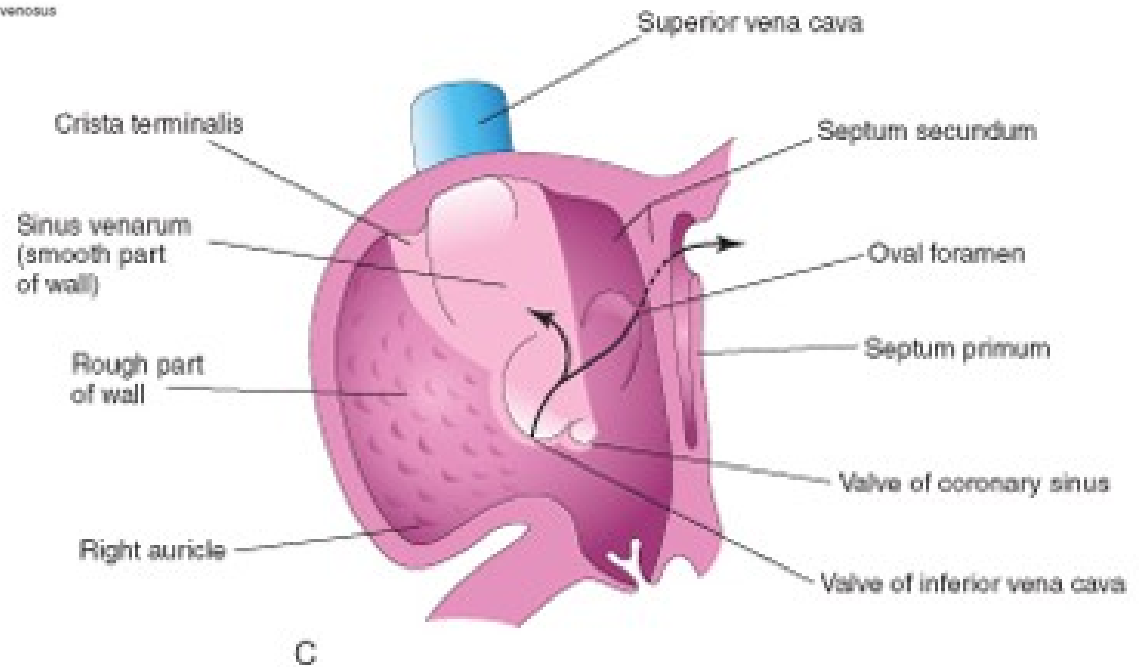
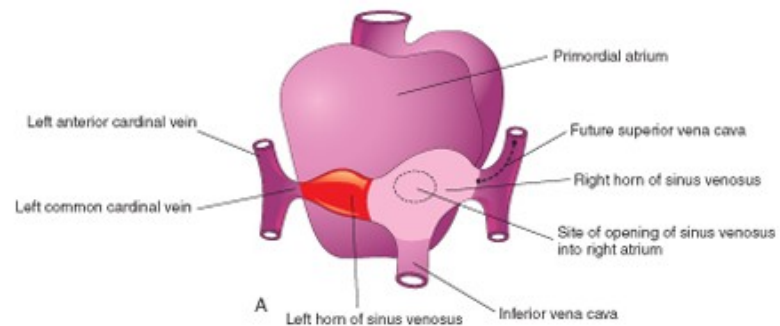
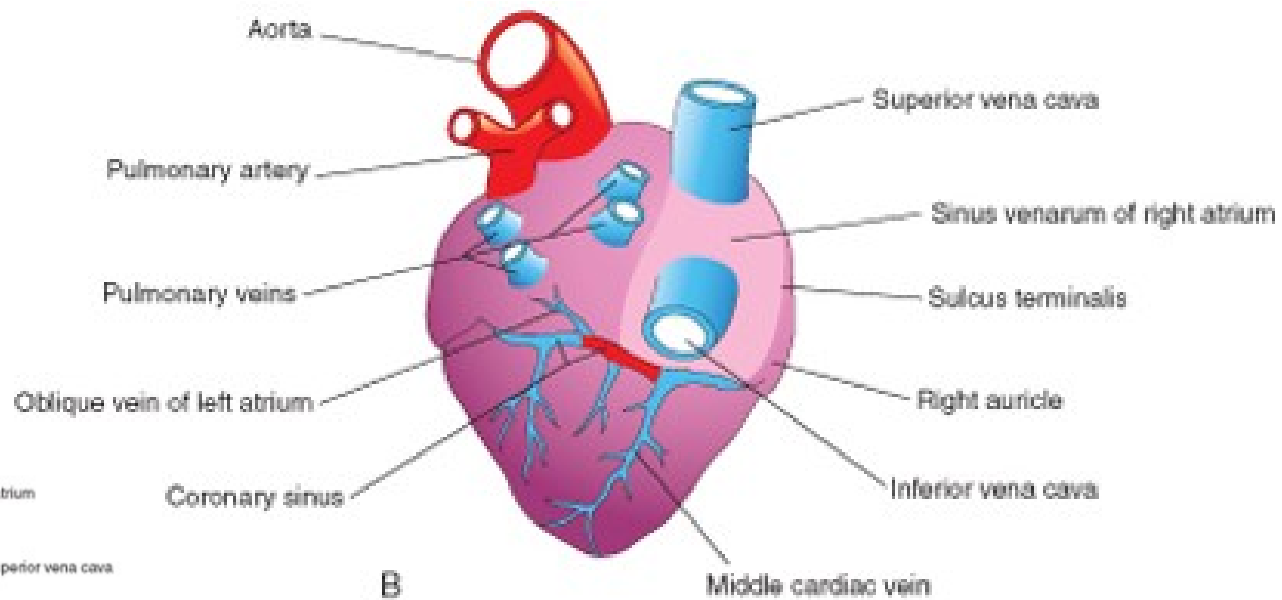
- Initially, the sinus venosus opens into the center of the dorsal wall of the primordial atrium through **sinoatrial orifice**
 - its right and left horns are of the same size
- Because of left to right shunts developing between the veins
 - the right horn of sinus venosus increases in size whereas the left horn becomes smaller.
 - SA orifice moves to the right



Changes in sinus venosus

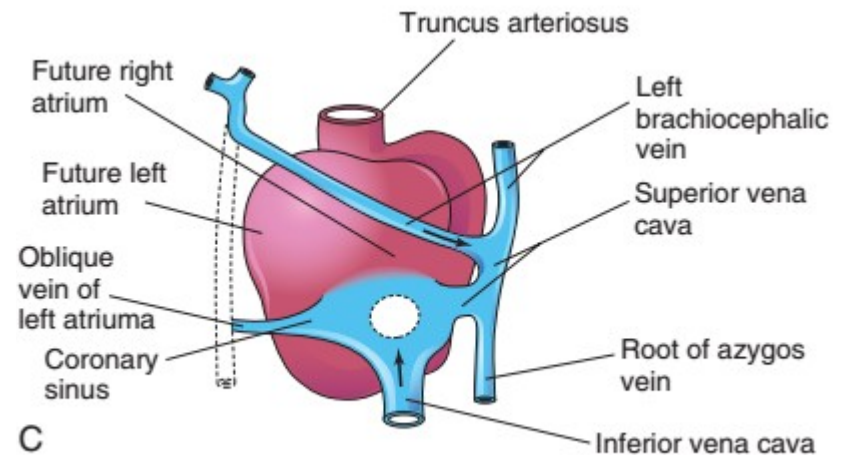
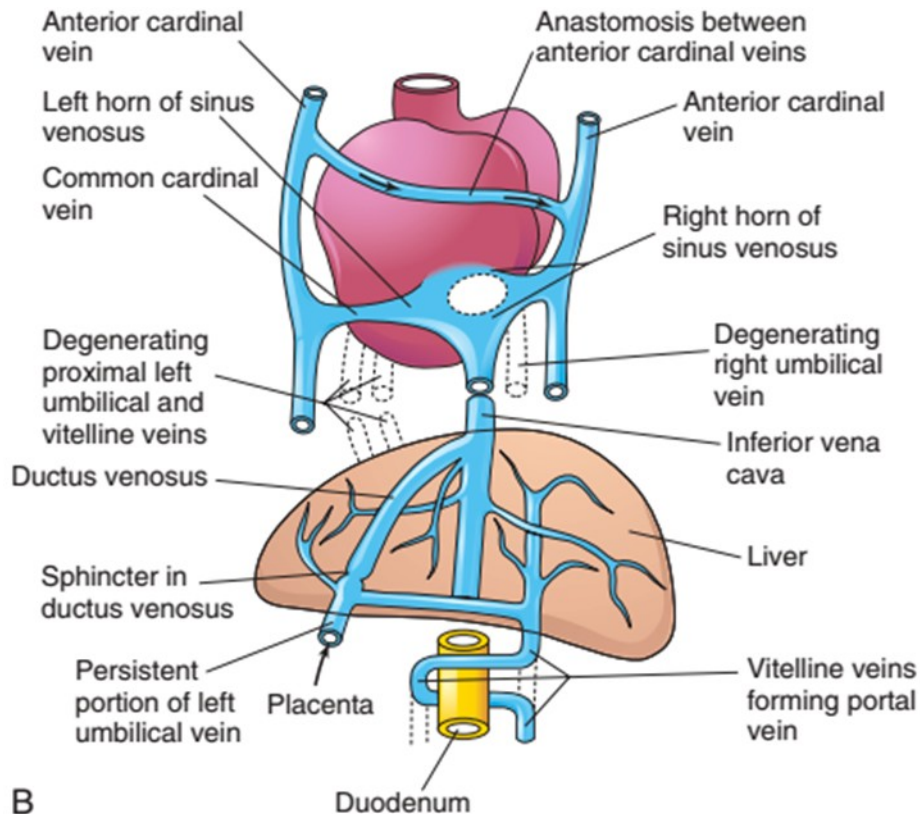
(Dorsal views of the developing heart at 7 wks and 8 wks)





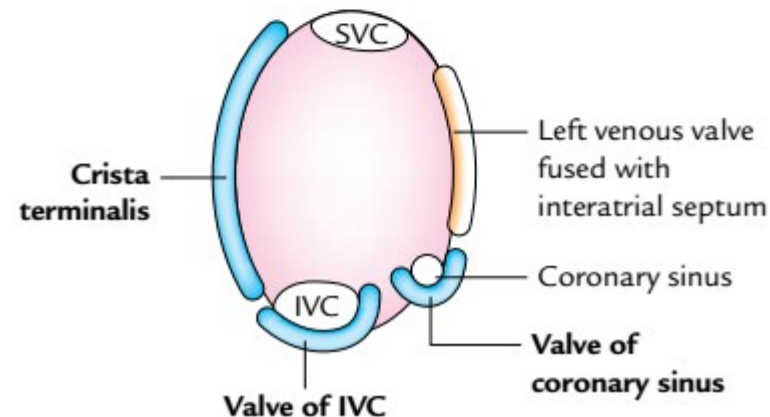
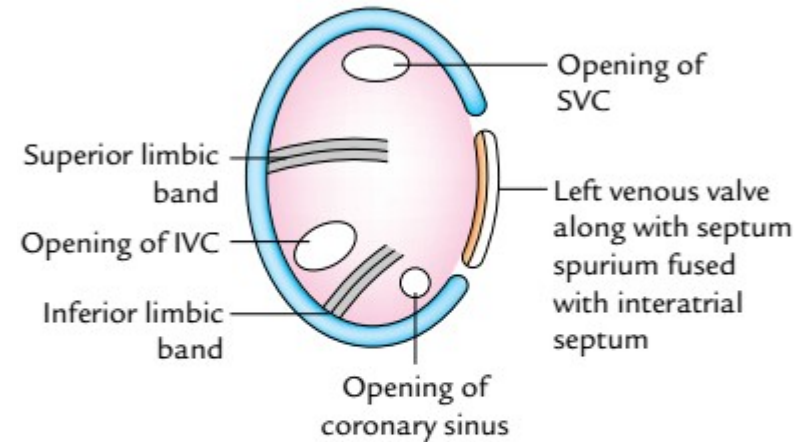
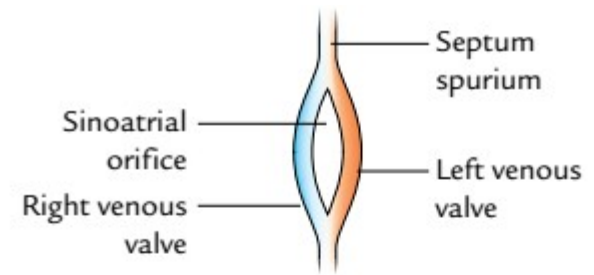
Fate of Sinus Venosus

- Right horn of sinus venosus receives all the venous blood through SVC and IVC
 - is incorporated into the wall of the right atrium
- The **left horn** becomes the **coronary sinus**.

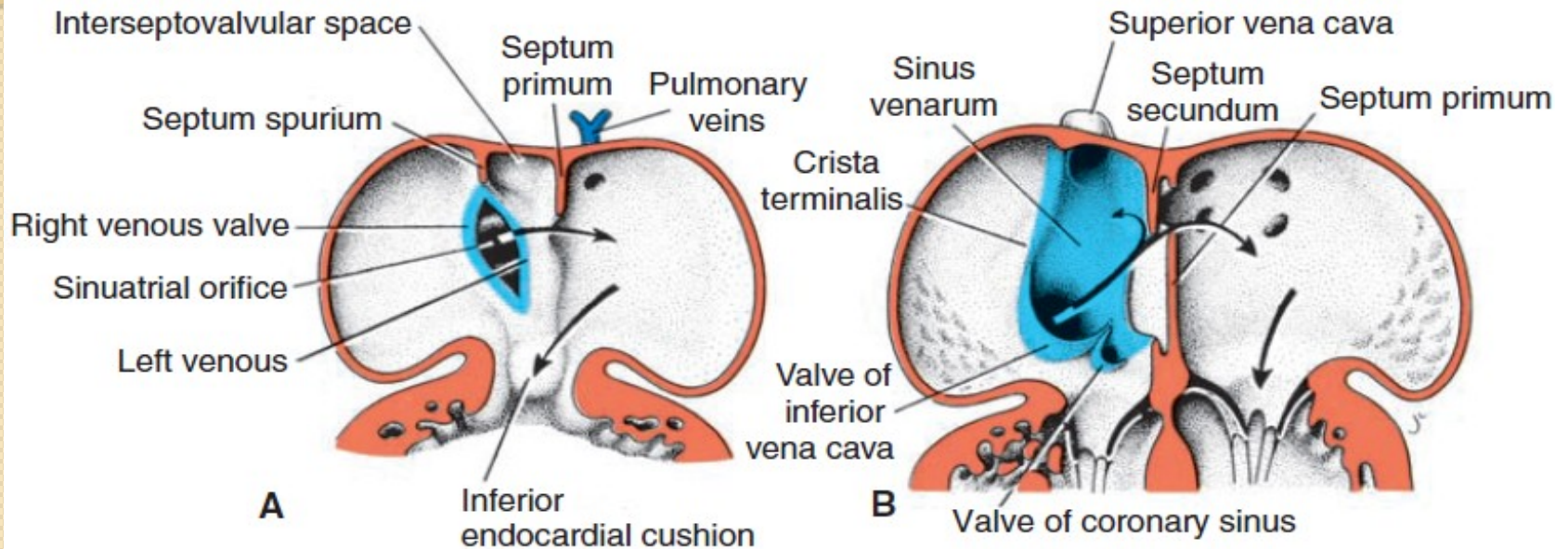


Fate of Sinus Venosus

- smooth part of the wall of the right atrium, sinus venarum, is derived from the sinus venosus
- The remainder and the right auricle has a rough appearance
 - derived from the primordial atrium.
- The smooth part and the rough part are demarcated internally by the crista terminalis, and externally by the sulcus terminalis
 - crista terminalis represents cranial part of the right SA valve
 - caudal part of the SA valve forms the valves of the IVC and coronary sinus.
- The left SA valve fuses with the septum secundum and is incorporated with it into the interatrial septum

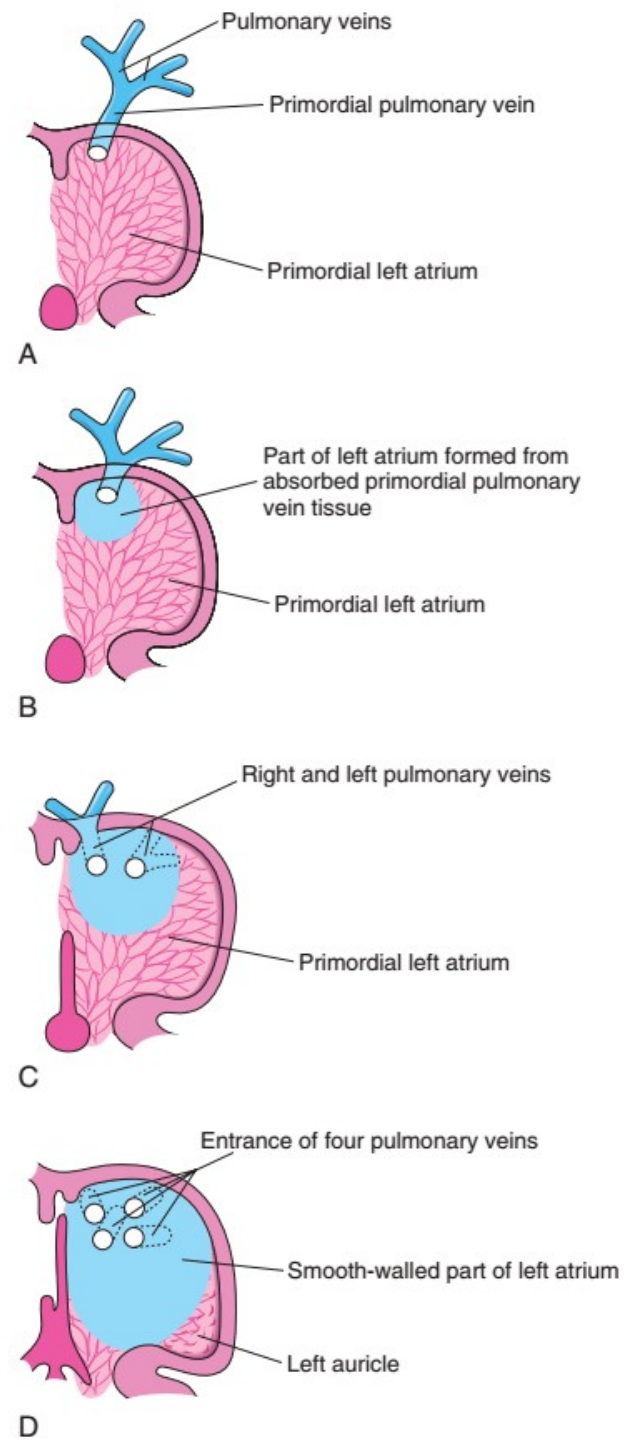


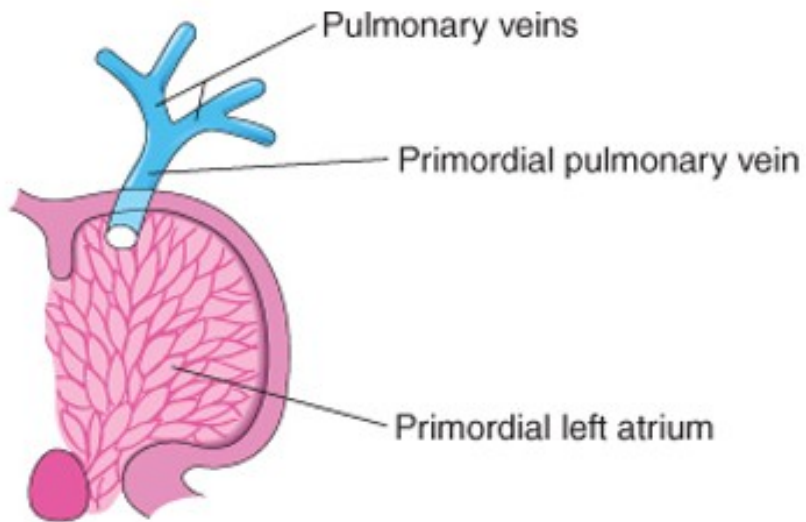
Fate of Sinus Venosus



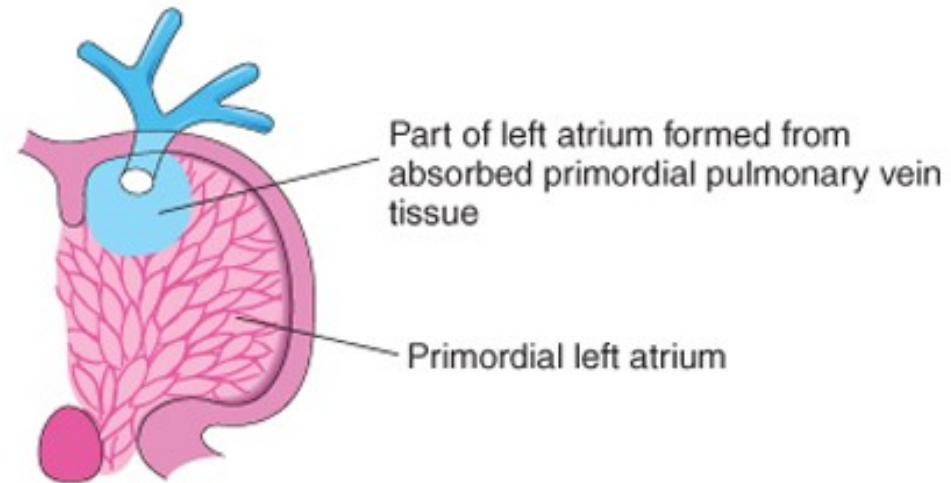
Formation of Pulmonary Veins

- Most of the wall of the left atrium is smooth
 - it is formed by incorporation of the primordial pulmonary vein.
- **Pulmonary vein** develops as an outgrowth of the dorsal atrial wall, just to the left of the septum primum.
- As the atrium expands, the primordial pulmonary vein and its main branches are incorporated into the wall of the left atrium.
- As a result, four pulmonary veins are formed

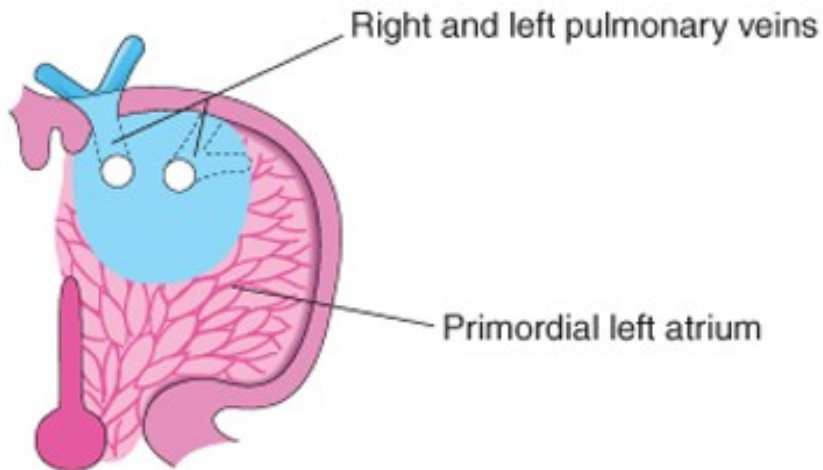




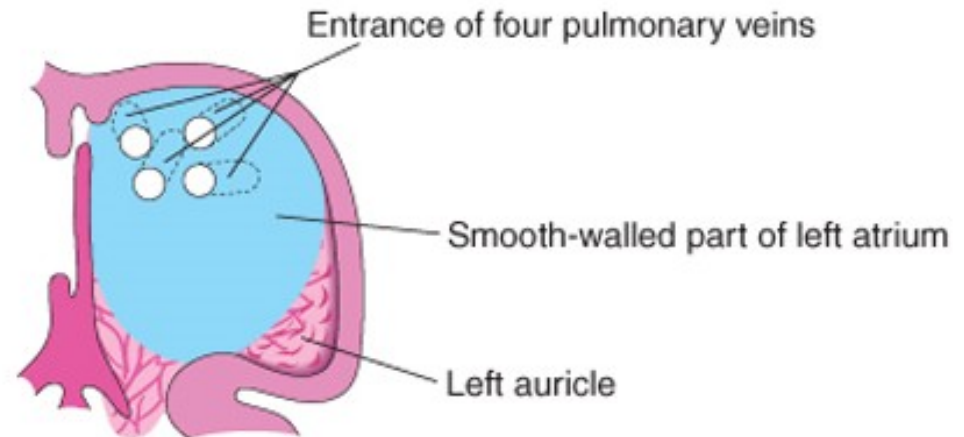
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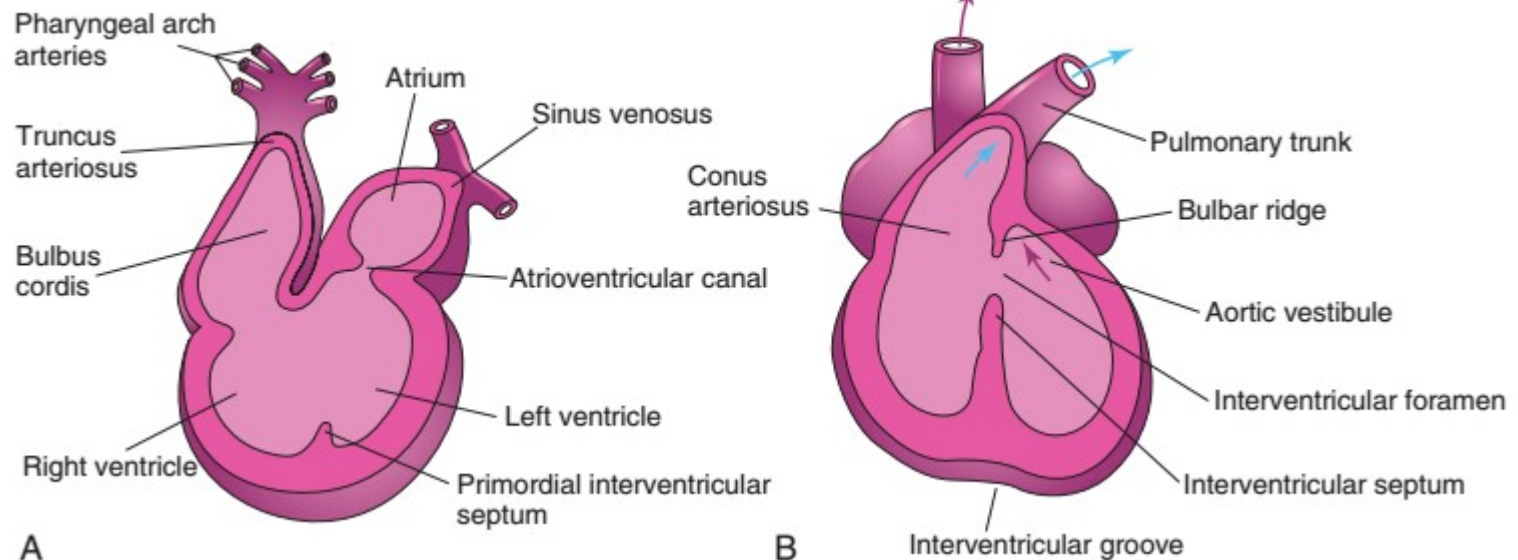
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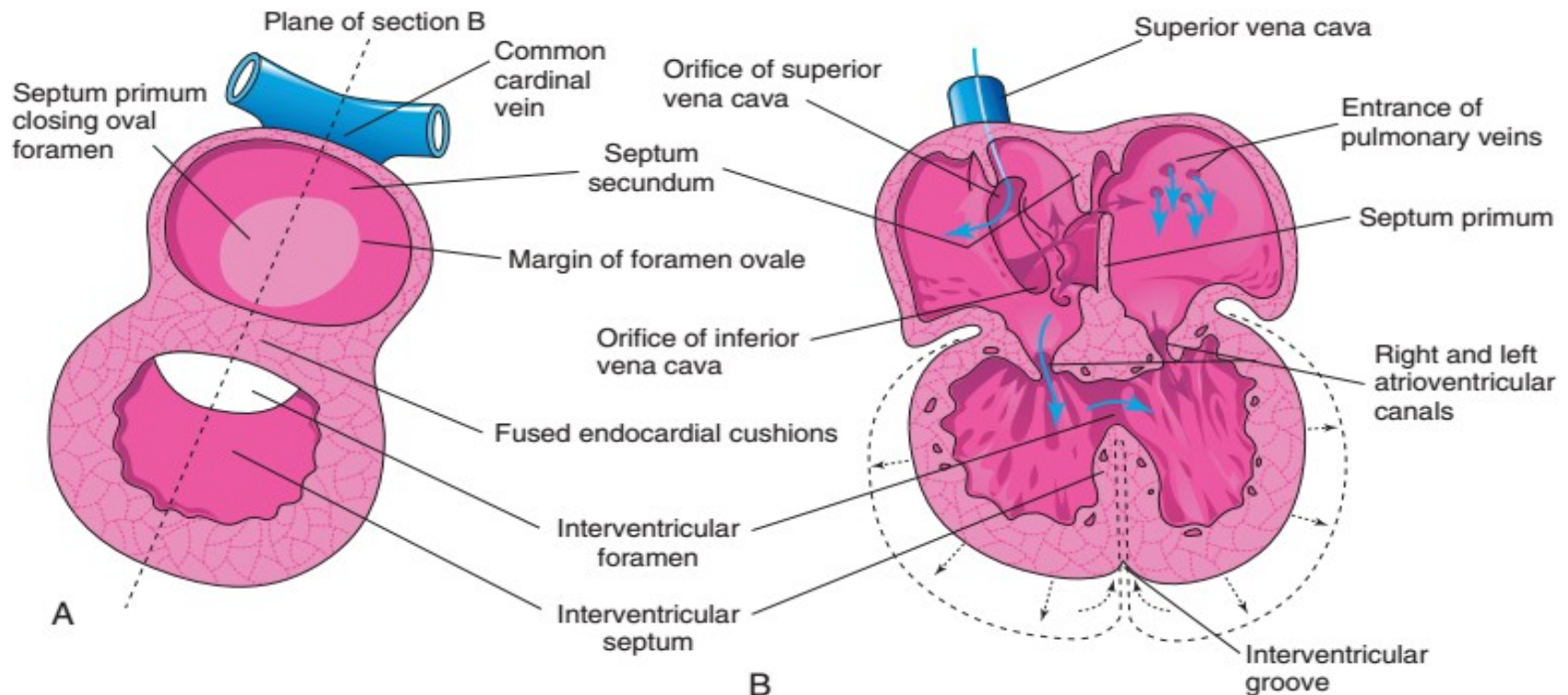
Partitioning of Primordial Ventricle

- A median muscular ridge, **interventricular septum**, forms in the floor of primitive ventricle (near its apex)
 - Myocytes from the primordial ventricles contribute to it
 - It forms the ***muscular part of the IV septum***
- The septum has a concave free edge.
- Increases in size due to
 - dilation of the ventricles away from the septum
 - active proliferation of myoblasts in the septum

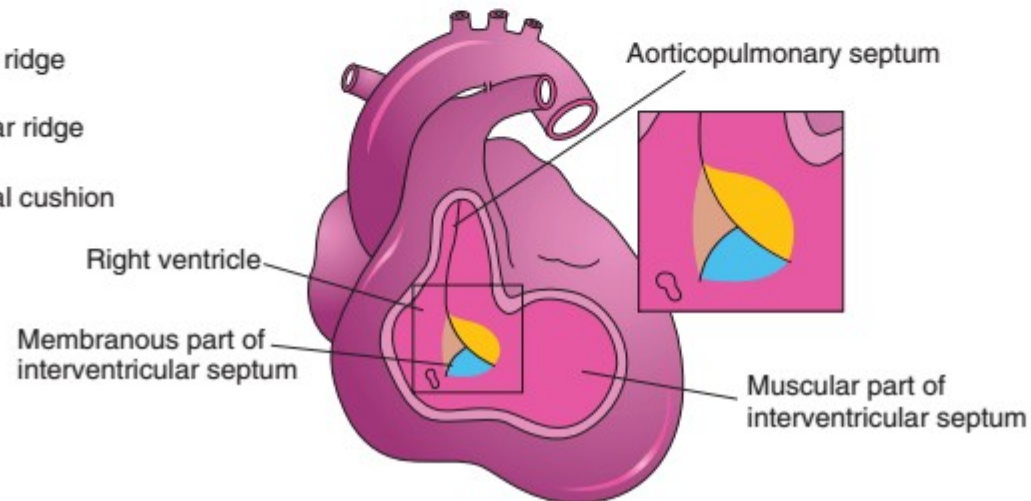
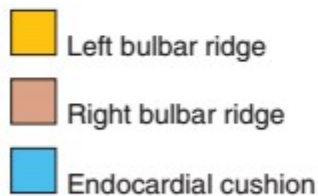
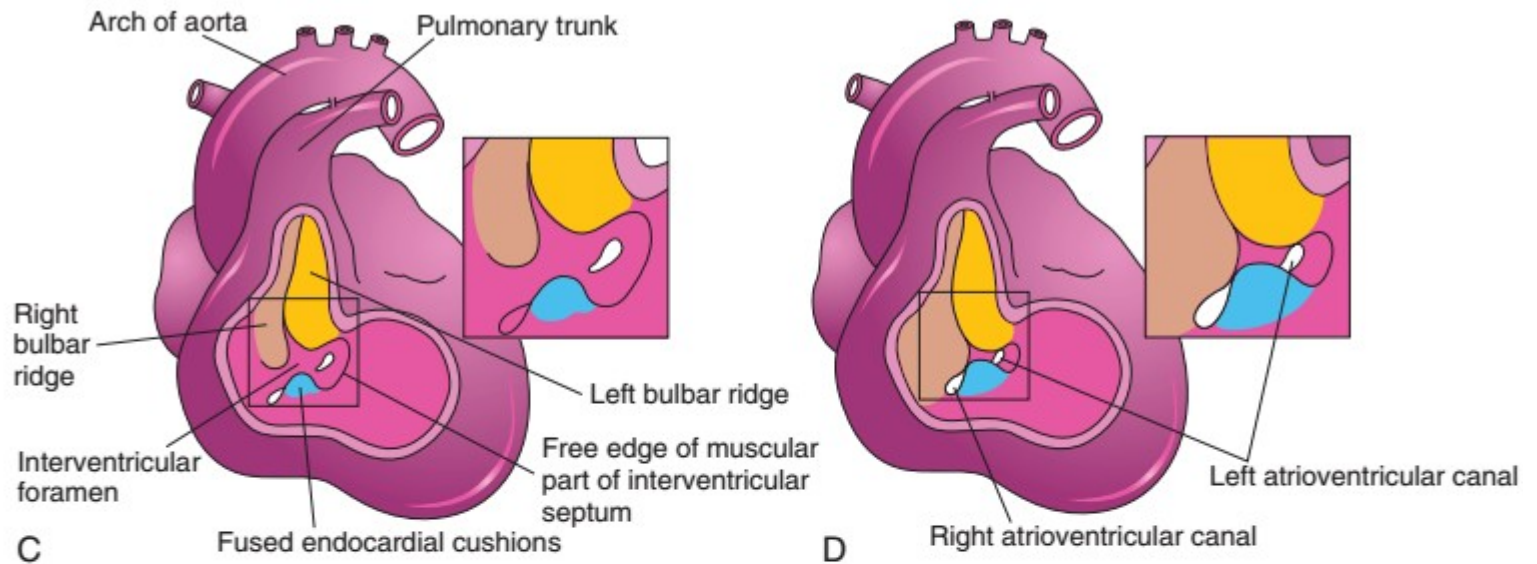


Partitioning of Primordial Ventricle

- Until 7th week, **interventricular foramen** is present
 - Opening between the free edge of the interventricular septum and the fused endocardial cushions.
- At the end of 7th week the foramen closes as the bulbar ridges fuse with the endocardial cushion

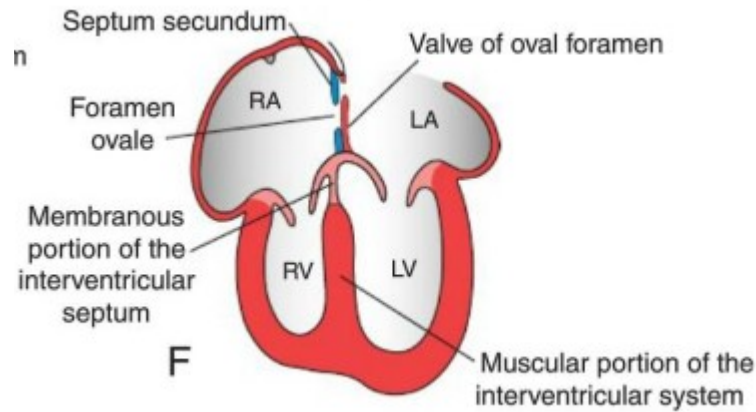


Partitioning of Primordial Ventricle



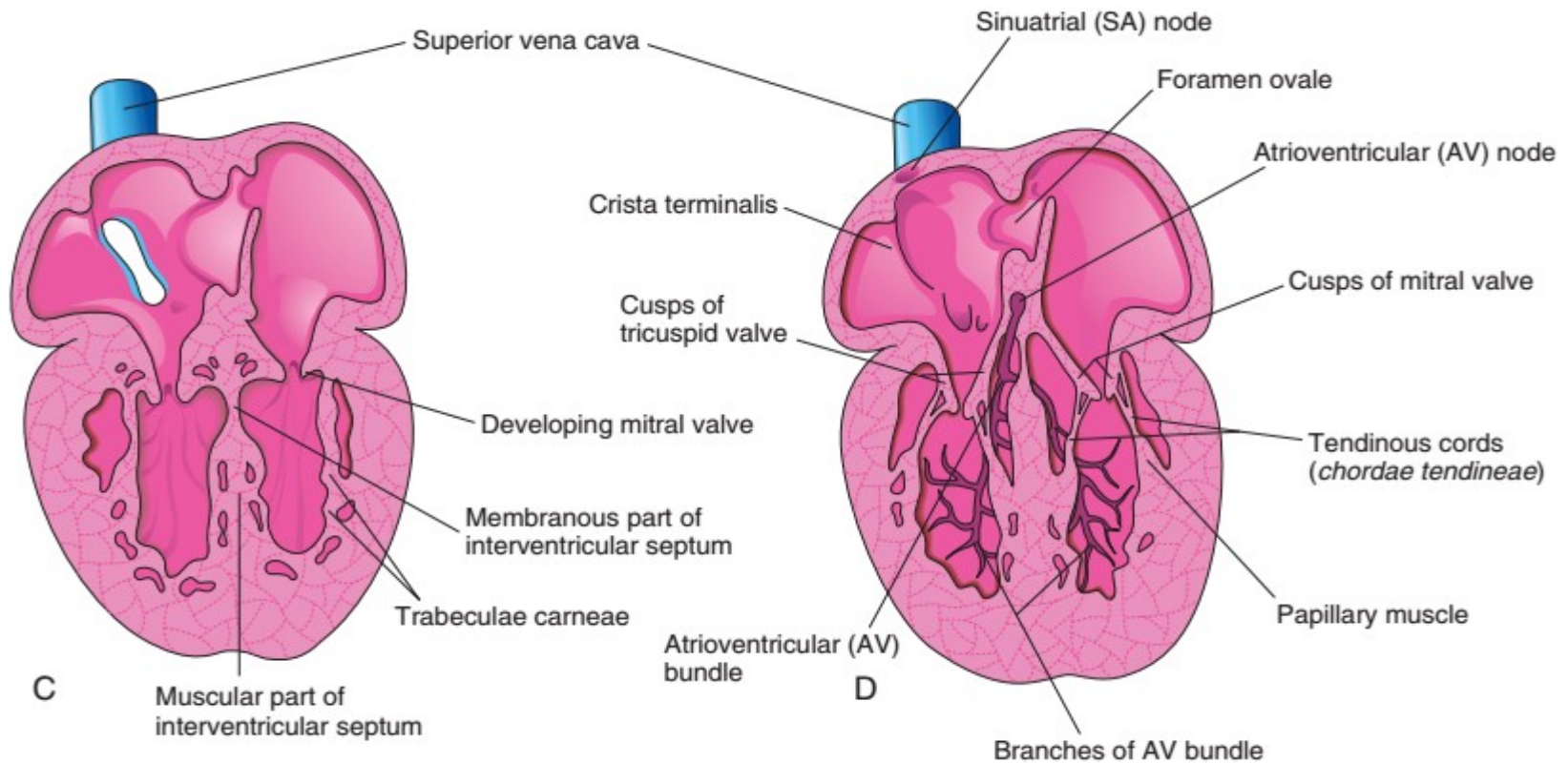
Partitioning of Primordial Ventricle

- Membranous part of IV septum is derived from **3 sources**:
 - Right bulbar ridge, left bulbar ridge, and AV cushions.
- extension of tissue from the right side of the endocardial cushion to the muscular part of the septum forms the ***membranous part of the IV septum***
 - Results in closure of interventricular foramen



Partitioning of Primordial Ventricle

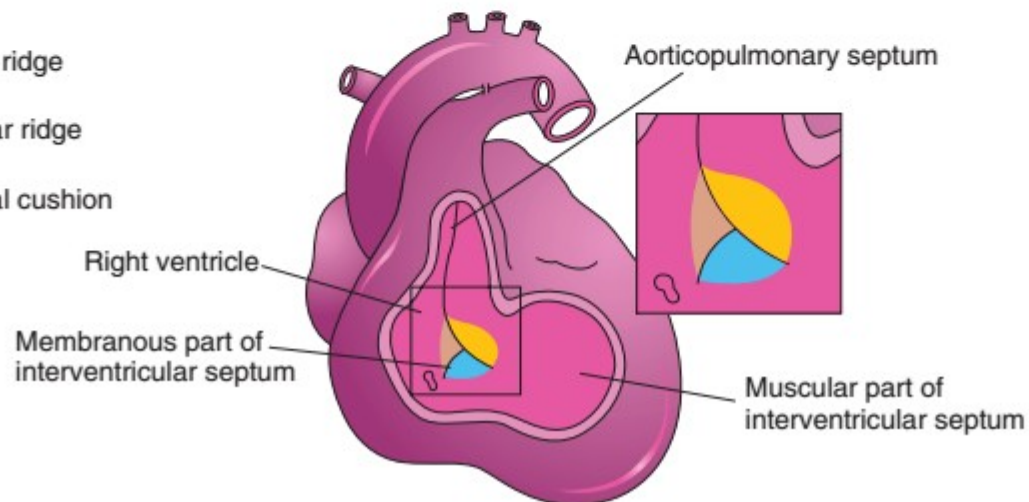
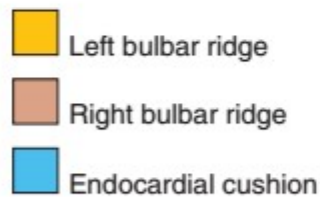
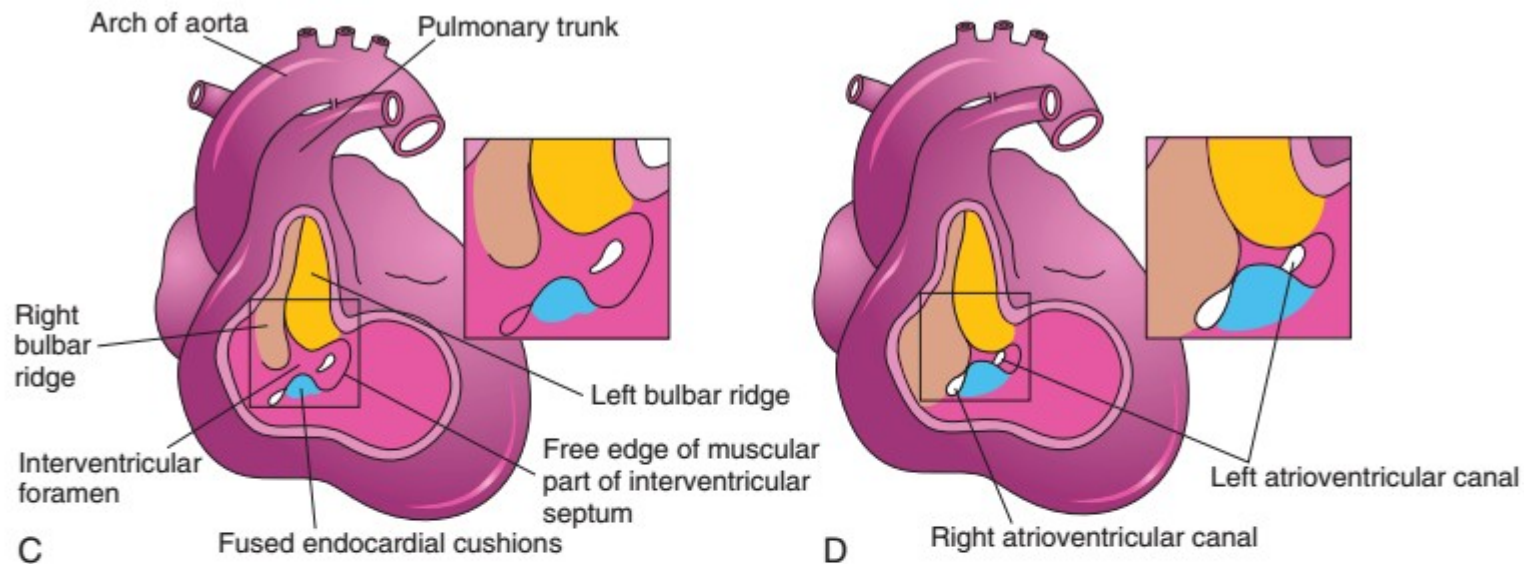
- Cavitation of the ventricular walls forms a spongy mass of muscular bundles, **trabeculae carneae**
- Some of these bundles become papillary muscles and chordae tendineae



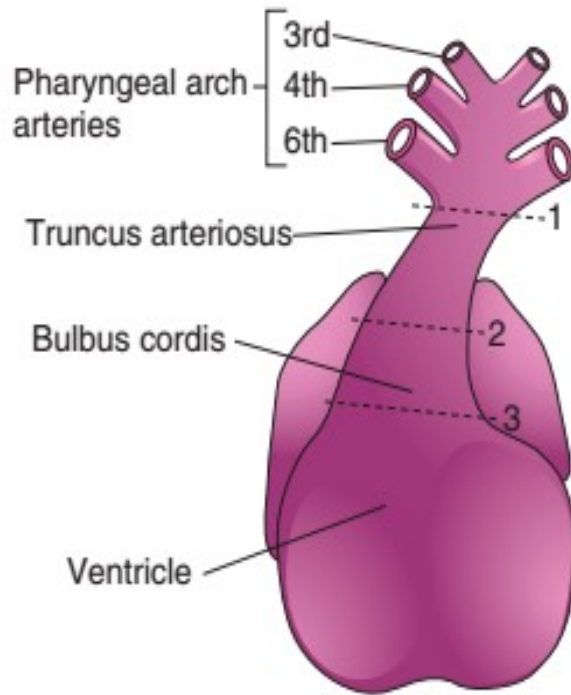
Partitioning of Bulbus Cordis and Truncus Arteriosus

- During the 5th week, active proliferation of mesenchymal cells in the walls of the bulbus cordis and truncus arteriosus
 - forms **bulbar ridges** and **truncal ridges**
- The ridges are derived from neural crest cells that migrate through the pharyngeal arches to the mesenchyme
- The bulbar and truncal ridges undergo a **180° spiraling**
 - the ridges fuse and form **aorticopulmonary septum**
 - This septum divides the bulbus cordis and TA into:
 - **aorta** and **pulmonary trunk**

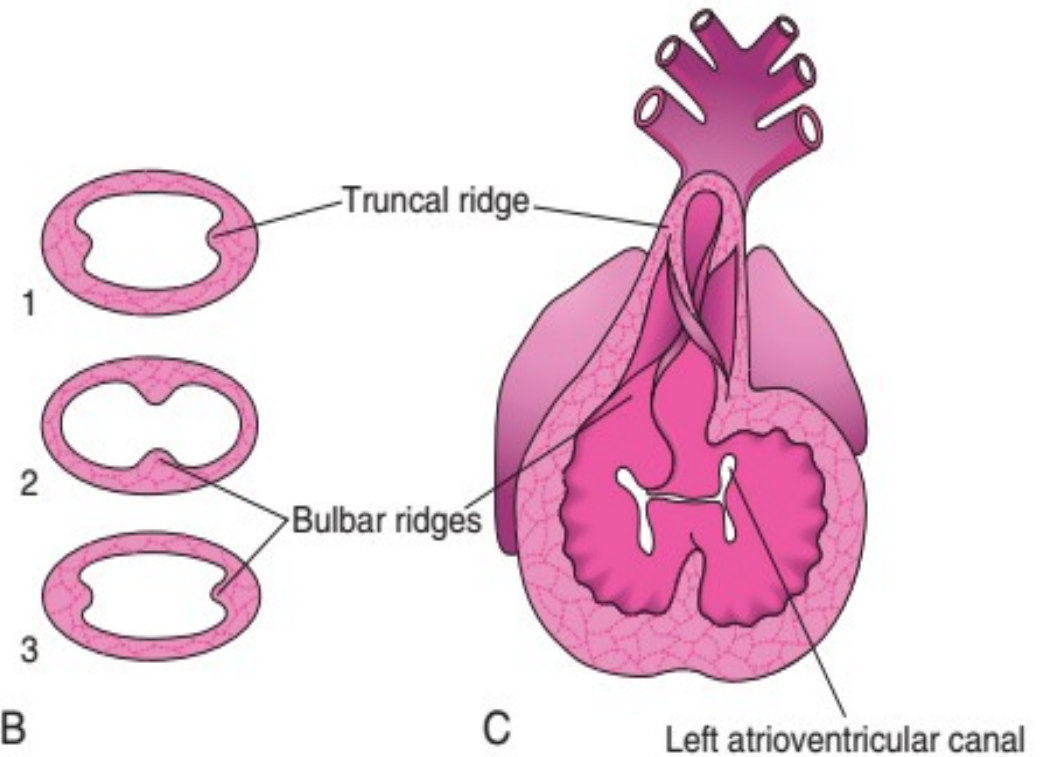
Formation of Aorticopulmonary septum



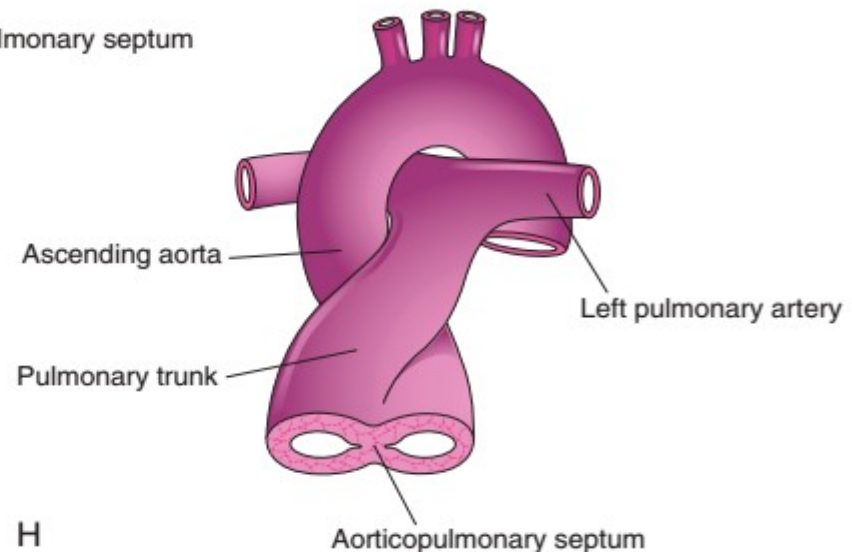
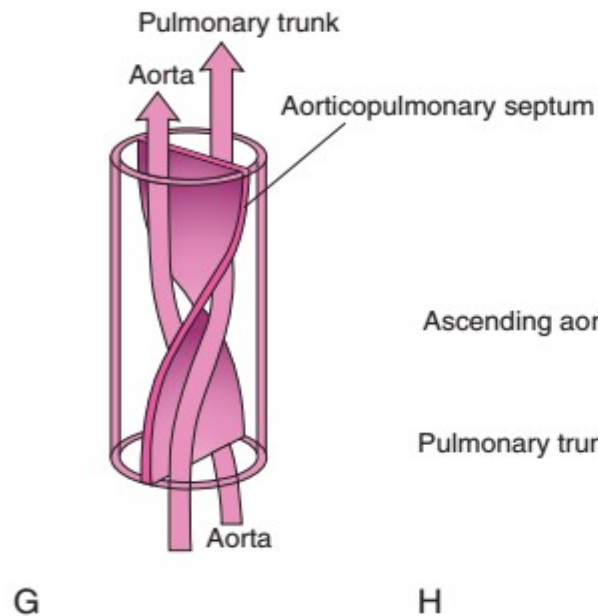
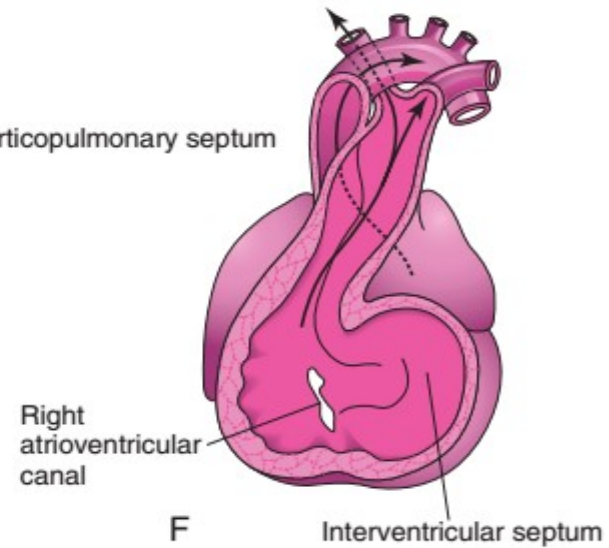
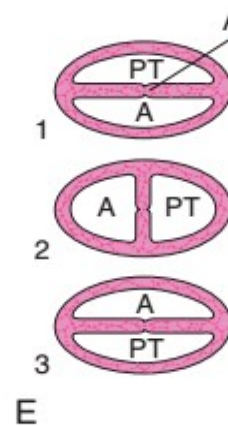
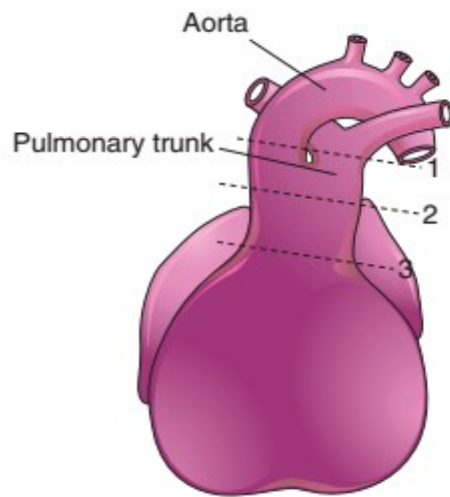
Formation of Aorticopulmonary septum



A



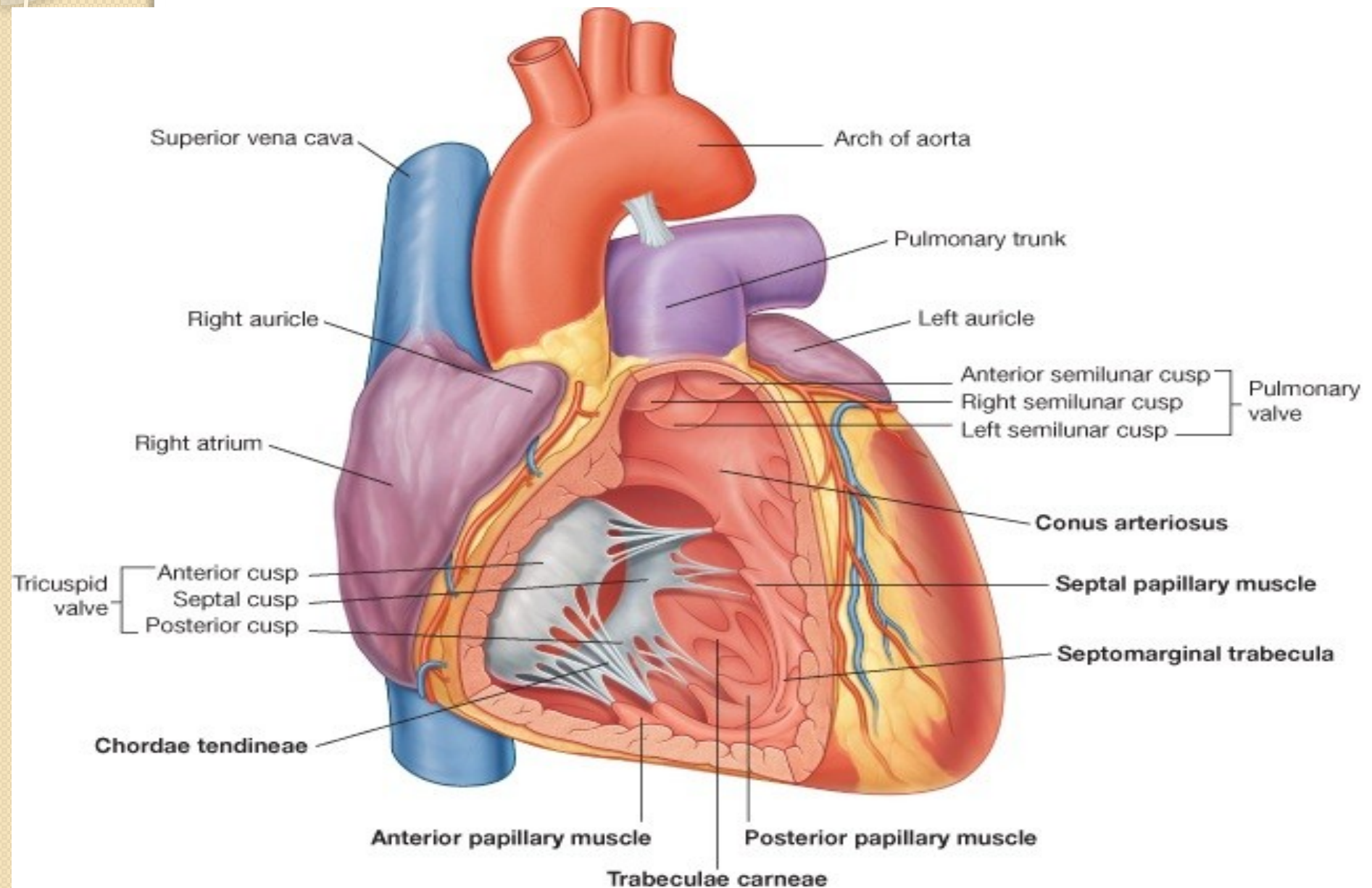
Formation of Aortopulmonary septum



Partitioning of Bulbus Cordis and Truncus Arteriosus

- The bulbus cordis is incorporated into the walls of the definitive ventricles:
 - In the **right ventricle**
 - bulbus cordis becomes the **conus arteriosus** (infundibulum) – origin of the pulmonary trunk.
 - In the **left ventricle**
 - bulbus cordis forms the walls of the **aortic vestibule**

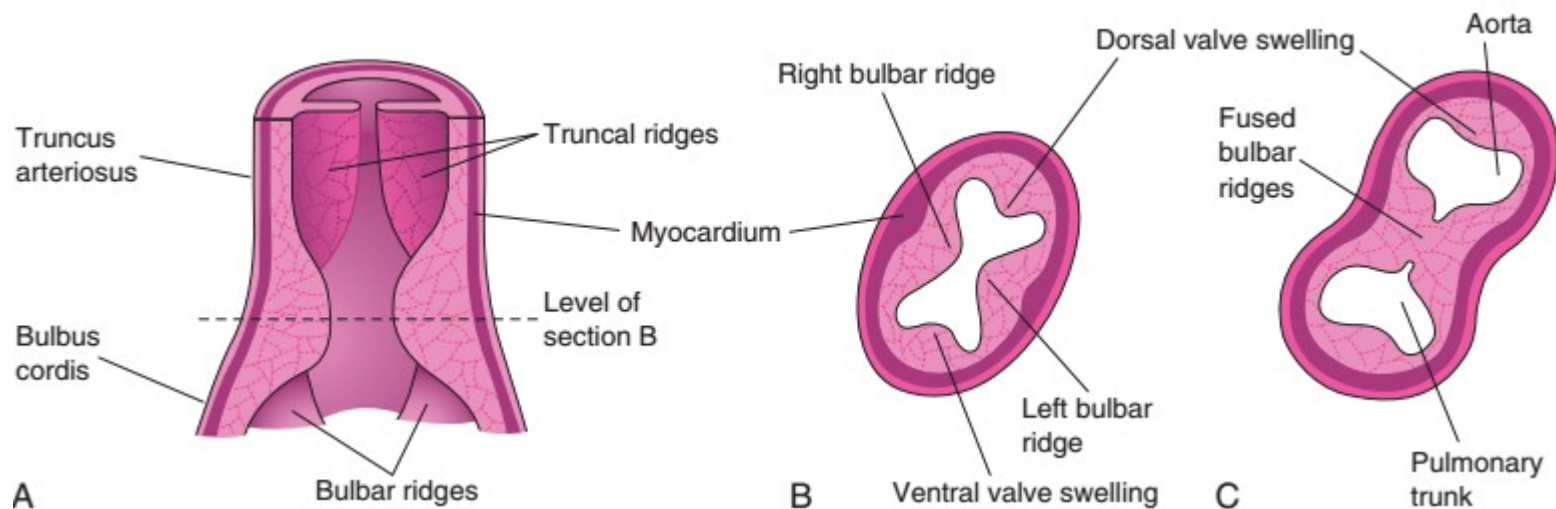
Right ventricle: internal features



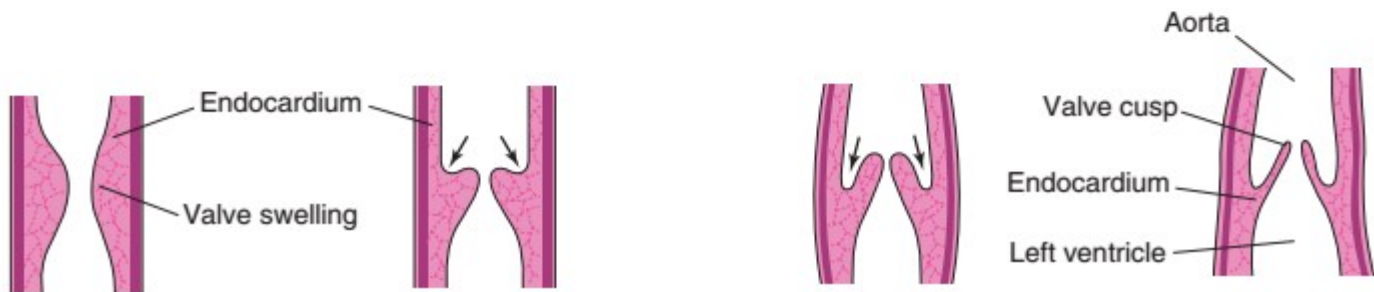
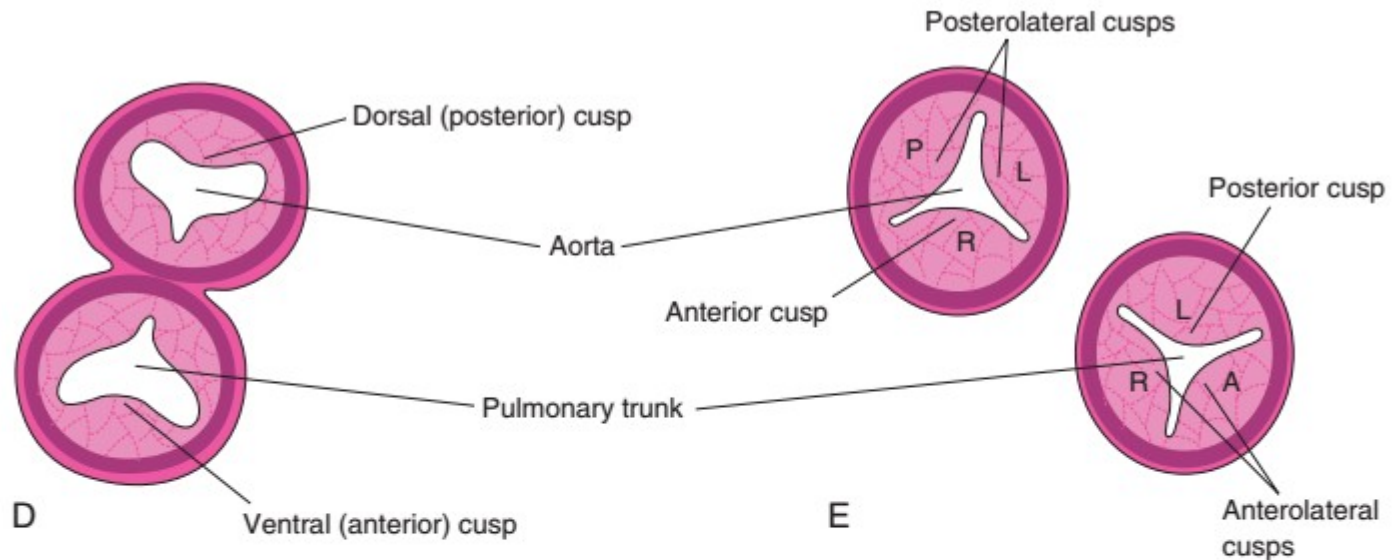
Development of Cardiac Valves

- **Pulmonary and Aortic Valves**

- begin to develop from **three swellings of subendocardial** tissue around the orifices of the aorta and pulmonary trunk.
- Cardiac precursor neural crest cells contribute to this tissue.
- These swellings are hollowed out and reshaped to form three thin-walled **cusps**.



Development of Semilunar Valves



Development of Cardiac Valves

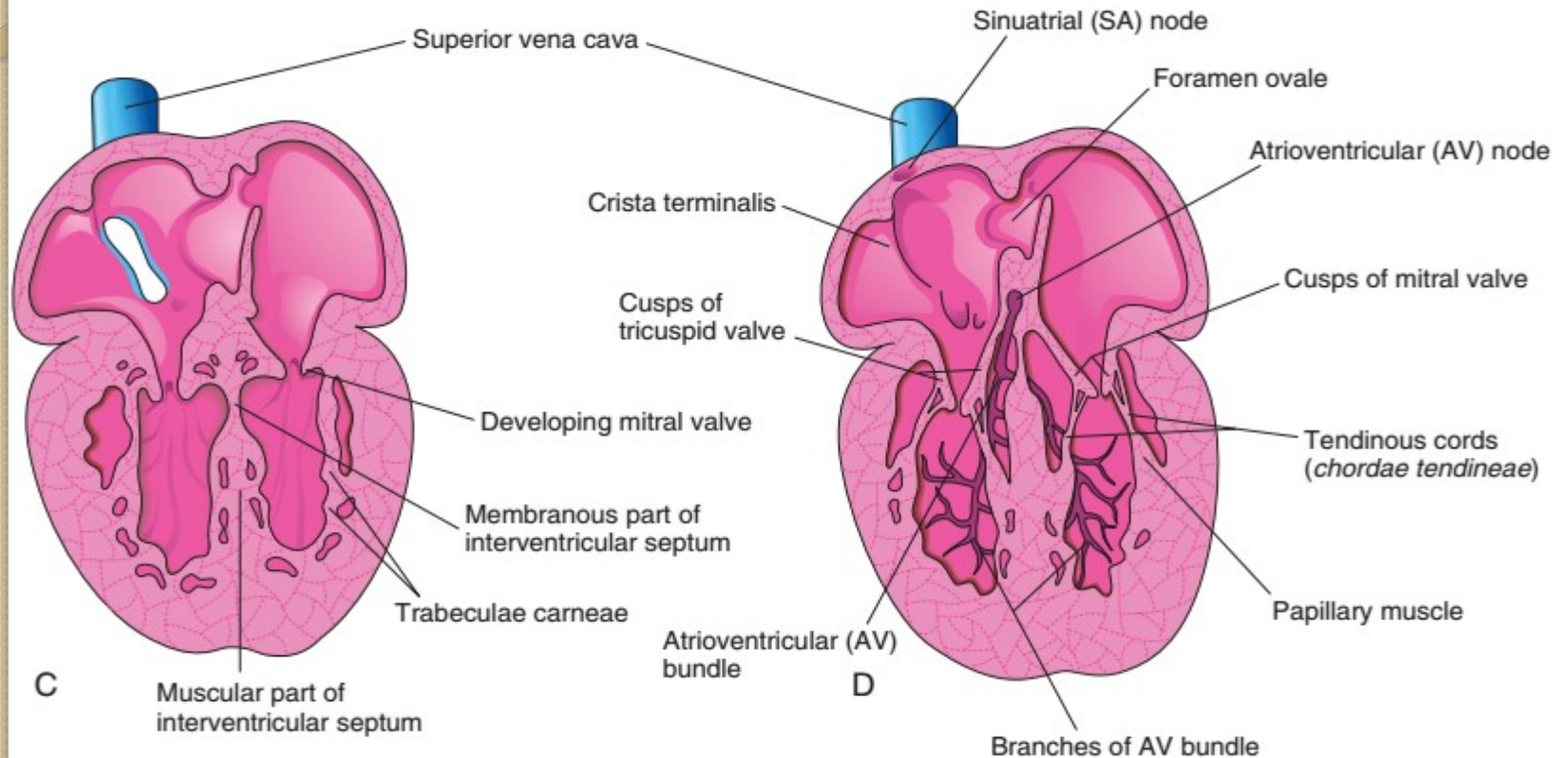
- **Tricuspid Valve**

- In the AV canal, subendocardial mesenchyme proliferates and forms three swellings (**anterior, posterior, and septal**).
- The swellings enlarge and meet each other in the lumen.
- free margins of these swellings do not fuse
- mesenchymal tissue degenerates & replaced by connective tissue
- ventricular surface of these swellings is hollowed and form **cusps**

- **Mitral Valve**

- It develops from two swellings: **anterior and posterior** in the same way as the tricuspid valve.

Development of AV Valves

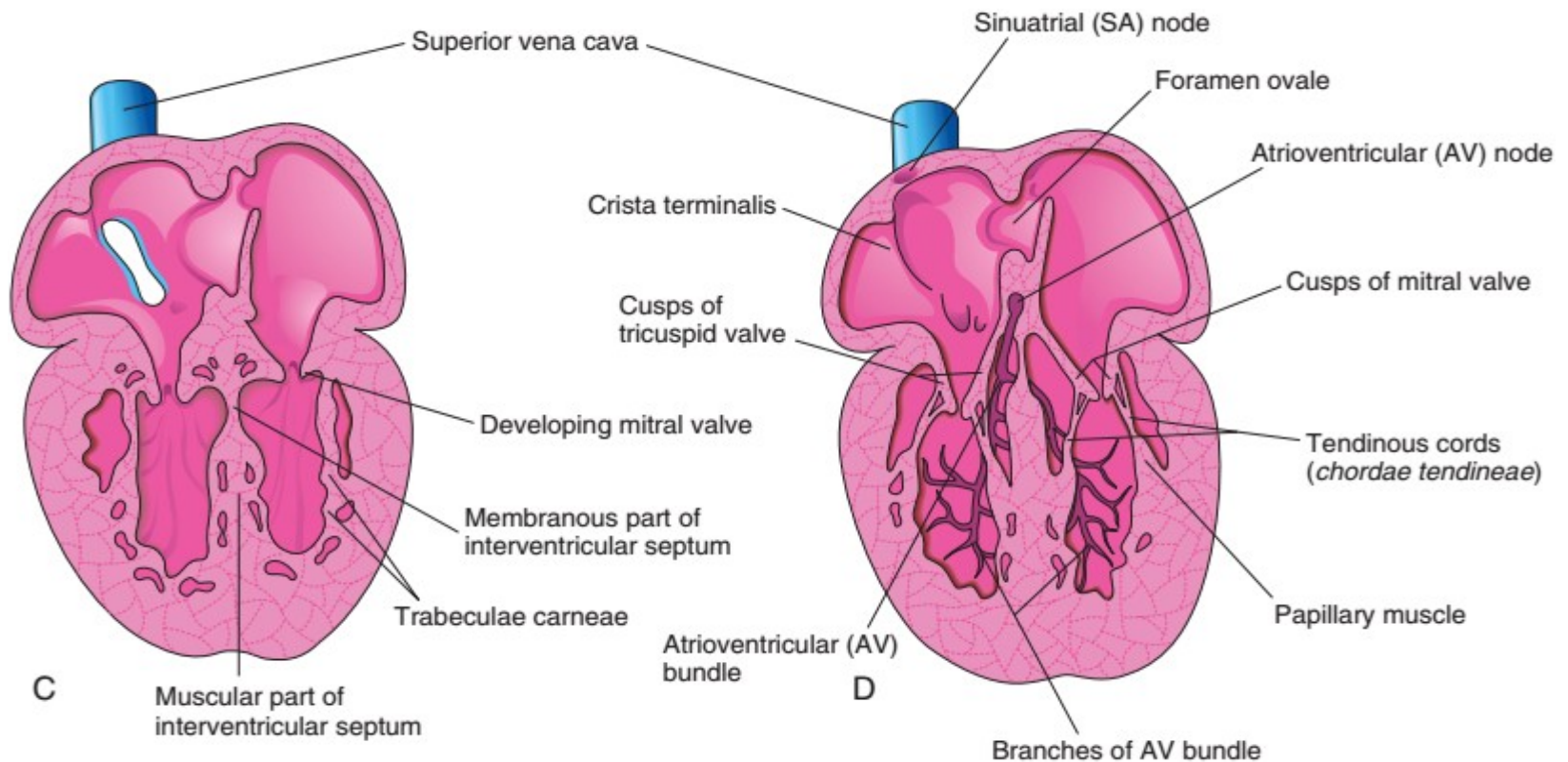


Conducting System of the Heart

- **SA node** develops during the fifth week.
 - located in the right atrium near the entrance of the SVC
 - Initially was in the right wall of sinus venosus but becomes incorporated into right atrium along with sinus venosus
- cells from sinus venosus left wall found in the base of the interatrial septum, near the opening of the coronary sinus.
 - Together with cells from the AV region, they form the **AV node** and the **AV bundle**, located just superior to the endocardial cushions.
- The fibers arising from the AV bundle pass from the atrium into the ventricle and split into **right and left bundle branches**
 - distributed throughout the ventricular myocardium

Conducting System of the Heart

- band of connective tissue grows in from the epicardium – forms part of the **cardiac skeleton**
 - separates muscle of the atria from that of the ventricles

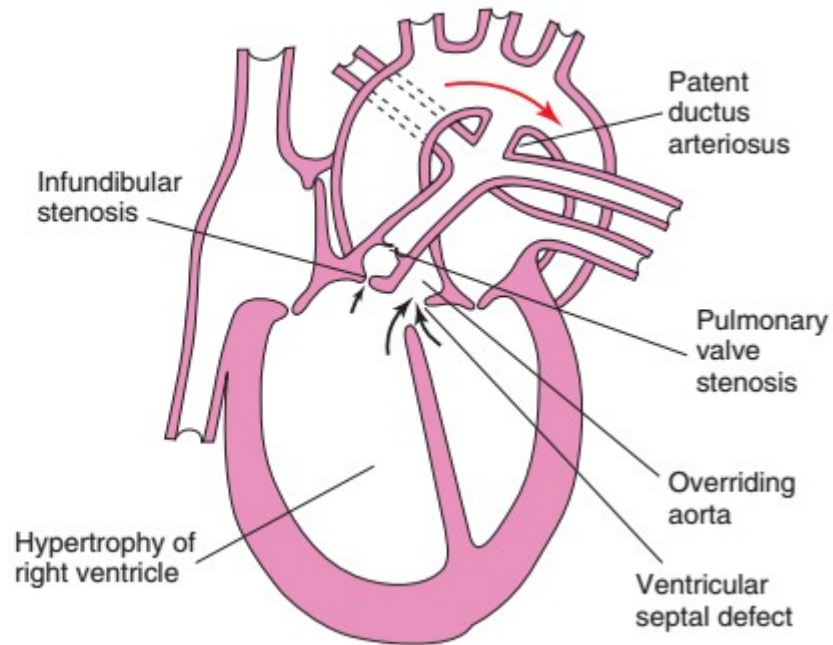
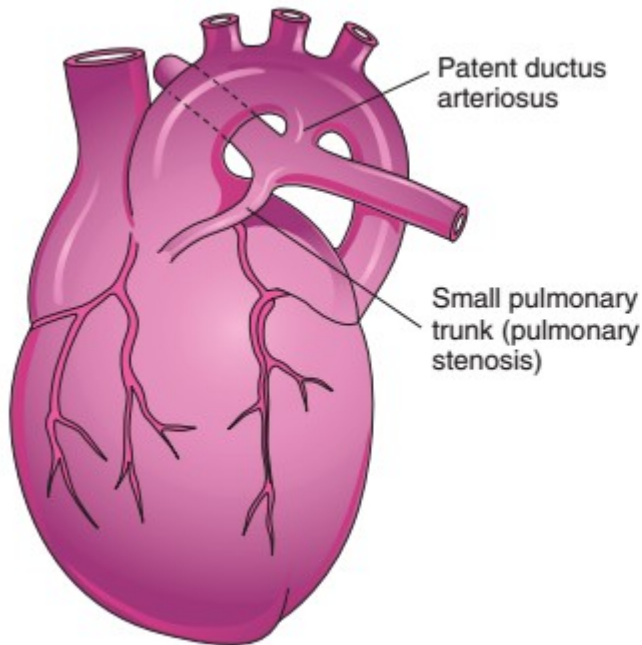


Clinical: Birth Defects of Heart (CHDs)

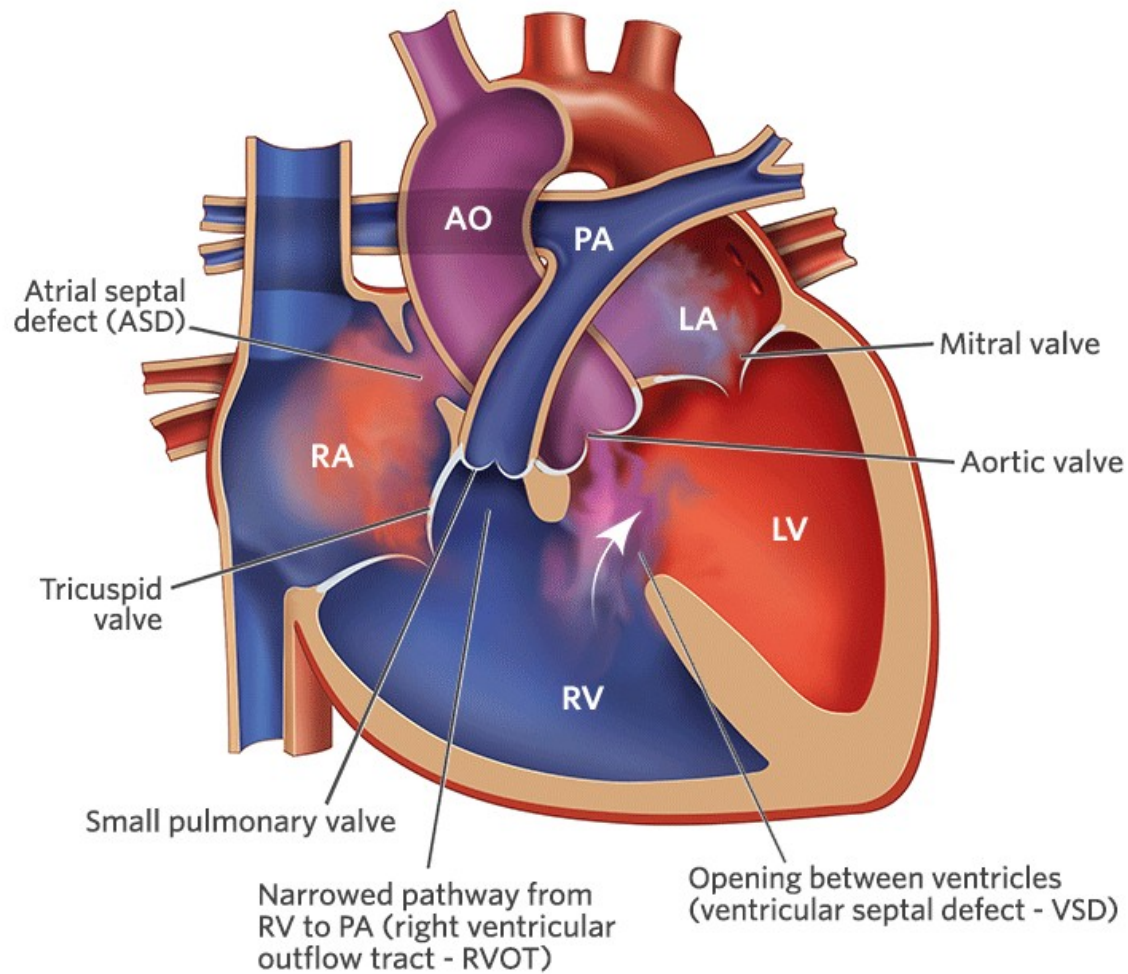
- Atrial septal defects (ASD) – frequent in females
- Ventricular septal defect (VSD) – most common type (frequent in males)
- Persistent Truncus Arteriosus – associated with VSD
- Transposition of the Great Arteries
- Aortic Stenosis

Clinical: Birth Defects of Heart (CHDs)

- **Tetralogy of Fallot** – group of 4 cardiac defects
 - Pulmonary artery stenosis
 - Ventricular Septal defect (VSD)
 - Dextroposition of the aorta (overriding aorta)
 - Right Ventricular hypertrophy



Tetralogy of Fallot (TOF) Interior View



- Oxygen-rich blood
- Oxygen-poor blood
- Mixed blood
- Mixed blood

AO: Aorta

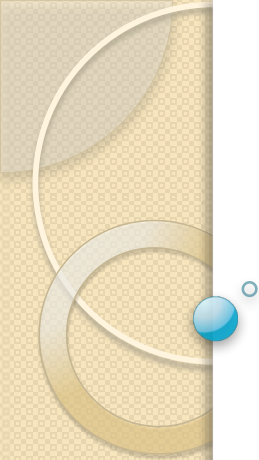
LA: Left atrium

RA: Right atrium

PA: Pulmonary artery

LV: Left ventricle

RV: Right ventricle



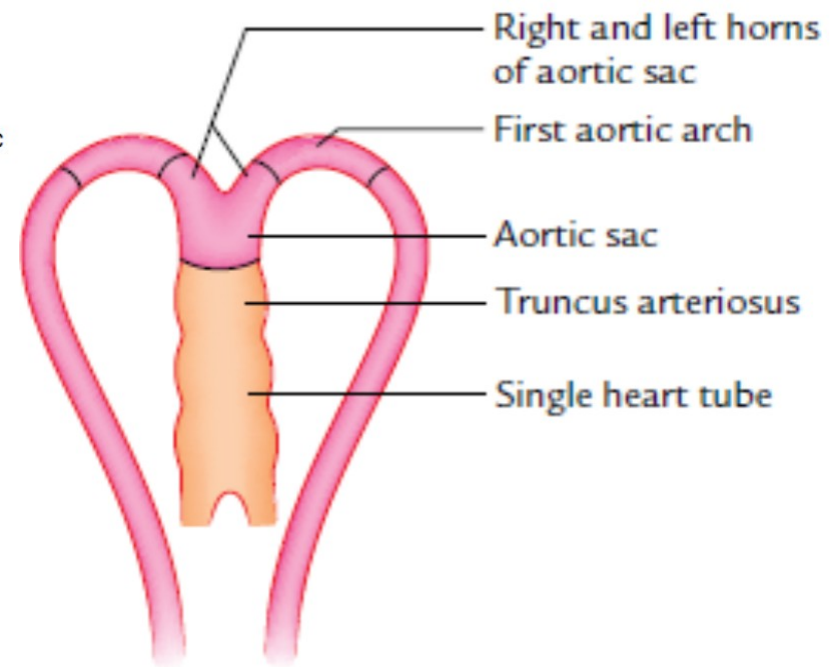
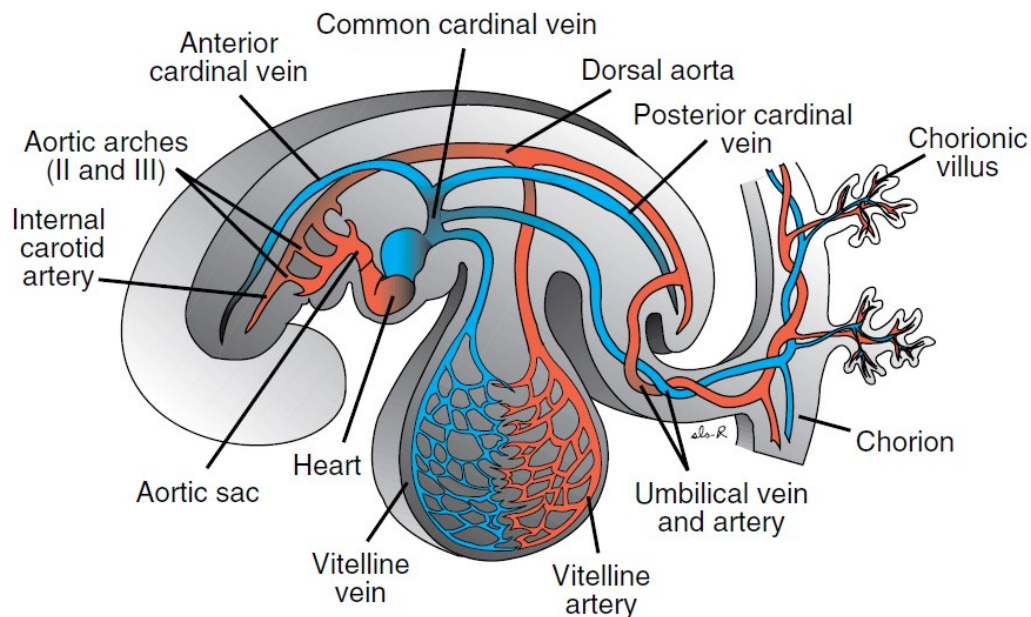
Development of vascular system

Development of vascular system

- The blood vessels develop from mesenchymal cells **derived from mesoderm**.
- Occurs by two mechanisms:
 - **Vasculogenesis** – formation of new vessels by union of angioblast cells.
 - dorsal aorta & cardinal veins are formed
 - **Angiogenesis** – vessels form by budding and branching from preexisting vessels. E.g. remaining vessels

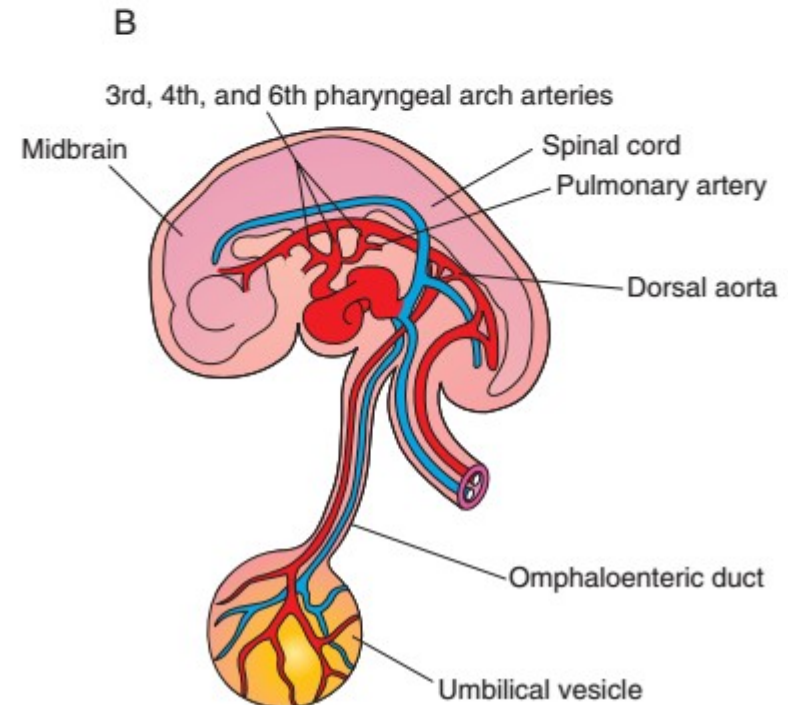
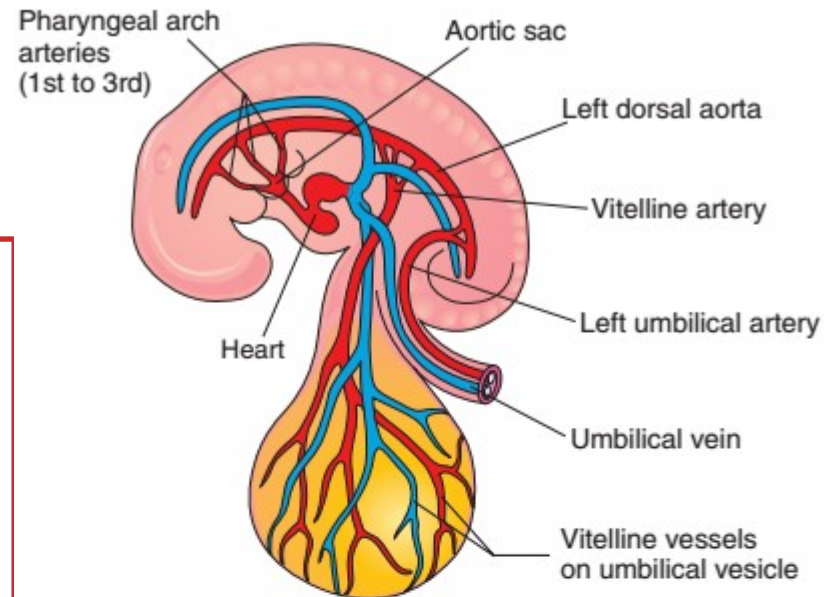
Development of Arterial system

- The arteries of body develop from two main sources:
 - **Pharyngeal (aortic) arch arteries** – give rise to the arteries of head and neck region.
 - **Dorsal aorta** – gives rise to the arteries of rest of the body

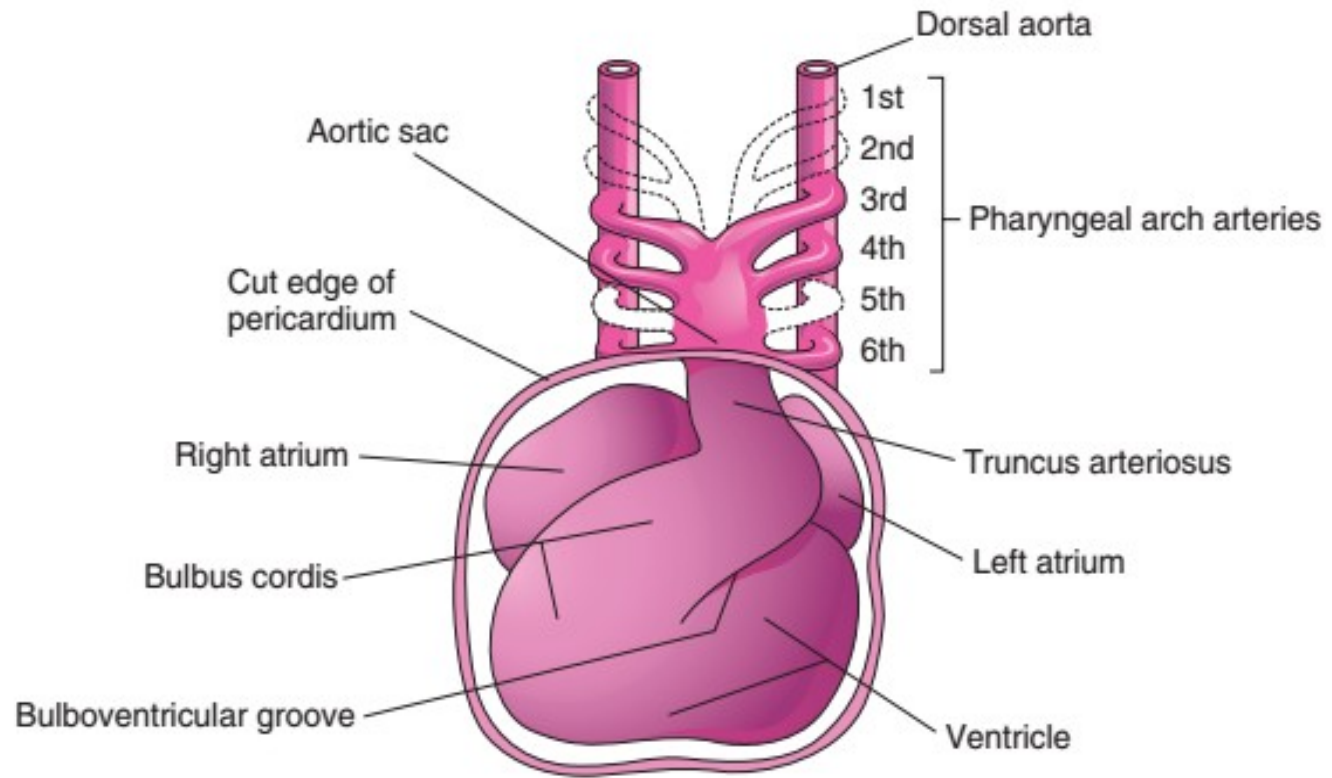


Pharyngeal Arch Arteries

- **Aortic sac** – distal part of the truncus arteriosus
- **Pharyngeal (aortic) arch arteries** arise from the aortic sac and terminate in the right and left dorsal aortae
 - embedded in mesenchyme of the pharyngeal arches
 - Neural crest cells contribute to their formation
 - **Six pairs** of arteries develop in a cranial-to-caudal sequence (The fifth arch and artery may not develop or regress)
 - numbered as I, II, III, IV, and VI



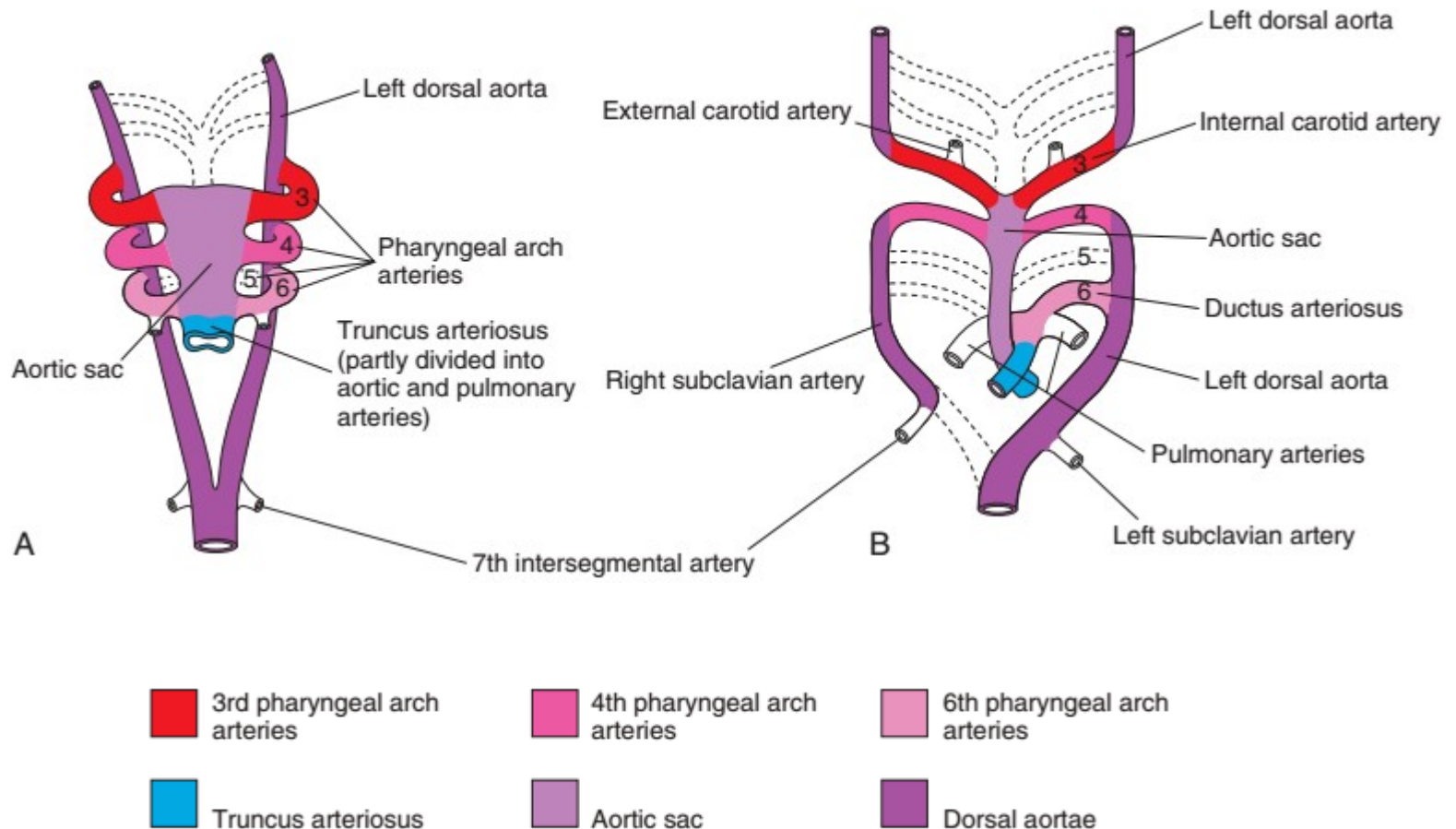
Pharyngeal arches and dorsal aorta



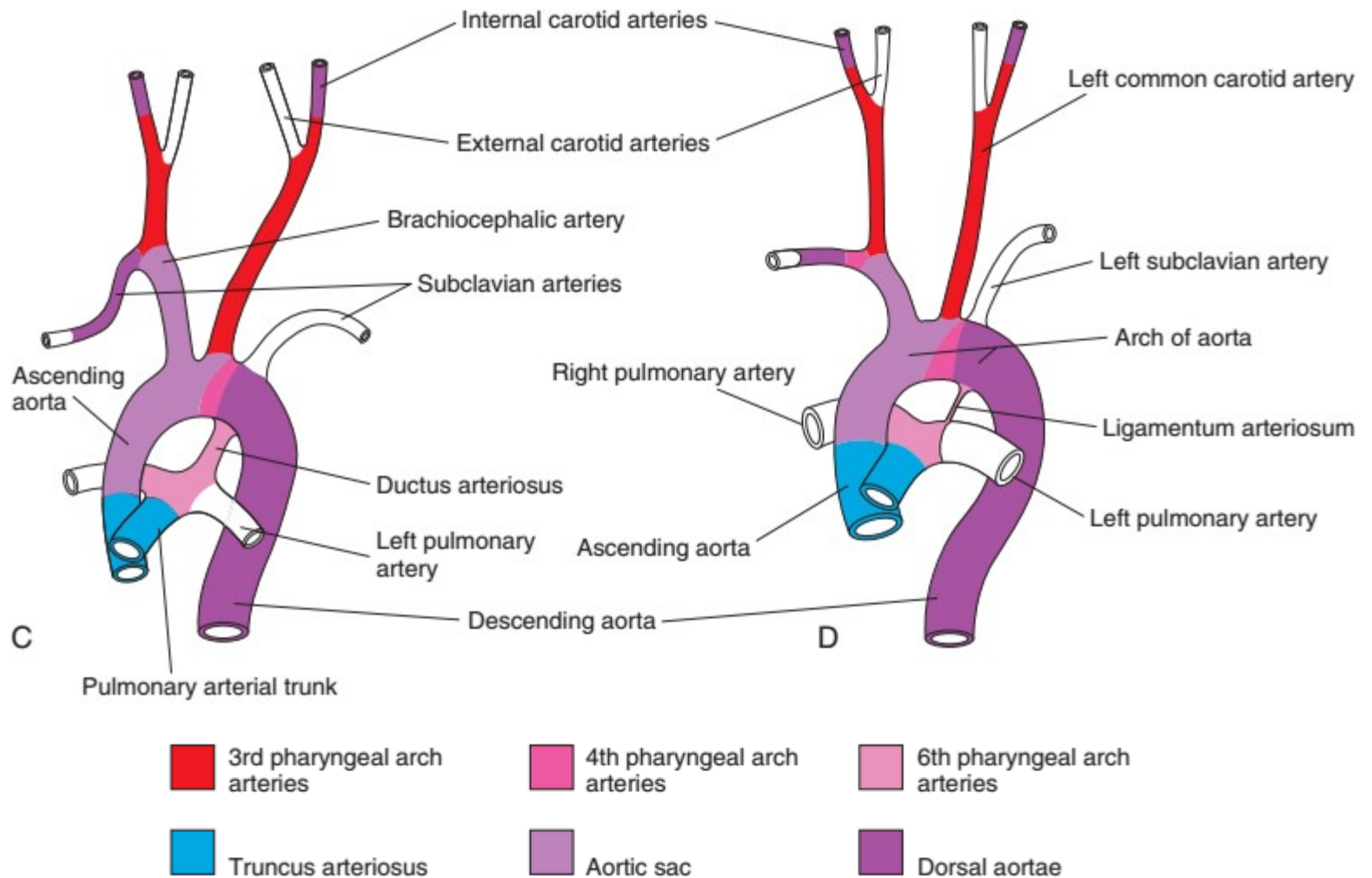
Derivatives of Pharyngeal Arch Arteries

- Derivatives of the **1st pharyngeal arch artery**
 - Mostly disappears, remnants give the **terminal part of maxillary artery**
- Derivatives of the **2nd pharyngeal arch artery**
 - Disappears, except, the embryological hyoid and stapedial arteries
- Derivatives of the **3rd pharyngeal arch artery**
 - Develops into:
 - **Common carotid artery**
 - **Root of internal carotid artery**
 - Each third arch artery gives off a bud that grows cranially to form the **external carotid artery**.

Derivatives of Pharyngeal Arch Arteries



Derivatives of Pharyngeal Arch Arteries

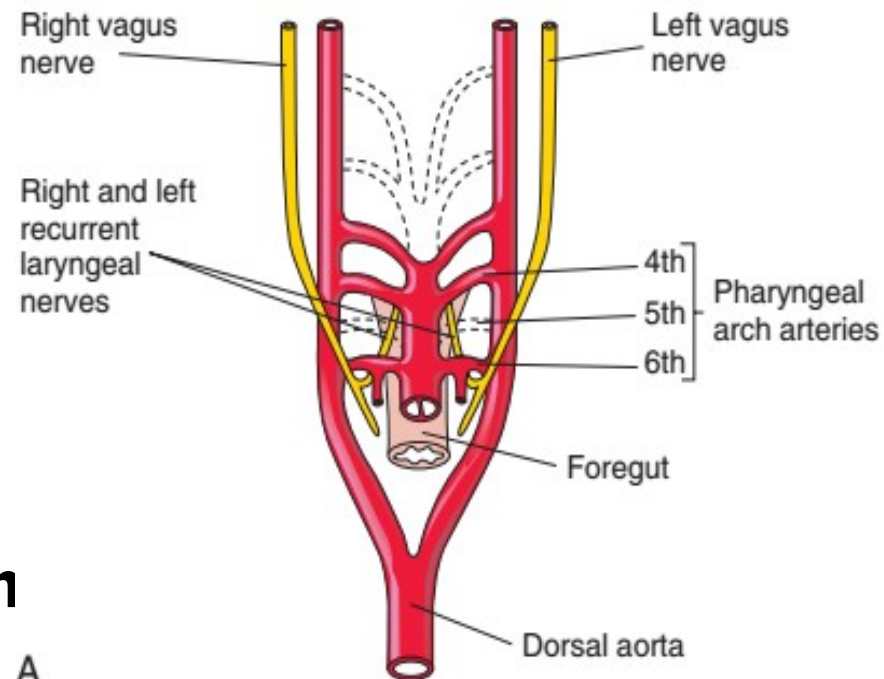


Derivatives of 4th Pharyngeal arch artery

- **4th Pharyngeal arch artery**
 - proximal part of the arch artery develops from the aortic sac
 - distal part is derived from the left dorsal aorta
- Derivatives of the **4th Pharyngeal arch artery**:
 - a. On the left side becomes **part of the arch of aorta**
 - b. On the right side becomes **proximal part of the right subclavian artery**
 - distal part of right subclavian artery forms from the right dorsal aorta and the right seventh intersegmental artery
 - **left subclavian artery** forms from the **left seventh intersegmental artery**

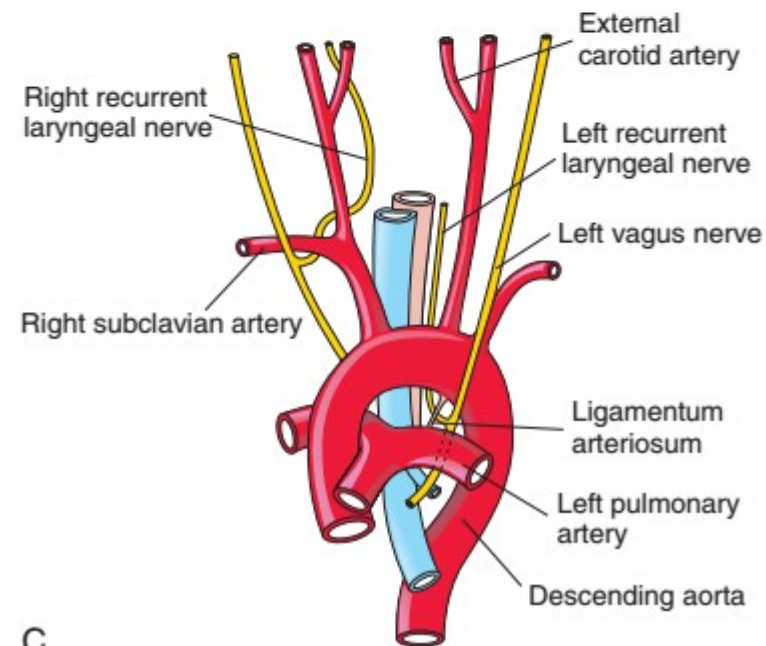
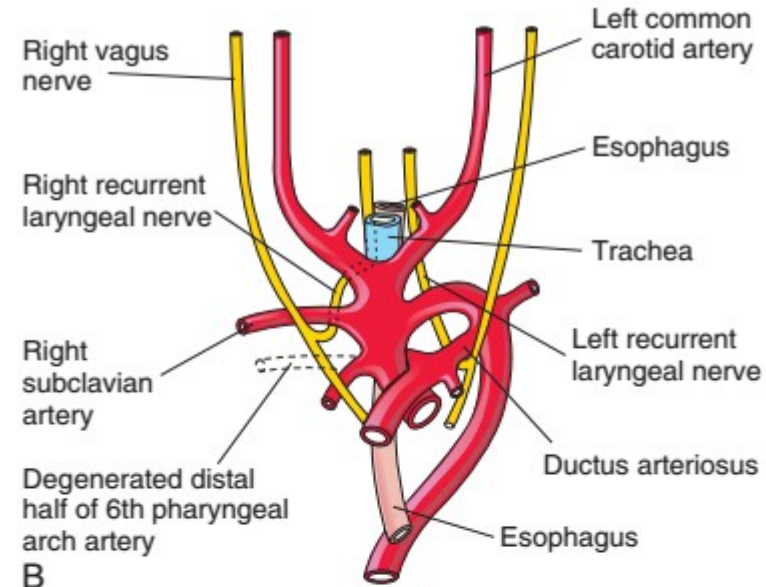
Derivatives of the 6th pharyngeal arch artery

- **6th pharyngeal arch artery** gives:
 - a. **Right & left pulmonary artery**, respectively from the right & left 6th pharyngeal arch arteries
 - b. **Ductus arteriosus** from the **left 6th pharyngeal arch artery**
 - after birth becomes **ligamentum arteriosum**
- Functional closure ductus arteriosus occurs within 24 hours after birth
- Anatomic closure and formation of the **ligamentum arteriosum** occur by the 12th postnatal week A



Derivatives of the 6th pharyngeal arch artery

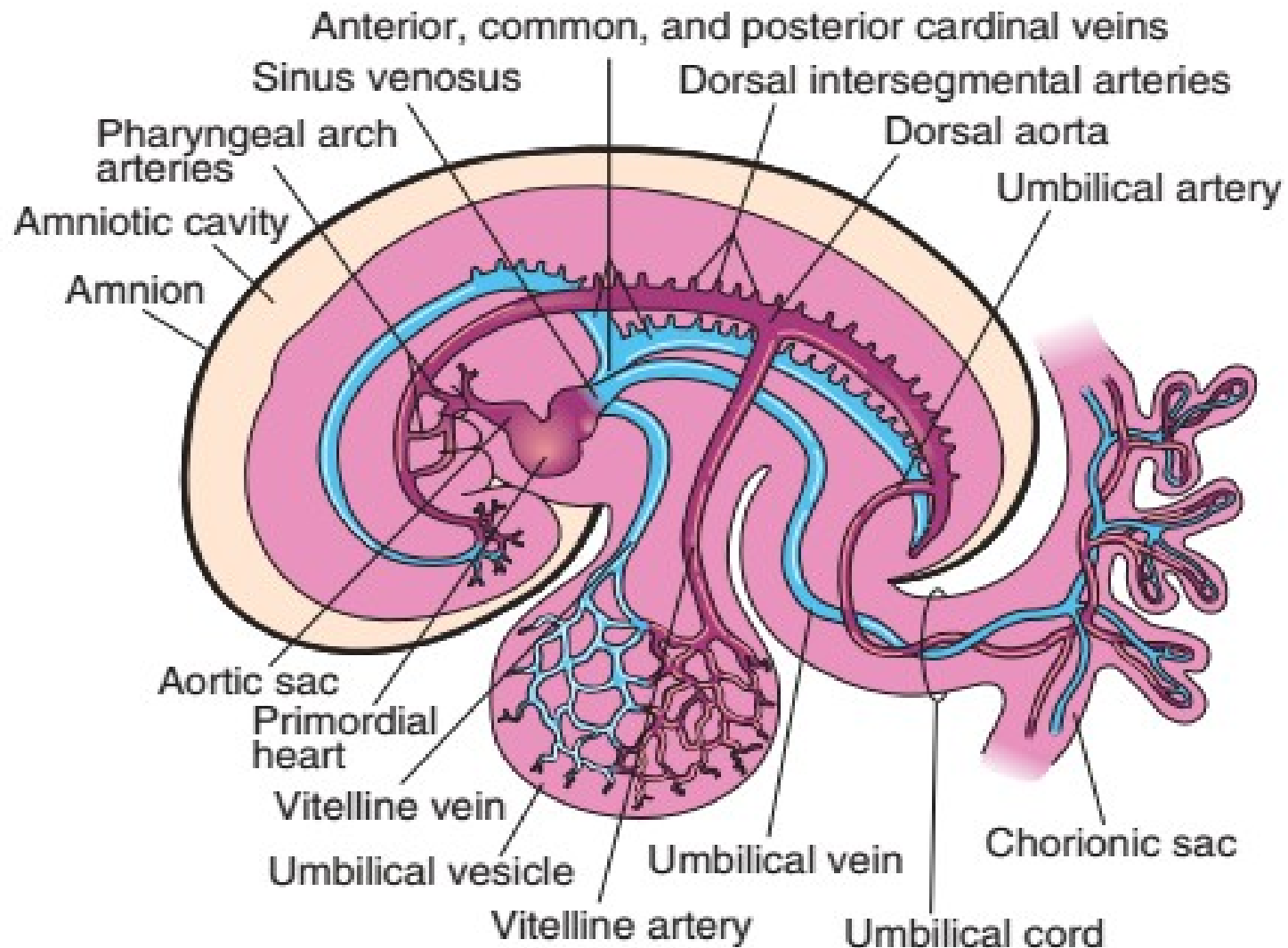
- On the right side, portion of the sixth arch artery beyond the artery to lung bud disappears
- On the left side, portion of the sixth arch artery beyond the artery to lung bud remains patent and forms the **ductus arteriosus**.



Dorsal aorta

- **Dorsal aorta**
 - first pair of arteries to develop
 - becomes continuous with two primitive heart tubes through the 1st pharyngeal arch artery
 - Initially, they run through the entire length of the embryo.
 - Later, the caudal portions **fuse to** form a single lower **thoracic/ abdominal aorta**.
 - Of the remaining paired dorsal aortae
 - **the right one regresses**
 - the **left one** becomes the **primordial aorta**.

Embryonic cardiovascular system – day 26



Intersegmental Arteries

- Thirty or more branches of the dorsal aorta
- pass between and carry blood to the somites and their derivatives.
- In the neck join to form the **vertebral artery**.
- In the thorax becomes **intercostal arteries**.
- In the abdomen become **lumbar arteries**
- Fifth pair of lumbar intersegmental arteries remains as the **common iliac arteries**.
- In the sacral region form the **lateral sacral arteries**.

Derivatives of vitelline arteries

- **Vitelline arteries**
 - unpaired ventral branches of the dorsal aorta supplying umbilical vesicle.
 - pass to the primordial gut, which forms from the incorporated part of the umbilical vesicle.
- Only **three vitelline artery** derivatives remain in adult:
 - **Celiac trunk** to the foregut
 - **Superior mesenteric artery** to the midgut
 - **Inferior mesenteric artery** to the hindgut

Derivatives of umbilical arteries

- **Umbilical arteries**

- paired ventral branches of the dorsal aorta which pass through the umbilical cord to the placenta.
- carry poorly oxygenated blood to the placenta.
- proximal parts – become **internal iliac arteries** and **superior vesical arteries**.
- distal parts – obliterated and becomes **medial umbilical ligaments**

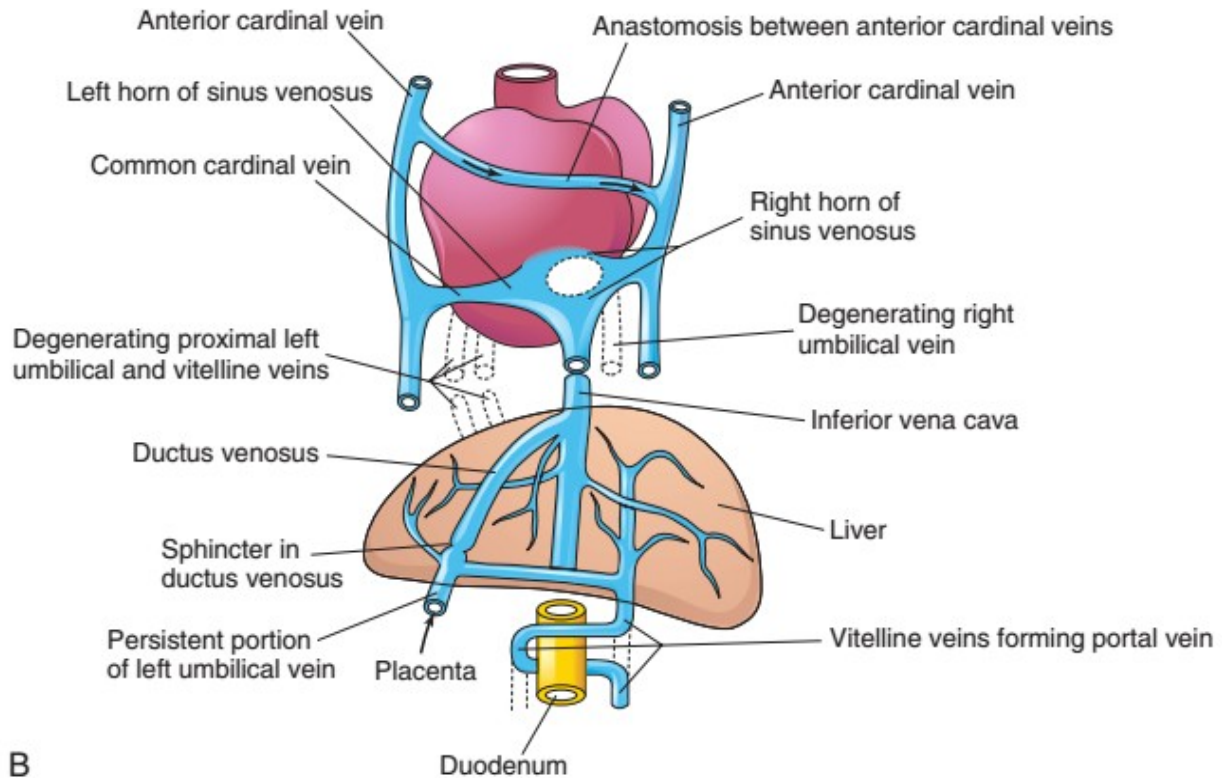
Development of Venous system

- The primitive veins of embryo are classified into two groups:
 - **Visceral Veins** – pairs of vitelline veins & umbilical veins
 - **Somatic Veins** – right and left anterior & posterior cardinal veins.
- Each horn of sinus venosus receives **three** paired primitive veins:
 - **Vitelline vein** from the **yolk sac (umbilical vesicle)**
 - **Umbilical vein** from the **placenta**
 - **Common cardinal vein** from the **body wall of the embryo**

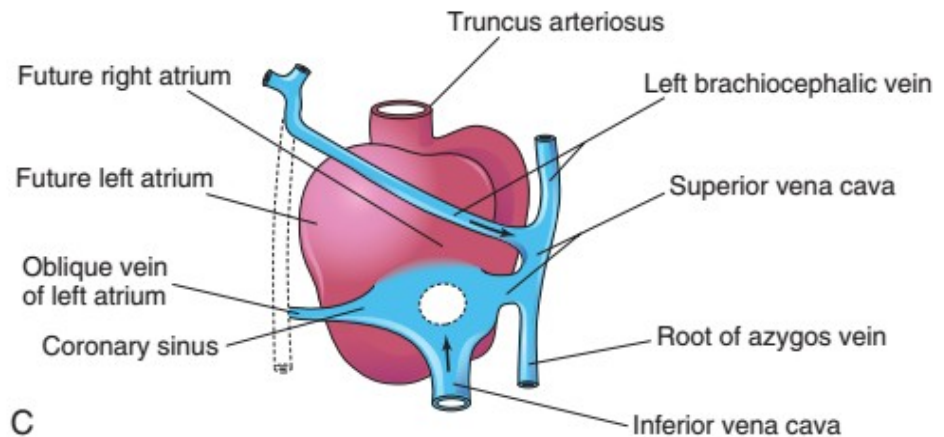
Fate of Vitelline veins

- arise from the yolk sac (umbilical vesicle) and pass to the sinus venosus
- **Left vitelline vein regresses**
- **Right vitelline vein forms most of the hepatic portal system, hepatic veins & a portion of IVC**
- The hepatic cords within the liver primordium anastomose around endothelium-lined spaces, the primordia of the hepatic sinusoids.
- These spaces later become linked to the vitelline veins.
- The **portal vein** develops from an anastomotic network of vitelline veins around the duodenum

Fate of Vitelline, umbilical & cardinal veins



B

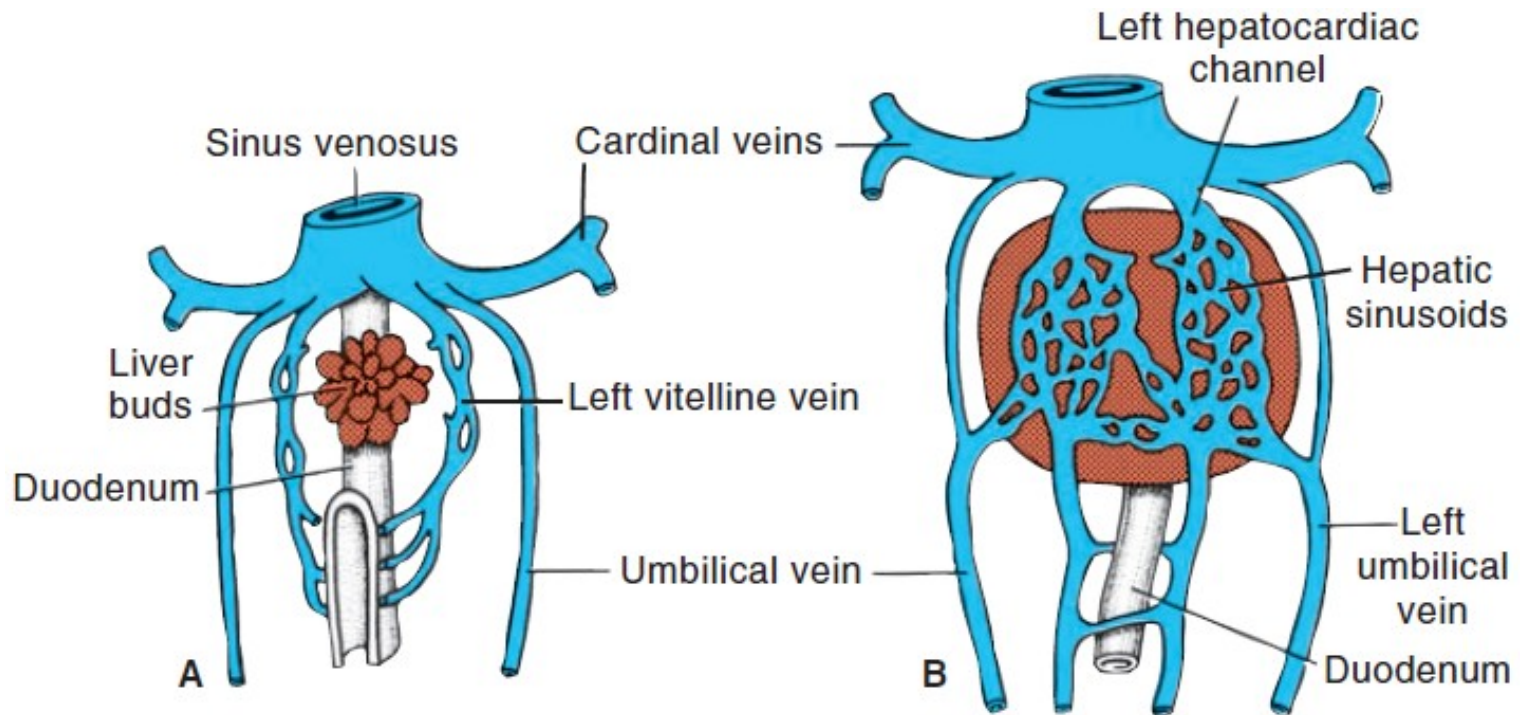


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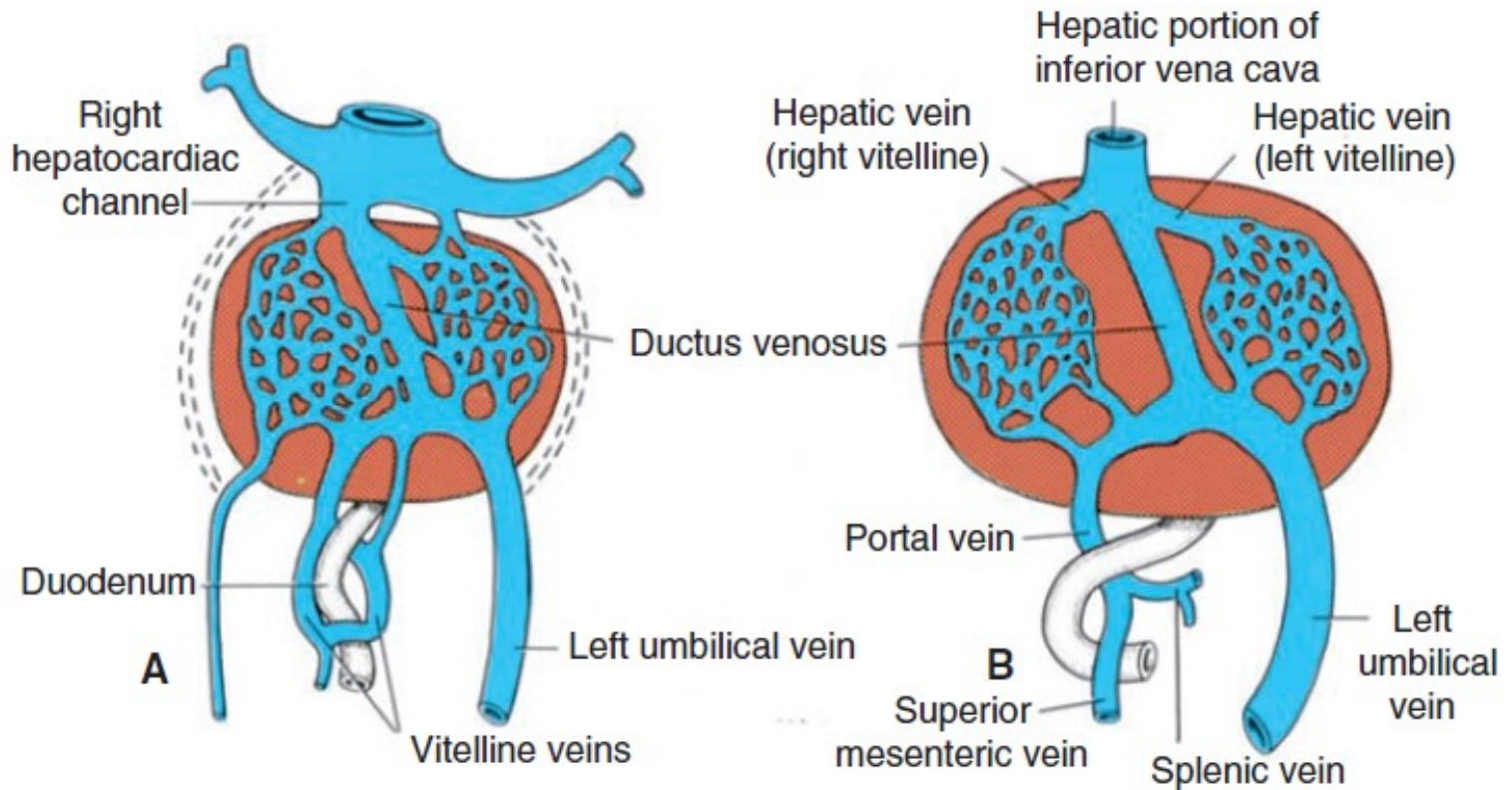
Fate of Umbilical veins

- run on each side of the liver and carry well-oxygenated blood from the placenta to the sinus venosus
- As the liver develops, the umbilical veins lose their connection with the heart and empty into the liver
 - The **right umbilical vein** degenerates during the 7th week
 - The **left umbilical vein**
 - cranial part between the liver and the sinus venosus degenerates
 - persistent caudal part becomes the **umbilical vein**
 - becomes the round ligament of the liver (**ligamentum teres**)
 - A large venous shunt—the **ductus venosus**—develops within the liver and connects the umbilical vein with the IVC.
 - becomes the **ligamentum venosum**

Fate of Vitelline, umbilical & cardinal veins



Fate of Vitelline, umbilical & cardinal veins



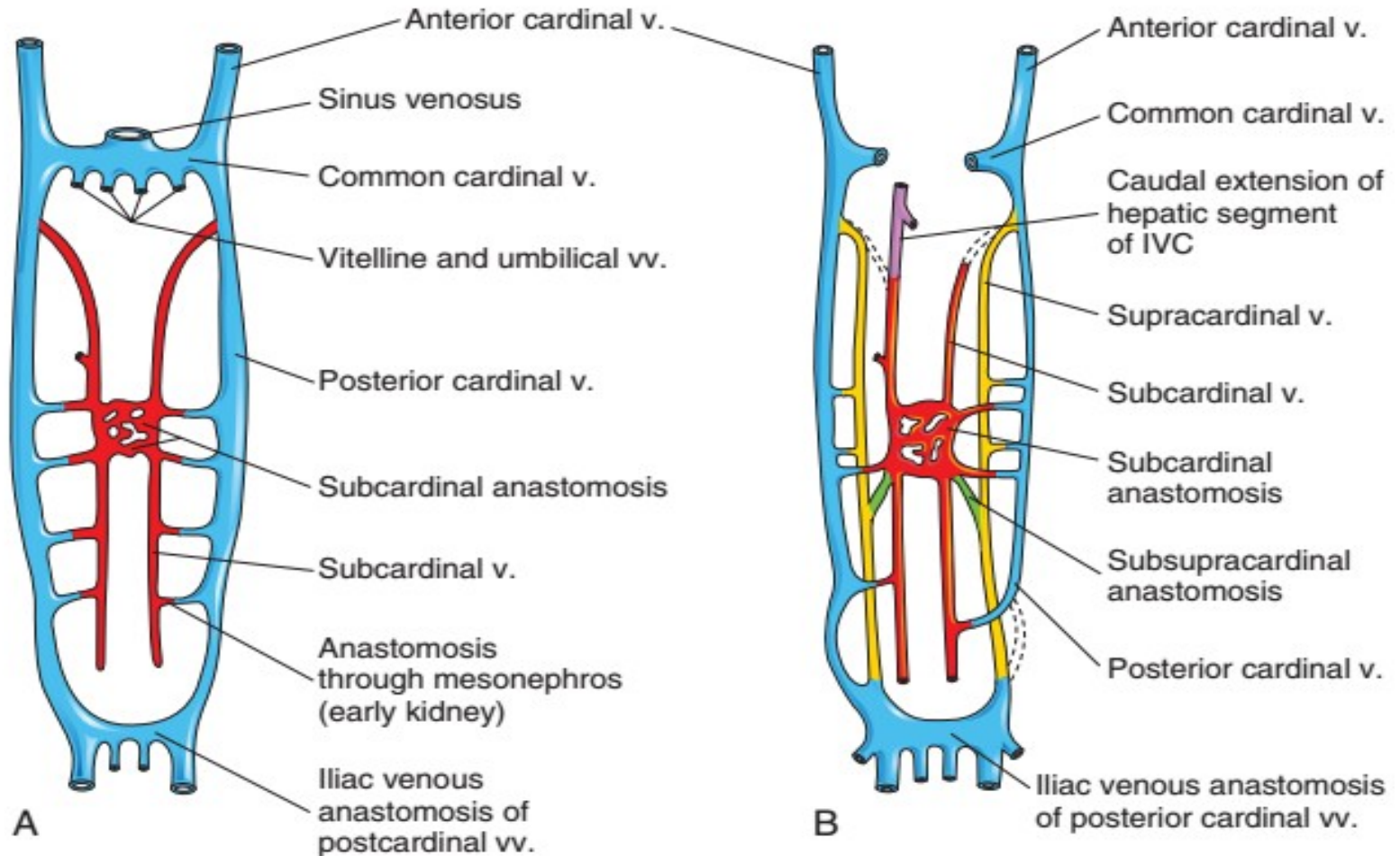
Fate of Cardinal veins

- The anterior and posterior cardinal veins drain the cranial and caudal parts of the embryo, respectively.
 - These join the **common cardinal veins**, which enter the sinus venosus.
- The left & right anterior cardinal veins are connected by an oblique anastomosis that shunts from left to the right
 - caudal part of the left anterior cardinal vein **degenerates**
 - The shunt becomes the **left brachiocephalic vein**
- **SVC** forms from the right anterior cardinal vein and the right common cardinal vein.
- Posterior cardinal veins in adults become the **root of the azygos vein** and the **common iliac veins**.

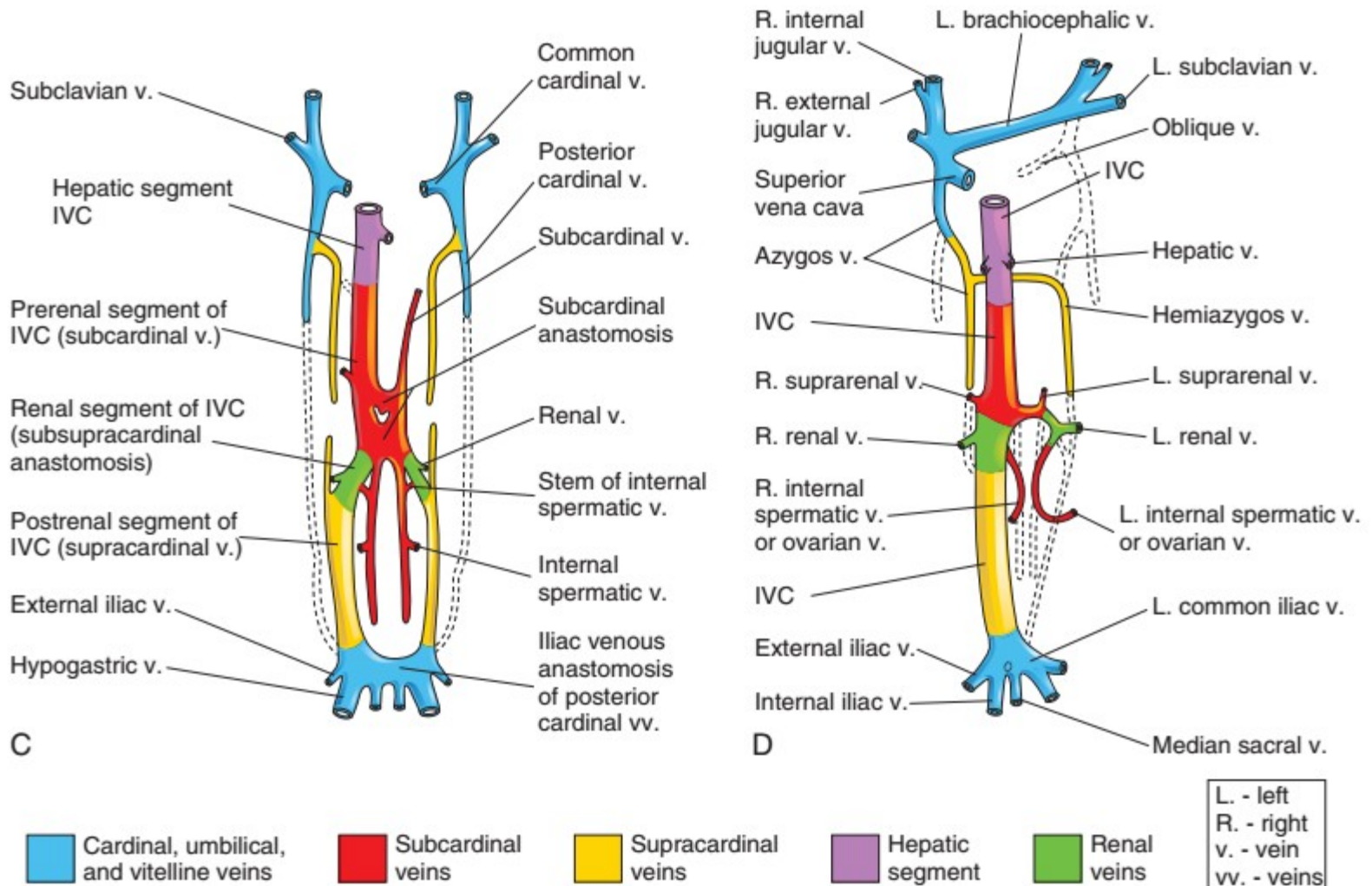
Fate of Cardinal veins

- The subcardinal and supracardinal veins gradually replace and supplement the posterior cardinal veins.
- **Subcardinal veins form**
 - stem of the left renal vein
 - suprarenal veins
 - gonadal veins (testicular and ovarian)
 - segment of the IVC
- **Supracardinal veins**
 - become disrupted in the region of the kidneys.
 - Cranial to the kidneys, anastomose to form the **azygos and the hemiazygos veins**.
 - Caudal to the kidneys, the left supracardinal vein degenerates but the right supracardinal vein becomes the inferior part of the IVC.

Fate of Vitelline, umbilical & cardinal veins



Fate of Vitelline, umbilical & cardinal veins



Development of Inferior Vena Cava

- The IVC is composed of **four main** segments:
 - **hepatic segment** derived from the hepatic vein (proximal part of the right vitelline vein) and hepatic sinusoids
 - **prerenal segment** derived from the right subcardinal vein
 - **renal segment** derived from the subcardinal–supracardinal anastomosis
 - **postrenal segment** derived from the right supracardinal vein

Clinical:

- Coarctation of aorta
 - Postductal coarctation
 - Preductal coarctation
- Patent Ductus Arteriosus (PDA)

